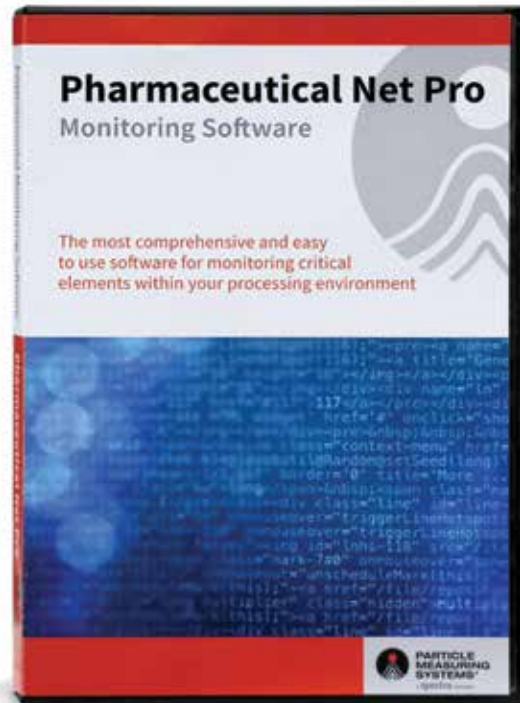




Pharmaceutical Net Pro

USER GUIDE

Pharmaceutical Net Pro

User Guide



HEADQUARTERS

5475 Airport Blvd
Boulder, Colorado 80301 USA
T: +1 303 443 7100, +1 800 238 1801

Instrument Service & Support

E: +1 800 557 6363

Customer Response Center

T: +1 877 475 3317
E: Info@pmeasuring.com

GLOBAL OFFICES

AUSTRIA

T: +43 1 71 728 285
E: pmsaustria@pmeasuring.com

BENELUX

T: +32 10 23 71 56
E: pmsbenelux@pmeasuring.com

BRAZIL

T: +55 11 5188 8227
E: pmsbrazil@pmeasuring.com

CHINA

T: +86 21 6113 3600
E: pmschina@pmeasuring.com

FRANCE

T: +33 (0)1 60 10 32 96
E: pmsfrance@pmeasuring.com

GERMANY

T: +49 6151 6671 632
E: pmsgermany@pmeasuring.com

ITALY

T: +39 06 9053 0130
E: pmsrl@pmeasuring.com

JAPAN

T: +81 3 5298 8175
E: pmsjapan@pmeasuring.com

KOREA

T: +82 31 286 5790
E: pmskorea@pmeasuring.com

MEXICO

T: +52 55 2271 5106
E: pmsmexico@pmeasuring.com

NORDIC

T: +45 707 028 55
E: pmsnordic@pmeasuring.com

PUERTO RICO

T: +1 787 718 9096
E: pmspuertorico@pmeasuring.com

SINGAPORE

T: +65 6496 0330
E: pmssingapore@pmeasuring.com

SWITZERLAND

T: +41 71 987 01 01
E: pmsswitzerland@pmeasuring.com

TAIWAN

T: +886-3-5525300 Ext: 301
E: pmstaiwan@pmeasuring.com

Pharmaceutical Net Pro User Guide
P/N 1000024849 Rev B
© 2019 Particle Measuring Systems, Inc.
All rights reserved.

Airnet[®], BioCapt[®], FacilityPro[®], and IsoAir[®] are registered trademarks of Particle Measuring Systems.

Microsoft[®] is a registered trademark of Microsoft Corporation.

All trademarks appearing in this manual are the property of their respective owners.

DO NOT REPRODUCE OR DISTRIBUTE

CONFIDENTIAL DOCUMENT

This confidential document contains proprietary information, which is protected by copyright. All rights are reserved. No part of this document may be reproduced, distributed, or transmitted in any form without the prior written consent of Particle Measuring Systems. The information contained in this document is subject to change without notice.

Quality Statement

The Quality Policy of Particle Measuring Systems is to strive to meet or exceed the needs and expectations of our customers, and to align the activities of all employees with the common focus of customer satisfaction through continuous improvement in the quality of our products and services.

Environmental Statement

There are no potential means by which asbestos exposure is possible because there are no materials in the design or manufacture of the FacilityPro containing asbestos.



This equipment must be properly disposed of at end-of-life by means of an authorized waste management system. Contact our Customer Response Center at (877) 475-3317 or (303) 443-7100 (International Telephone +1 3034437100) for dismantling and disposal information.

Manual Conventions

WARNING

A warning in the text is used to notify the user of the potential for bodily injury or death.

CAUTION

A caution in the text is used to highlight an item that if not done, or incorrectly done, could damage the instrument and/or any materials or devices affected by the instrument.

— — NOTICE — —

A notice in the text is an instructional communication regarding requirements or policies issued by Particle Measuring Systems.

NOTE: A note in the text is used to highlight an item that is of operational importance to the user.

It is important that you observe cautions and warnings while performing the procedures described in this manual. Caution and warning labels are located on and inside the instrument to alert you to potentially hazardous conditions. Please familiarize yourself with this information.

This page is intentionally left blank.

Brief Table of Contents

List of Figures	xiii
List of Tables	xvii
Chapter 1: Overview	1-1
Chapter 2: Installation	2-1
Chapter 3: Get Started	3-1
Chapter 4: Recipes	4-1
Chapter 5: Sampling	5-1
Chapter 6: Sensors	6-1
Chapter 7: Alarms	7-1
Chapter 8: Reports	8-1
Chapter 9: Real Time Trend	9-1
Chapter 10: Run Time Logic	10-1
Chapter 11: User Management	11-1
Chapter 12: System Tasks	12-1
Chapter 13: Aventino Setup Options	13-1
Appendix A: Statistical Data Analyzer	A-1
Index	Index-1

This page is intentionally left blank.

Table of Contents

Quality Statement - - - - -	ii
Environmental Statement - - - - -	ii
Manual Conventions - - - - -	iii
List of Figures - - - - -	xiii
List of Tables - - - - -	xvii
Chapter 1: Overview - - - - -	1-1
Configuration Options - - - - -	1-2
Software Architecture - - - - -	1-2
Server Software Description - - - - -	1-4
Interface - - - - -	1-4
Electronic Signature - - - - -	1-5
Electronic Signature window - - - - -	1-5
Specifications - - - - -	1-6
Chapter 2: Installation - - - - -	2-1
Prerequisites - - - - -	2-1
Pharmaceutical Net Pro Installation - - - - -	2-1
Chapter 3: Get Started - - - - -	3-1
System Start-Up - - - - -	3-1
Components of the Pharmaceutical Net Pro software screen - - - - -	3-2
Header - - - - -	3-2
Central area - - - - -	3-2
Buttons row - - - - -	3-2
Main Page Window - - - - -	3-5
Log In and Log Out - - - - -	3-6
Log in window - - - - -	3-6
Chapter 4: Recipes - - - - -	4-1
Recipes Editor Window - - - - -	4-1
Create a New Recipe - - - - -	4-5
Edit an Existing Recipe - - - - -	4-5
Delete an Existing Recipe - - - - -	4-6
Chapter 5: Sampling - - - - -	5-1
Sampling Editor Window - - - - -	5-1
Schedule a Sampling - - - - -	5-3
Sampling on input editor dialog box - - - - -	5-3
Abort a Sampling Event - - - - -	5-5

Edit a Scheduled Sampling-----	5-5
Delete a Scheduled Sampling-----	5-5
Configuring Custom Fields to Apply to a Sampling-----	5-6
Custom fields configuration window-----	5-6
Chapter 6: Sensors-----	6-1
Particle counter Window-----	6-1
Particle Sensors Values-----	6-3
Biocapt Window-----	6-6
Biocapt Values-----	6-6
Analog Inputs Window-----	6-9
Analog Inputs-----	6-9
Chapter 7: Alarms-----	7-1
Alarms Window-----	7-1
Acknowledge an Alarm-----	7-3
Chapter 8: Reports-----	8-1
Overview-----	8-1
Reports available through the Report Generator-----	8-1
Reports available through the Sampling Report window-----	8-1
Report Window-----	8-2
Configuration of Tags Catalog-----	8-3
Tags Catalog tabs-----	8-4
Sensor Type-----	8-4
Sensors Alarms-----	8-5
Sensors-----	8-6
Data Types-----	8-7
Sensor Groups-----	8-8
Data Type Groups-----	8-9
Recipes-----	8-10
Tags Control-----	8-11
Pens Color Selection-----	8-12
History Automation Configuration-----	8-13
Configure Sensor Group Tags-----	8-16
Configure Recipe Tags-----	8-18
Configure Data Type Group Tags-----	8-19
Creating Reports through the Report Window-----	8-20
Create a report “by group”-----	8-21
Set a date and time range for a report-----	8-24
Select a data subset-----	8-25
Select a group of modules to include in the report-----	8-25
Select report(s) to generate-----	8-25
Select an interval, specify the type for the report-----	8-26
For a Trend Report, set calculation parameters-----	8-26
Execute the report-----	8-27
Create a report “by sampling”-----	8-28
Set a date and time range for a report-----	8-30
Select a recipe, a sampling, and filters-----	8-31

Select a report to generate	8-32
Select an interval, specify the type for the report	8-33
For a Trend Report, set calculation parameters	8-33
Execute the report	8-33
Change the View in the Report Displayed in the Report Window	8-35
Print a Report from the Report Window	8-37
Export a Report from the Report Window	8-37
Report Examples	8-38
Audit Report examples – Select by Group – Audit Report	8-38
Header information – Audit Report – Select by Group	8-38
Report body – Audit Report – Select by Group	8-39
Footer information – Audit Report – Select by Group	8-39
Last page – Audit Report – Select by Group	8-39
Audit Report examples – Select by Sampling Audit Report	8-40
Header information – Audit Report – Select by Sampling	8-40
Report body – Audit Report – Select by Sampling	8-41
Footer information – Audit Report – Select by Sampling	8-41
Last page – Audit Report – Select by Sampling	8-41
Data Report examples – Select by Group	8-42
Header information – Data Report – Select by Group	8-42
Report body – Data Report – Select by Group	8-42
Report summary – Data Report – Select by Group	8-43
Footer information – Data Report – Select by Group	8-43
Last page – Data Report – Select by Group	8-43
Data Report examples – Select by Samplings	8-44
Header information – Data Report – Select by Samplings	8-44
Report body – Data Report – Select by Samplings	8-45
Report summary – Data Report – Select by Samplings	8-45
Footer information – Data Report – Select by Samplings	8-45
Last page – Data Report – Select by Samplings	8-45
Trend Report examples – Select by Samplings	8-46
Header information – Trend Report – Select by Samplings	8-46
Header information - Trend Report Exported PDF - Select by Group -	8-47
Footer information – Trend Report Exported PDF	8-47
Last page – Trend Report	8-47
Sampling Report	8-48
Sampling Report tab	8-48
Create a Sampling Report	8-49
Change the view in report displayed in the Sampling Report window - -	8-49
Print a Sampling Report	8-49
Export a Sampling Report from the Sampling Report Window-	8-50
Sampling Report components-	8-50
Header information – Sampling Report	8-50
Report body – First section – Sampling Report	8-51
Report body – Second section – Sampling Alarms – Sampling Report	8-51
Footer information – Sampling Report	8-51
Last page	8-51

Chapter 9: Real Time Trend	9-1
Real Time Trend Window	9-1
Starting and Stopping a Real Time Trend	9-3
Editing Pens of the Recipe	9-4
Chapter 10: Run Time Logic	10-1
Available Tags pane	10-2
Analog Input Folder	10-3
BioCapt (BC) Folder	10-4
Digital Input (DI) Folder	10-5
Digital Output (DO) Folder	10-5
Virtual Digital Input/Output (DX) Folder	10-6
I/O Module Folder	10-6
Particle Counter Folder	10-6
Pumps Group Folder	10-8
Pneumatic Module Folder	10-9
System Folder	10-9
Usage for Tag pane	10-11
Editor pane	10-11
Rule Editor	10-12
Rule Type	10-12
Example Rule Editor Rule Creation	10-13
Creating a Child Rule	10-13
Sampling on Input Editor	10-14
Tower Lights	10-15
Example Tower Light Rule Creation	10-17
Testing Output Activation	10-18
Chapter 11: User Management	11-1
Default Users	11-1
“Nobody” user	11-1
User Groups	11-1
Group functions	11-2
Emergency User	11-3
Managing Users	11-3
User Management window	11-3
Create a New User	11-5
Modify a User definition	11-6
Delete a User	11-6
Chapter 12: System Tasks	12-1
System management Window	12-1
Sensors Editor	12-5
Sensors Editor window	12-5
Backup and Restore	12-8
Daily backup configuration	12-8
Manual backup	12-9
Copy backup files	12-9
Close the Pharmaceutical Net Pro software application	12-9

Restart the System - - - - -	12-9
Shutdown the System - - - - -	12-10
Shutdown a Disconnected Client - - - - -	12-10
Change the Display Language - - - - -	12-11
Reset the Buffer - - - - -	12-12
Limits Editor - - - - -	12-13
Limits Editor window - - - - -	12-13
Limits Editor History window - - - - -	12-16
Limits Editor Configuration window - - - - -	12-18
Analog Inputs Limits - - - - -	12-20
Chapter 13: Aventino Setup Options - - - - -	13-1
Opening the Aventino Wizard Application - - - - -	13-1
New Installation - - - - -	13-2
Upgrade from Previous Version - - - - -	13-12
Restore from Backup - - - - -	13-12
Modify Configuration - - - - -	13-13
Restore FacilityPro - - - - -	13-13
Appendix A: Statistical Data Analyzer - - - - -	A-1
Licensing the Application - - - - -	A-1
Running the Application - - - - -	A-1
Application Window - - - - -	A-2
Loading CSV Files - - - - -	A-2
Display Configuration - - - - -	A-4
Axis Adjustment - - - - -	A-4
Zoom - - - - -	A-5
Interpolation - - - - -	A-5
Saving a Configuration - - - - -	A-6
Loading a Previous Configuratoion - - - - -	A-6
Virtual Pens - - - - -	A-6
Markers and Sensor Display Toggling - - - - -	A-7
Daily Sensor Chart - - - - -	A-7
Creating a Report - - - - -	A-7
Language - - - - -	A-8
Index - - - - -	Index-1

This page is intentionally left blank.

List of Figures

Chapter 1: Overview	1-1
Figure 1-1 Architecture of Pharmaceutical Net Pro software system	1-2
Figure 1-2 Pharmaceutical Net Pro network (with FacilityPro 3010 Processor Module as an example)	1-3
Figure 1-3 Electronic Signature window	1-6
Chapter 2: Installation	2-1
Figure 2-1 Pharmaceutical Net Pro Installer	2-1
Figure 2-2 Installation status	2-2
Figure 2-3 Network card popup	2-2
Figure 2-4 Hard disk popup	2-2
Figure 2-5 Assign computer name window	2-3
Figure 2-6 Local or domain user option	2-3
Figure 2-7 License Manager	2-4
Chapter 3: Get Started	3-1
Figure 3-1 Pharmaceutical Net Pro Main Page window	3-1
Figure 3-2 Log in window	3-6
Chapter 4: Recipes	4-1
Figure 4-1 Recipes Editor window	4-1
Figure 4-2 Recipes Editor confirm delete dialog box	4-6
Chapter 5: Sampling	5-1
Figure 5-1 Sampling Editor window	5-1
Figure 5-2 Sampling on input editor dialog box	5-3
Figure 5-3 Custom fields configuration window	5-6
Chapter 6: Sensors	6-1
Figure 6-1 Pharmaceutical Net Pro Particle counter window	6-2
Figure 6-2 Sampling point – Particle sensors	6-4
Figure 6-3 Biocapt window	6-6
Figure 6-4 Sampling point – Biological	6-7
Figure 6-5 Analog Inputs window	6-9
Chapter 7: Alarms	7-1
Figure 7-1 Alarms window	7-1
Chapter 8: Reports	8-1
Figure 8-1 Report window, Select by Group tab fields	8-2

Figure 8-2	Tags catalog window, Sensor Type tab	8-4
Figure 8-3	Tags catalog window, Sensors Alarms tab	8-5
Figure 8-4	Tags catalog window, Sensors tab	8-6
Figure 8-5	Tags catalog window, Data Types tab	8-7
Figure 8-6	Tags catalog window, Sensor Groups tab	8-8
Figure 8-7	Tags catalog window, Data Type Groups tab	8-9
Figure 8-8	Tags catalog window, Recipes tab	8-10
Figure 8-9	Tags catalog window, Tags Control tab	8-11
Figure 8-10	Tags catalog window, Pens Color Selection tab	8-12
Figure 8-11	Tags catalog window, History Automation Config tab	8-13
Figure 8-12	Tags catalog window, Sensor Groups tab	8-16
Figure 8-13	Select sensors in group window	8-17
Figure 8-14	Tags catalog window, Recipes tab	8-18
Figure 8-15	Tags catalog window, Data Type Groups tab	8-19
Figure 8-16	Parameter fields in the Report window for Select by Group reports	8-21
Figure 8-17	Calendar popup	8-24
Figure 8-18	Input a comment to print on reports dialog box	8-27
Figure 8-19	Parameter fields in the Report window for Select by Sampling reports	8-28
Figure 8-20	Calendar popup	8-30
Figure 8-21	Input a comment to print on reports dialog box	8-33
Figure 8-22	Audit Report window controls for changing the view of a generated report	8-35
Figure 8-23	Insert alarm ID dialog box	8-36
Figure 8-24	Select pages to print dialog box	8-37
Figure 8-25	Example of an Audit Report Select by Group displayed in the Report window	8-38
Figure 8-26	Example of an Audit Report – Select by Sampling displayed in the Report window	8-40
Figure 8-27	Example Data Report – Select by Group displayed in the Report window	8-42
Figure 8-28	Example Data Report – Select by Samplings displayed in the Report window	8-44
Figure 8-29	Example Trend Report displayed in the Report window	8-46
Figure 8-30	Sampling Report window	8-48
Chapter 9:	Real Time Trend	9-1
Figure 9-1	Real Time Trend window	9-2
Chapter 10:	Run Time Logic	10-1
Figure 10-1	Run Time Logic Editor window	10-1
Figure 10-2	Available tag selection displays its usage	10-11
Figure 10-3	Editor pane contains all edited rules organized by type	10-11
Figure 10-4	Rule Editor window	10-12
Figure 10-5	Example Rule created with Rule Editor	10-13
Figure 10-6	Child rule creation in Rule Editor	10-14
Figure 10-7	Sampling on input editor window	10-14
Figure 10-8	TowerLightEditForm window	10-15
Figure 10-9	Example Tower Light rule	10-17
Chapter 11:	User Management	11-1
Figure 11-1	Pharmaceutical Net Pro Users Management window	11-3

Chapter 12: System Tasks	12-1
Figure 12-1 FacilityPro System management window	12-2
Figure 12-2 Sensors Editor window	12-5
Figure 12-3 FacilityPro Tools Language Switch window	12-11
Figure 12-4 Limits Editor window	12-13
Figure 12-5 Limits Editor History window displayed in front of the Limits Editor window	12-16
Figure 12-6 Limits Editor Configuration window displayed in front of the Limits Editor window	12-18
Figure 12-7 Analog Inputs Limits window	12-20
Chapter 13: Aventino Setup Options	13-1
Figure 13-1 Aventino Setup Wizard	13-2
Figure 13-2 New installation and entering user credentials	13-3
Figure 13-3 New installation and entering user credentials	13-3
Figure 13-4 Processor module type selected from license	13-4
Figure 13-5 Database restore	13-4
Figure 13-6 System hardware configuration	13-5
Figure 13-7 System sampling parameters	13-5
Figure 13-8 Sampler calibration configuration	13-6
Figure 13-9 Legacy pneumatic/sensor module configuration	13-6
Figure 13-10 Pneumatic module configuration with individual samplers	13-7
Figure 13-11 Pneumatic module configuration with individual samplers	13-8
Figure 13-12 Pneumatic module configuration with individual samplers	13-8
Figure 13-13 Setting number of pumps in group	13-9
Figure 13-14 I/O board and module configuration	13-9
Figure 13-15 Configuration Save Page	13-10
Figure 13-16 FacilityPro configuration file transfer and restart	13-10
Figure 13-17 Database initialization	13-11
Figure 13-18 Creation of Pharmaceutical Net Pro software pages	13-11
Figure 13-19 Testing processor availability	13-12
Appendix A: Statistical Data Analyzer	A-1
Figure A-1 Statistical Data Analyzer window	A-2
Figure A-2 Select Folder button	A-2
Figure A-3 Select Folder window	A-3
Figure A-4 SensorsList.CSV directory	A-3
Figure A-5 Expand All button	A-4
Figure A-6 Global stats display	A-4
Figure A-7 Step interpolation	A-5
Figure A-8 Save Config. button	A-6
Figure A-9 Load Config. button	A-6
Figure A-10 Show Virtual Pens button	A-6
Figure A-11 Virtual Pens window	A-6
Figure A-12 Daily sensor chart window	A-7
Figure A-13 Display language adjustment	A-8

This page is intentionally left blank.

List of Tables

Chapter 1: Overview	1-1
Chapter 2: Installation	2-1
Chapter 3: Get Started	3-1
Table 3-1 Components of the Pharmaceutical Net Pro screen	3-2
Table 3-2 Channel Status Legend	3-5
Chapter 4: Recipes	4-1
Table 4-1 Recipes Editor window components	4-2
Chapter 5: Sampling	5-1
Table 5-1 Sampling Editor window components	5-2
Table 5-2 Sampling on input editor dialog box	5-4
Table 5-3 Components of the Custom fields configuration window	5-6
Chapter 6: Sensors	6-1
Table 6-1 Particle Sensors window EBI Channel view components	6-3
Table 6-2 Biological Sensors window B0 Remote Channel view components	6-7
Table 6-3 Analog Inputs window view components	6-10
Chapter 7: Alarms	7-1
Table 7-1 Alarms window components	7-2
Chapter 8: Reports	8-1
Table 8-1 Report window buttons	8-2
Table 8-2 Components of the History Automation Config tab	8-13
Table 8-3 Select by Group tab Report window components	8-21
Table 8-4 Select by Sampling mode Report window components	8-28
Table 8-5 General and report-specific controls in the Report window	8-35
Chapter 9: Real Time Trend	9-1
Table 9-1 Components of the Real Time Trend window	9-2
Chapter 10: Run Time Logic	10-1
Table 10-1 Buttons of the Run Time Logic Editor screen	10-2
Table 10-2 Analog Input Folder	10-3
Table 10-3 BioCapt (BC) Folder	10-4
Table 10-4 Digital Input (DI) Folder	10-5

Table 10-5	Digital Output (DO) Folder-----	10-5
Table 10-6	Digital Output (DO) Folder-----	10-6
Table 10-7	I/O Module Folder-----	10-6
Table 10-8	Particle Counter (PC) Folder-----	10-6
Table 10-9	Pumps Group Folder-----	10-8
Table 10-10	Pumps Group Folder-----	10-9
Table 10-11	System Folder-----	10-9
Table 10-12	Effect on output of turning all input on/off-----	10-12
Table 10-13	Effect on output of turning any input on/off-----	10-12
Table 10-14	Tower Light Outputs and Inputs-----	10-16
Chapter 11: User Management	-----	11-1
Table 11-1	Default users-----	11-1
Table 11-2	Default User Groups' functions-----	11-2
Table 11-1	User Management window components-----	11-4
Chapter 12: System Tasks	-----	12-1
Table 12-1	System management window components-----	12-2
Table 12-2	Sensors Editor window components-----	12-6
Table 12-3	Components of the Limits Editor window-----	12-14
Table 12-4	Components of the Limits Editor History window-----	12-17
Table 12-5	Analog Inputs Limits window view components-----	12-20
Chapter 13: Aventino Setup Options	-----	13-1
Appendix A: Statistical Data Analyzer	-----	A-1

Chapter 1

Overview

Pharmaceutical Net Pro is software that interfaces with two FacilityPro processors and provides a simple monitoring and reporting tool. Pharmaceutical Net Pro can be configured with the following hardware components:

- FacilityPro 1010, 3010 and 5010 processor modules
 - The FacilityPro 1010 allows configuration with:
 - 8, 16 or 32 biological or particle sensors
 - 1 ENODE with a maximum of six modules
 - The FacilityPro 3010 allows configuration with:
 - 16 biological or particle sensors
 - Standard: 16 digital outputs/8 digital inputs
 - Optional: An additional 16 digital outputs/8 digital inputs
 - Optional: 8 analog inputs
 - The FacilityPro 5010 allows configuration with:
 - 64 biological or particle sensors
 - up to 192 analog sensors (+32 virtual points)
 - up to 128 digital outputs
 - up to 64 digital inputs
- Pharmaceutical Net Pro View Client PCs (HMI)
 - 2 or 5 clients
- Samplers, monitors, and/or sensors:
 - BioCapt® microbial samplers
 - Remote Vacuum Unit (RVU) microbial samplers
 - IsoAir® 310P particle sensor
 - Airnet® particle counters
 - Analog sensors (such as temperature, pressure, relative humidity)

Configuration Options

The configuration option determines the maximum availability of sensors, DI/DO and analog I/O for the system.

System size

SMALL	6 sensors	8 digital in/16 digital out	8 analog in
MEDIUM	12 sensors	16 digital in/32 digital out	48 analog in
LARGE	128 sensors	64 digital in/128 digital out	192 analog in + 32 virtual
CLIENTS	2 or 5		

System hardware

FACILITYPRO PROCESSORS	5010, 3010, 1010
-------------------------------	------------------

Software Architecture

Pharmaceutical Net Pro software architecture is illustrated in **Figure 1-1**.

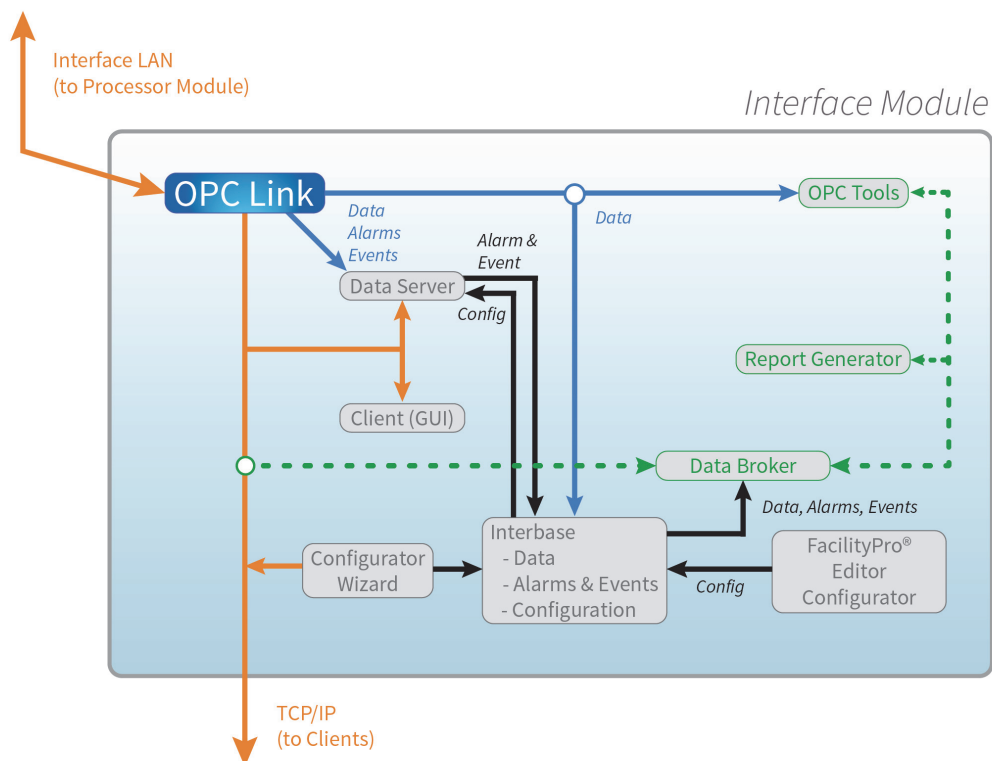


Figure 1-1 Architecture of Pharmaceutical Net Pro software system

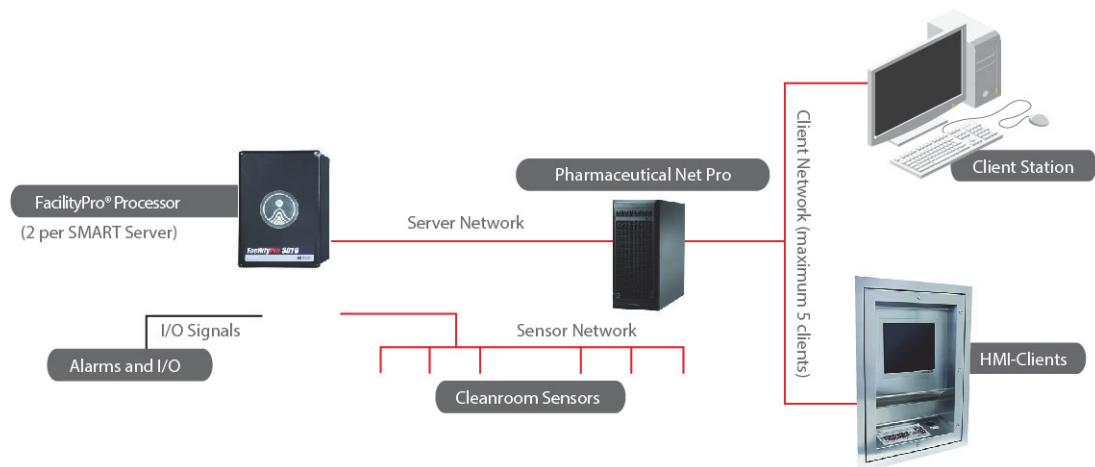


Figure 1-2 Pharmaceutical Net Pro network
(with FacilityPro 3010 Processor Module as an example)

WARNING

It is recommended to use only dedicated LAN networks. Avoid connection to internal office networks or Internet (permitted only for remote assistance purposes).

Server Software Description

Pharmaceutical Net Pro software provides the interface and supervisory control for the FacilityPro 1010/3010/5010 Environmental Monitoring System.

The provided server should contain:

- Data Server runtime
- Data Broker – Data interface between software and tag database
- OPC (OLE for Process Control) – OPC standard communication driver

The database should be managed by a factory secure password.

Interface

Pharmaceutical Net Pro application software:

- Pharmaceutical Net Pro Release
- Report Generator – Extracts data from DB and generating data reports, audits and trends
- PMS Custom Shell – Launcher software with graphical interface showing progression bar and a button to access the OS with a local administrator password for maintenance purposes
- OPC Link – OPC driver to communicate to Pharmaceutical Net Pro
- Sensor Editor – Sensor setup
- Recipes Editor – Recipe setup
- Sampling Editor – Setting up sampling recipes
- Limits Editor – Setup of alarm and warning limits for each class of sensor
- Run Time Logic Editor – DI/DO logic management
- Aventino Backup Service – Performs system backup
- PMS Sampling Report Automation – Periodically checks for finished samplings and automatically exports to a user-defined filepath
- PMS History Automation Service – Daily extraction of sensor data (statistical and sampling) within a user-defined group
- Aventino Data Server – Manages alarms written to the database
- SMART Editor – Configures and customizes graphical pages
- PMS License Manager – Licenses the software

Electronic Signature

The Electronic Signature feature is a control that applies password protection for the performance of an operation or a change of critical values inside the system.

Operations include:

- User Event (See **User Event button** on page 3-3 and **User Event button** on page 3-3)
- Recipe management (see **Create a New Recipe** on page 4-5 and **Edit an Existing Recipe** on page 4-5)
- Begin or stop sampling (see **Schedule a Sampling** on page 5-3 and **Edit a Scheduled Sampling** on page 5-5)
- Manual Switch ON/OFF a particle counter, biological sensor, or a direct digital output (see **Chapter 6** Sensors on page 6-5 and page 6-8)
- Alarm management (see **Acknowledge an Alarm** on page 7-3)
- Stopping real time trending (see **Chapter 9** Real Time Trend)
- Add, modify, or delete a user (see **Chapter 11** User Management)
- Change the descriptions of individual sampling points (see **Sensors Editor** on page 12-5 in **Chapter 12** System Tasks)

Electronic Signature window

In the **Electronic Signature** window (see **Figure 1-3**), the user confirms the event by entering a valid username and password.

The system stores the following event data for Electronic Signature events:

- Operation type
- Old and modified values
- Account
- Comment

Event data can be accessed through the **Audit Trail** from the **Report** window (see **Reports available through the Report Generator** on page 8-1).

Figure 1-3 Electronic Signature window

Note that the text displayed in the top field of the **Electronic Signature** window is a brief description of the action being performed provided by the system.

Specifications

The minimum recommended hardware specifications for Pharmaceutical Net Pro are listed below.

Pharmaceutical Net Pro Specifications	
Processor	Intel® Core Processor i3 2.2 GHz
Operating System	Windows® Server 2016 Standard English (64 bit) Version: 10.0.14393, Build: 14393
Operating System (Client)	Windows 10
RAM	Minimum 8 GB
Hard Disk C	1 TB (for OS)
Hard Disk D	1 TB (for Data)
Network Card	Minimum 1 network card 2 are needed to support clients 3 are needed to support clients and remote assistance

CAUTION

Ensure Windows Update is turned off to allow the software to function as intended.
Specific Windows updates compatible with Pharmaceutical Net Pro include:

Windows 10 Professional EN

- Cumulative update 1803
- KB4458469
- KB4100347
- KB4456655
- KB4343669

Windows Server 2016 Standard EN

- KB4457127
- KB4091664
- KB4132216
- KB3192137

The software may be impacted if other Windows updates are installed.

This page is intentionally left blank.

Chapter 2 Installation

This chapter describes the installation steps of Pharmaceutical Net Pro.

Prerequisites

Ensure the PC has the minimum **Specifications** found on page 1-6. The operating system installed should be Windows Server 2016 Standard. The hard disks or partitions available should be named C:\ and D:\.

Pharmaceutical Net Pro Installation

- 1 Insert the Pharmaceutical Net Pro disk. Initiate **PharmNetProInstaller.exe** located on the disk.
Once the installer window opens, click **Next**.

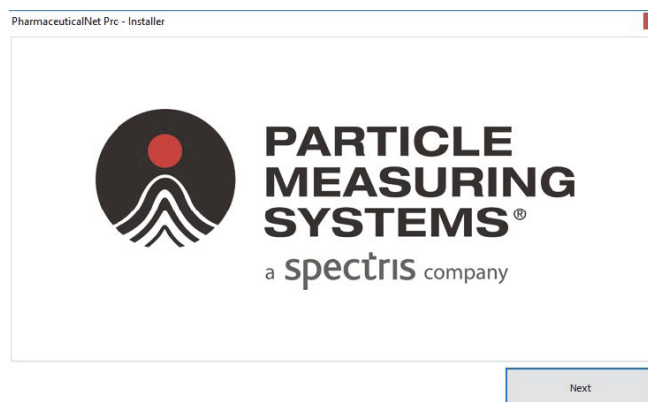


Figure 2-1 Pharmaceutical Net Pro Installer

- 2 The initial status of installation is displayed. Click **Next**.

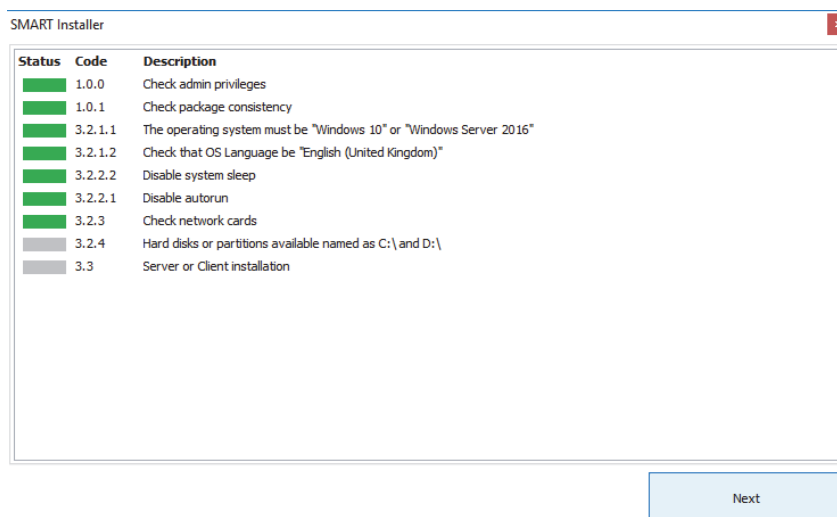


Figure 2-2 Installation status

- 3** Click **OK** when the network card information popup appears.

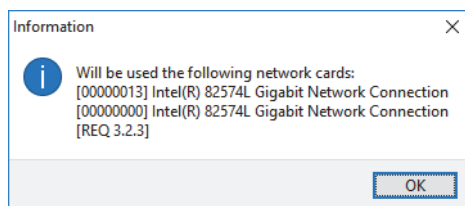


Figure 2-3 Network card popup

- 4** Click **OK** when the hard disk information popup appears.

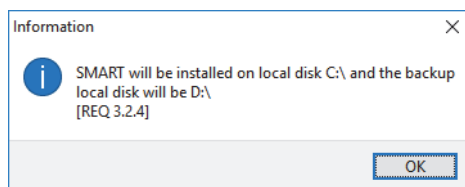


Figure 2-4 Hard disk popup

- 5** Select **Pharmaceutical Net Pro Server** or **Pharmaceutical Net Pro Client** depending on how the PC will be used and click **OK**.
- 6** If you wish to change the computer name, do so with the **Assign computer name** window. The system will automatically restart after clicking **OK**.

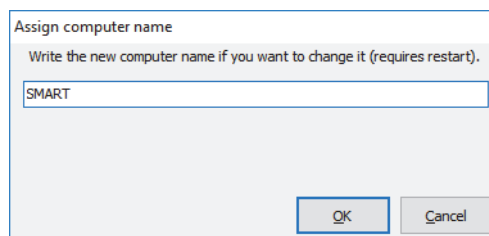


Figure 2-5 Assign computer name window

- 7** Select **Create new Local User** and use it and click **OK**.
The option **Run as Local User (pick one)** requires an administrator created on the PC.

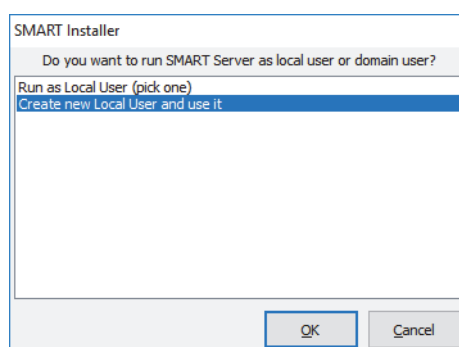


Figure 2-6 Local or domain user option

- 8** If creating a local user:
- Enter the username and click **OK**.
 - Enter the password and click **OK**.
 - Enter the password again and click **OK**.
 - The system automatically restarts.
- 9** Use the following network information for the FIELD network card:
- IPv4 address: 192.168.0.2
 - Subnetmask: 255.255.255.0
- 10** The installer will go step-by-step.
If the system is not connected to the internet, skip **Check Windows Defender** (click **OK**).
- 11** Once **Matrikon Explorer Installation** starts, it might be necessary to wait several minutes for this step to complete.
- 12** Select the desired language in the **Language Switch** window, then close it.



Figure 2-7 License Manager

13 Once the step **License Manager** step has been reached, refer to your specific system configuration:

a. System size license types:

- Small: Maximum of 6 sampling points allowed (either biological or particle), with 8 DI, 16 DO and 8 Analog
- Medium: Maximum of 12 sampling points allowed (either biological or particle), with 16 DI, 32 DO and 48 Analog
- Large: Unlimited sampling points allowed (either biological or particle), with unlimited DI, DO and Analog

b. Client number license types:

- Allow only SERVER: 0 clients
- Allow 2 physical Clients: 1 to 2 clients
- Allow 5 physical Clients: 3 to 5 clients

14 The following steps are for the **System size license**:

- a.** Click **Activate PharmaceuticalNet-Pro Server**.
- b.** Choose the activation type based on whether internet is available or not.
- c.** Enter the License ID and Password found on the disk or downloaded to the computer connected to the internet.
- d.** Click **Continue** when finished.

15 The following steps are for the **Clients number license**:

- a.** Click **Activate PharmaceuticalNet-Pro CLIENTS**.
- b.** Choose the activation type based on whether internet is available or not.
- c.** Enter the License ID and Password found on the disk or received through email.
- d.** Click **Continue** when finished.

16 System size and clients are now active. Close the **License Manager**.

17 Click **OK** when the installation application confirms setup is finished.

18 The system will automatically restart.

This page is intentionally left blank.

Chapter 3 Get Started

Pharmaceutical Net Pro runs on both the server and client PCs, if present.

System Start-Up

At power-up, the Pharmaceutical Net Pro software application launches automatically and displays the **Main Page** window (see **Figure 3-1**) with the default user logged in (username **Nobody**).

NOTE: To work with the Pharmaceutical Net Pro software interface you will need to log out the user **Nobody** and log in as an authorized user (see Log In and Log Out on page 3-6).

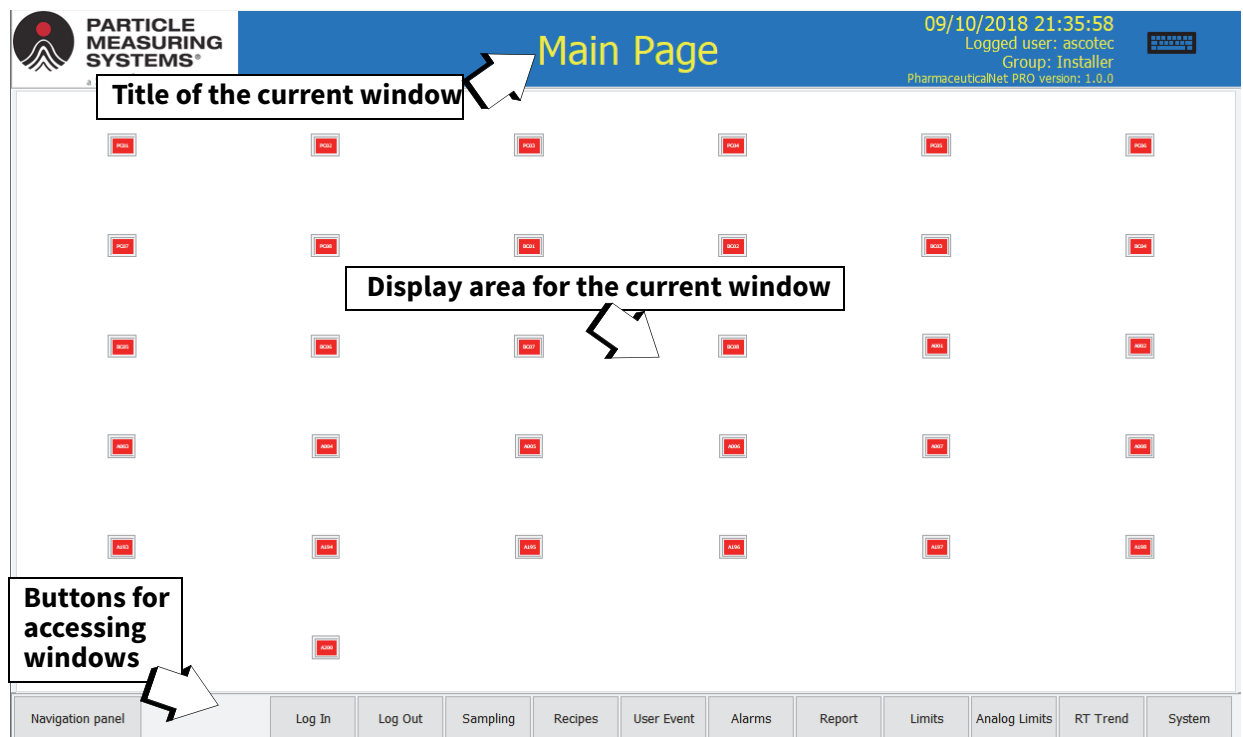


Figure 3-1 Pharmaceutical Net Pro **Main Page** window

Components of the Pharmaceutical Net Pro software screen

Header

The top of the Pharmaceutical Net Pro **Main Page** window displays the following information:

- **Logo**
- **Window title** – Name of the window currently displayed
- **Date and time** – Current date and time, in the same format as the OS
- **Logged user** – User Name of the user currently logged in
- **Group** – Role (i.e., Administrator, Supervisor, Power User, User, Emergency) assigned to the user currently logged in

Central area

The central area of the Pharmaceutical Net Pro screen displays one of several windows used to monitor or configure the FacilityPro Environmental Monitoring System. At start-up, the **Main Page** window displays in the central area of the Pharmaceutical Net Pro screen. Navigating to a different window requires clicking one of the buttons at the bottom of the Pharmaceutical Net Pro screen (described in **Table 3-1**, below).

Buttons row

The bottom of the Pharmaceutical Net Pro screen (see **Figure 3-1** on page 3-1) has rows of buttons for accessing those windows and for logging in or logging out. Access is governed by the logged-in user's privileges.

The components of the Pharmaceutical Net Pro screen are described in **Table 3-1**.

Table 3-1 Components of the Pharmaceutical Net Pro screen

Component	Description
Window title area	The title of a window displays across the top of the screen.
Logged user and Group area	In the upper-right corner, the Logged user and Group are displayed for the user currently logged in. The user "Nobody" is logged in at start-up.
Central area: current window	One of the windows used to monitor or configure the FacilityPro Environmental Monitoring System.

Table 3-1 Components of the Pharmaceutical Net Pro screen (Continued)

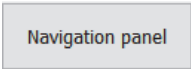
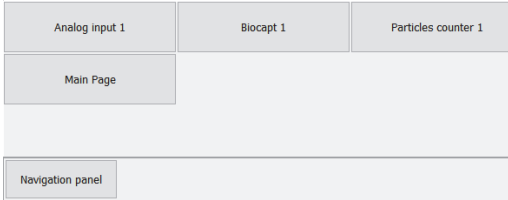
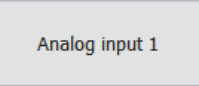
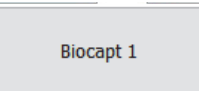
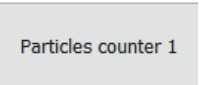
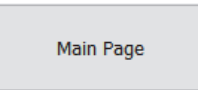
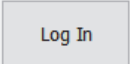
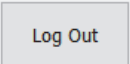
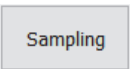
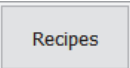
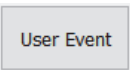
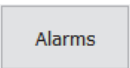
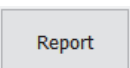
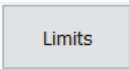
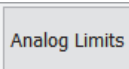
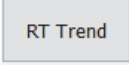
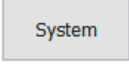
Component	Description
 Navigation panel button	Click this button to display the following options: Analog input, Biocapt, Particle counter, Main Page. 
 Analog input button	Click this button to display the Analog input window (see Analog Inputs Window on page 6-9).
 Biocapt button	Click this button to display the Biocapt window (Biocapt Window on page 6-6).
 Particle counter button	Click this button to display the Particles counter window (Particle counter Window on page 6-1).
 Main Page button	Click this button to display the Main Page window (described in Main Page Window on page 3-5).
 Log-in button	Click this button to display the Login window (described in Log In and Log Out on page 3-6).
 Log-out button	Click this button to log out the current user (described in For more information on managing users, see Chapter 11 User Management. on page 3-6).
 Sampling button	Click this button to display the Sampling window (described in Chapter 5 Sampling).
 Recipes button	Click this button to display the Recipes window (described in Chapter 4 Recipes).
 User Event button	Click this button to open the Electronic Signature window to log a custom user event.
 Alarms button	Click this button to display the Alarms window (described in Chapter 7 Alarms).
 Report button	Click this button to display the Report window (described in Chapter 8 Reports).

Table 3-1 Components of the Pharmaceutical Net Pro screen (Continued)

Component		Description
	Limits button	Click this button to open the Limits Editor window (described in Limits Editor on page 12-13 in Chapter 12 System Tasks).
	Analog Limits button	Click this button to display the Analog Inputs Limits window (Analog Inputs Limits on page 12-20).
	RT Trend button	Click this button to open the RT Trend window (described in Chapter 9 Real Time Trend).
	System button	Click this button to open a page with options for data backup, restart of the installation Wizard, shutdown of the software, user management, sensor editor, language switch, VNC Viewer (described in Chapter 12 System Tasks).

Main Page Window

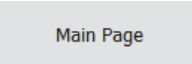
The **Main Page** window (see **Figure 3-1**) is a display-only window that contains a table an overview of the status of the sensor modules connected to the FacilityPro Environmental Monitoring System.

The color of a rectangle indicates the current status of the sensor.

- See **Table 3-2** on page 3-5 for descriptions of what a particular color indicates about the status of a module.

>> To open the Main Page window:

1 Click  the **Navigation panel** button at the bottom of the Pharmaceutical Net Pro screen.

2 Click  the **Main Page** button in the pop-up window.

The **Main Page** window appears showing the floorplan and sensor locations (see **Figure 3-1**).

Table 3-2 Channel Status Legend

State	Color	
Idle	white	
Starting/Stopping/Waiting/Cleaning/Holding	light green	
Sampling	green	
Pause	cyan (light blue)	
Delayed start	purple	
Alarm	red	
Alert	yellow	

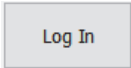
Log In and Log Out

At system start-up, the Pharmaceutical Net Pro **Main Page** window displays with the default user (username **Nobody**) logged in. In order to interact with the Pharmaceutical Net Pro application, a user is expected to log in with their username and password, as the default user has limited access to the features in the Pharmaceutical Net Pro application.

Management of user privileges, usernames, and passwords for access to Pharmaceutical Net Pro features is described in **Chapter 11** User Management.

Log in window

>> To log in to FacilityPro:

- 1 Click  the **Log In** button at the bottom of the Pharmaceutical Net Pro screen.

The **Log in** window appears (see **Figure 3-2**).

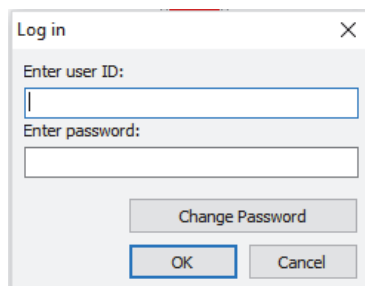
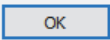


Figure 3-2 Log in window

- 2 In the **Enter user ID** field, enter your username.
- 3 In the **Enter password** field, enter your password.
- 4 Click  the **OK** button.

Your username will display at the top right corner of the FacilityPro screen.

NOTE: If another user is currently logged in, the **Log in** window displays the **Log-out** button, which you need to click before you can log in.

For more information on managing users, see **Chapter 11** User Management.

>> To log out the current user:

- 1 Click the **Log-out** button at the bottom of the Pharmaceutical Net Pro screen.
The **Nobody** username will display at the top right corner of the Pharmaceutical Net Pro screen.

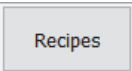
Chapter 4

Recipes

The **Recipes Editor** window (see **Figure 4-1**) is used to create, edit, and delete recipes for running the FacilityPro system. A recipe defines a group of sensors selected to sample at the same time.

Recipes Editor Window

>> To open the Recipes Editor window:

- Click  the **Recipes** button at the bottom of the Pharmaceutical Net Pro screen.
- The **Recipes Editor** window appears (see **Figure 4-1**).

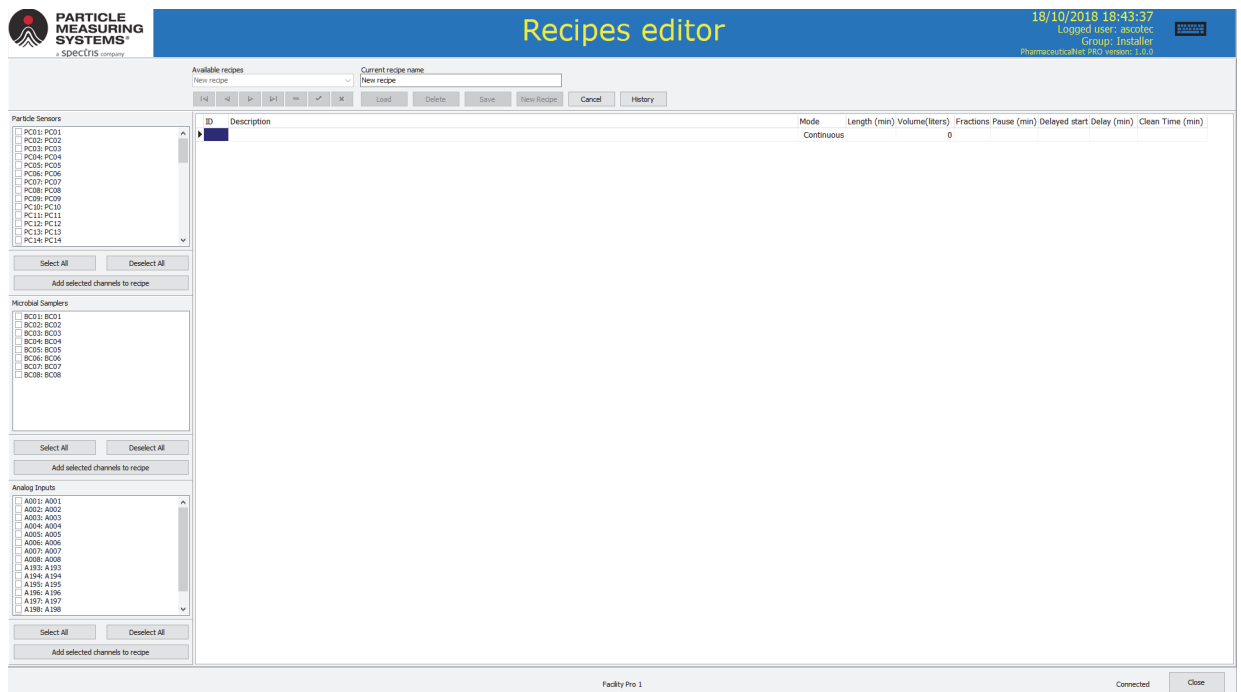


Figure 4-1 Recipes Editor window

The components of the **Recipes Editor** window are described in **Table 4-1** on page 4-2.

Table 4-1 Recipes Editor window components

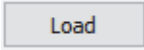
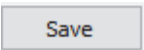
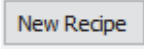
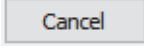
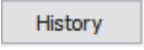
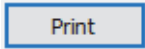
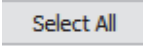
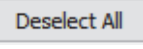
Component	Description
Available recipes dropdown list	This list contains the names of all currently defined recipes. When you select the recipe you want to work with, the recipe name displays in the Current recipe name field and the recipe definition displays in the spreadsheet area.
 Load button	Click this button to view the recipe selected in the Available recipes field so you can modify its definition.
 Delete button	Click this button to delete the recipe selected in the Current recipe name field.
 Save button	Click this button to save a recipe that is new or is being modified.
 New Recipe button	Click this button to begin defining a new recipe.
 Cancel button	Click this button to exit the <i>new recipe mode</i> or the <i>modify recipe mode</i> and enable the Available recipes field and  the Load button.
 History button	Select a recipe from the Available recipes field, then click this button to open a report window that shows the definition of the selected recipe and has a  Print button for sending a copy of the report to the printer.
Particle Sensors area	Contains a check box for each sensor in the system.
50 sensor n check boxes	☒ When checked, the sensor can be added to a recipe. ☐ When not checked, the sensor will not be added to a recipe.
 Select All button	Click this button to quickly place a checkmark ☒ in all the sensor check boxes.
 Deselect All button	Click this button to quickly remove the checkmark ☐ from all the sensor check boxes.
 Add selected channels to recipe button	After a recipe has been selected from the Available recipes field or  the New Recipe button has been clicked, AND one or more sensor check boxes is ☒ checked, click this button to add the checked sensors to the recipe.

Table 4-1 Recipes Editor window components (Continued)

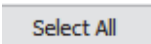
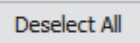
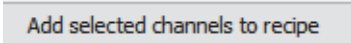
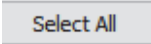
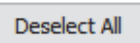
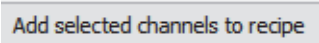
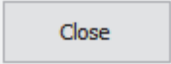
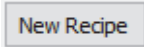
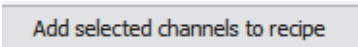
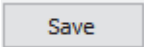
Component	Description
Microbial Samplers area BC: BC channel n check boxes  Select All button  Deselect All button  Add selected channels to recipe button	Contains a check box for each BIO sampler in the system.  When checked, the BIO sampler can be added to a recipe.  When not checked, the BIO sampler will not be added to a recipe. Click this button to quickly place a checkmark  in all the BIO sampler check boxes. Click this button to quickly remove the checkmark  from all the BIO sampler check boxes. After a recipe has been selected from the Available recipes field or  the New Recipe button has been clicked, AND one or more BIO sampler check boxes is checked , click this button to add the checked BIO samplers to the recipe.
Analog Inputs area A0 channel n check boxes  Select All button  Deselect All button  Add selected channels to recipe button	Contains a check box for each Analog sensor in the system.  When checked, the Analog sensor can be added to a recipe.  When not checked, the Analog sensor will not be added to a recipe. Click this button to quickly place a checkmark  in all the Analog sensor check boxes. Click this button to quickly remove the checkmark  from all the Analog sensor check boxes. After a recipe has been selected from the Available recipes field or  the New Recipe button has been clicked, AND one or more Analog sensor check boxes is checked , click this button to add the checked BIO sensors to the recipe.
Recipes spreadsheet fields	Each row shows information and data entry fields for a single sensor
ID column	Displays the short identification number of a sensor.
Description column	Sensor description.

Table 4-1 Recipes Editor window components (Continued)

Component	Description
Mode column	Select the sampling mode for the recipe: • Continuous • Time • Volume
Length (min) column	When the Mode is set to Time , enter the duration of the sampling in minutes.
Volume (liters) column	When the Mode is set to Volume , enter the amount of air to be sampled (in liters).
Fractions column	For a Biological Module, it is possible to divide the sample into different fractions. Enter the desired number of fractions (range: 1 to 20 for RVUs, and 1 to 60 for MiniCapt Remote).
Pause (min) column	Enter the number of minutes to pause before sampling.
Delayed start column	Select Yes to set a delay on the start-up of sampling, then enter the delay in the Delay (min) column.
Delay (min) column	When Yes is entered in the Delayed start column, enter the length of the delay (range 1 to 480 minutes).
Clean Time (min) column	It is possible to set the cleaning time at the start of the sampling. • Only pump is driven in order to clean the vacuum line. • During this time, the system does not handle warnings and alarms.
Status bar	Displays the status of the connection to the FacilityPro Processor Module: • Connected – Indicates Pharmaceutical Net application is connected to the FacilityPro Processor Module. • Disconnected – Indicates Pharmaceutical Net Pro software application is disconnected from the FacilityPro Processor Module.
 Close button	Click this button to close the Recipes Editor window and return to the Main Page window.

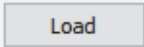
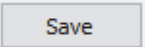
Create a New Recipe

>> To create a new recipe:

- 1 Log in as a user with Administrator privileges.
- 2 Open the **Recipes Editor** window as described on page 4-1.
- 3 Click  the **New Recipe** button.
- 4 In the **Current recipe name** field, enter a unique name for the new recipe.
- 5 Place a checkmark  in the check box for each sensor to include in the new recipe.
- 6 Click  the **Add selected channels to recipe** button.
- 7 Configure samples for single points:
- 8 Click  the **Save** button.
The **Recipe** electronic signature window appears.
- 9 Perform the electronic signature to save the recipe (see **Electronic Signature** on page 1-5).

Edit an Existing Recipe

>> To edit a recipe:

- 1 Open the **Recipes Editor** window as described on page 4-1.
- 2 From the **Available recipes** dropdown list, select the recipe you want to edit.
- 3 Click  the **Load** button.
The system loads the recipe's configuration.
- 4 Make the edits to the recipe.
(Use the procedures suggested in Create a New Recipe, above).
- 5 When editing is complete, click  the **Save** button.
The **Recipe** electronic signature window appears.
- 6 Perform the electronic signature to save the recipe (see **Electronic Signature** on page 1-5).

Delete an Existing Recipe

>> To delete a recipe:

- 1 Open the **Recipes Editor** window as described on page 4-1.
- 2 From the **Available recipes** dropdown list, select the recipe you want to delete.
- 3 Click  the **Delete** button.

The system displays a confirmation window (see **Figure 4-2**).

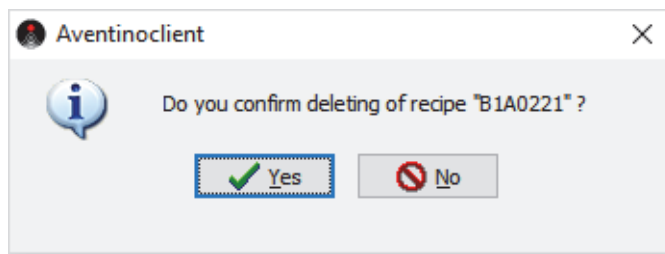
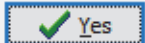


Figure 4-2 Recipes Editor confirm delete dialog box

- 4 To confirm the deletion, in the confirmation window:
 - a. Click  the **Yes** button.
 - b. Perform the electronic signature to remove the recipe from the system (see **Electronic Signature** on page 1-5).

Chapter 5

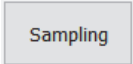
Sampling

The **Samplings editor** window (see **Figure 5-1**) is used to schedule sampling times for running specific recipes.

Sampling Editor Window

The **Sampling Editor** window allows you to schedule a sampling and displays a list of all samplings planned or currently running.

>> **To open the Samplings editor window:**

- Click  the **Sampling** button at the bottom of the Pharmaceutical Net Pro screen.

The **Samplings editor** window appears (see **Figure 5-1**).

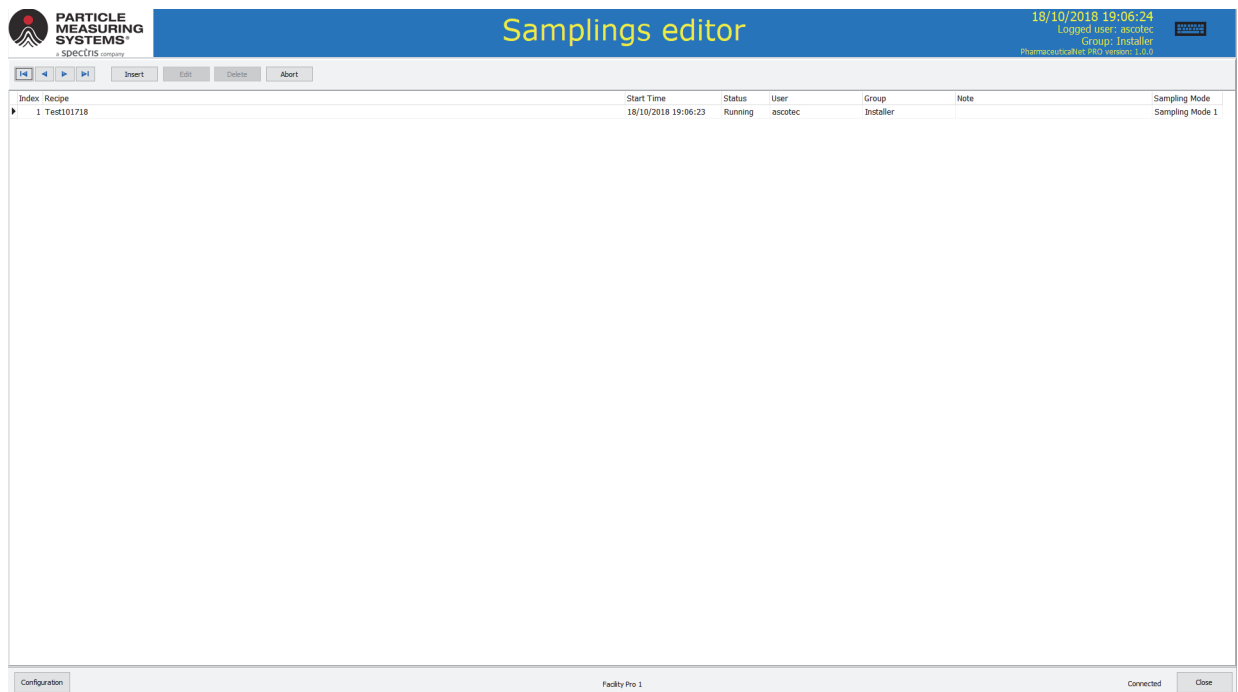
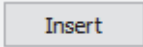
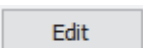
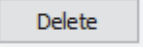
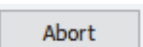
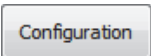
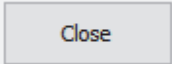


Figure 5-1 Sampling Editor window

The components of the **Sampling Editor** window are described in **Table 5-1** on page 5-2.

Table 5-1 Sampling Editor window components

Component	Description
 First Record button	Click this button to view the first record.
 Previous Record button	Click this button to view the previous record.
 Next Record button	Click this button to view the next record.
 Last Record button	Click this button to view the last record.
 Insert button	Click this button to open the Sampling on input editor dialog box where you can schedule a new sampling run (see Schedule a Sampling on page 5-3).
 Edit button	Click this button to open the Sampling on input editor dialog box where you can modify a scheduled sampling (see Edit a Scheduled Sampling on page 5-5).
 Delete button	Click this button to delete the recipe selected in the Current recipe name field.
 Abort button	Click this button to stop the sampling run.
 Configuration button	Click this button to open the Custom fields configuration window where you can manage definitions available to apply to a sampling (see Configuring Custom Fields to Apply to a Sampling on page 5-6).
 Close button	Click this button to close the Sampling Editor window and return to the Main Page window.
Samplings Table	The Samplings Table lists information for the samplings currently scheduled.
Index column	The sequential number automatically assigned when the sampling was scheduled.
Recipe column	The name of the recipe that will run during the sampling.
Start Time column	The date and time the sampling will take place in the same format as the OS.
Status column	The current state of the sampling: • Idle – The sampling program has not been run. • Running – Indicates the sampling program is currently running. • Stopping – The sampling program is stopping.
User column	The name of the user that scheduled the sampling.
Group column	The name of the group that contains the user that scheduled the sampling.
Note column	Comments included when the sampling was created.
Mode column	The mode assigned to the sampling. Ex. Operational, At Rest, Normal Operation, Custom.

Schedule a Sampling

Scheduling a sampling using the **Samplings editor** window involves defining the fields of the **Sampling on input editor** dialog box such as:

- Selection of an existing recipe
- Batch number
- Date and time the sampling should begin
- Mode for the sampling
- Comments

and performing the Electronic signature

Sampling on input editor dialog box

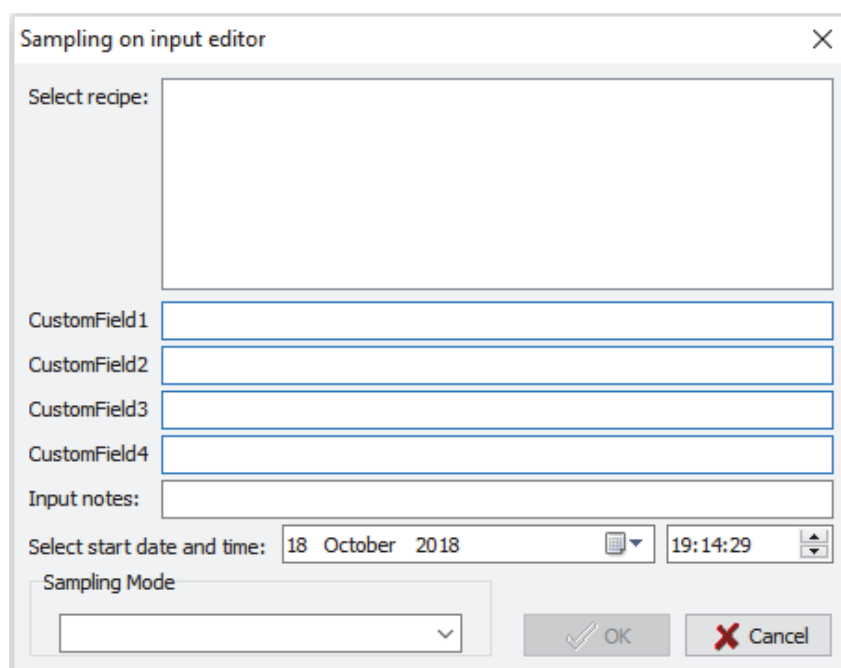
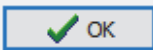
The image shows a software dialog box titled "Sampling on input editor" with a close button (X) in the top right corner. The dialog contains several input fields: a large text area labeled "Select recipe:"; four smaller text fields labeled "CustomField1", "CustomField2", "CustomField3", and "CustomField4"; a text field labeled "Input notes:"; a date and time selection field labeled "Select start date and time:" showing "18 October 2018" and "19:14:29"; and a "Sampling Mode" section with a dropdown menu. At the bottom right are "OK" and "Cancel" buttons.

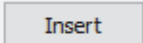
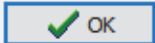
Figure 5-2 Sampling on input editor dialog box

The components of the **Sampling on input editor** dialog box for scheduling a sampling are described in **Table 5-2**.

Table 5-2 Sampling on input editor dialog box

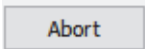
Component	Description
Select recipe list	Lists the recipes available so you can select the one to use in the sampling.
Custom fields	Lists custom fields to be selected for the sampling.
Input notes field	Defines notes to include in the Sampling Report.
Select start date and time field	The date field defines the day of the year to begin the sampling. Click in the field to display a calendar from which you can select the desired date. The time field defines the time of day to begin the sampling.
Sampling mode list	Lists the operating modes so you can select the one to apply to the sampling (ex: Operational, At Rest, Normal Operation, Custom).
 OK button	Click this button to save the new sampling schedule and close the Sampling editor dialog box.
 Cancel button	Click this button to close the Sampling editor dialog box without scheduling a sampling.

>> To select and start an existing sampling recipe:

- 1 Click  the **Insert** button at the top of the **Sampling Editor** window.
The **Sampling editor** dialog box appears (see **Figure 5-2** on page 5-3).
- 2 In the **Select recipe** field of the **Sampling editor** dialog box, select the recipe you want to use for the sampling.
- 3 If desired, in a **Custom** field, enter a value to include in the Sampling Report.
Optional; see **Custom fields configuration window** on page 5-6.
- 4 If desired, in the **Input notes:** field, enter notes to include in the Sampling Report.
- 5 From the **Select start date and time:** fields, select the day and time the recipe should begin sampling.
- 6 From the **Sampling Mode** dropdown list, select the operating mode in which to start the sampling (ex: **Operational, At Rest, Normal Operation, Custom**).
- 7 Click  the **OK** button in the **Sampling editor** dialog box.
- 8 Perform the electronic signature to start the sampling (see **Electronic Signature** on page 1-5).
The new sampling definition appears in the **Samplings Table** list.

Abort a Sampling Event

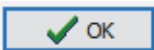
>> To abort a sample:

- 1 Open the **Sampling Editor** window as described on page 5-1.
- 2 In the **Recipe** column, select the sampling you want to abort.
- 3 Click  the **Abort** button.
- 4 Perform the electronic signature to abort the sampling (see **Electronic Signature** on page 1-5).
The sampling run is stopped.

Edit a Scheduled Sampling

You can change a sample that has already been programmed. Time, date and sampling mode are changeable and new record can be saved with an electronic signature.

>> To edit a sampling event in the Sampling Editor window:

- 1 Open the **Sampling Editor** window as described on page 5-1.
- 2 In the **Recipe** column, select the sampling you want to edit.
- 3 Click  the **Edit** button.
The **Sampling editor** dialog box appears (see **Figure 5-2** on page 5-3).
- 4 In the **Sampling editor** dialog box, make the changes to the scheduled departure of the recipe.
- 5 Click  the **OK** button in the **Sampling editor** dialog box.
- 6 Perform the electronic signature to update the configuration (see **Electronic Signature** on page 1-5).

Delete a Scheduled Sampling

>> To delete a sampling event in the Sampling Editor window:

- 1 Open the **Sampling Editor** window as described on page 5-1.
- 2 In the **Recipe** column, select the sampling you want to delete.
- 3 Click  the **Delete** button.
The sampling definition disappears from the table in the **Sampling Editor** window.

Configuring Custom Fields to Apply to a Sampling

A user with **administrator** privileges can use the **Custom fields configuration** window (see **Figure 5-3**) to define fields that are available in the **Sampling editor** dialog box (see **Sampling on input editor dialog box** on page 5-3), the dialog box where you schedule a sampling.

Custom fields configuration window

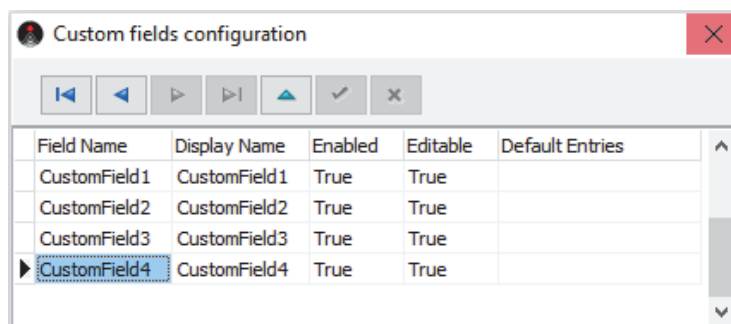


Figure 5-3 Custom fields configuration window

Table 5-3 describes the components of the **Custom fields configuration** window.

Table 5-3 Components of the **Custom fields configuration** window

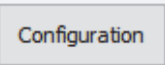
Attribute	Description
 Close Window button	Click this button to close the Custom fields configuration window.
 View First Row button	Click this button to view the first row in the table.
 View Previous Row button	Click this button to view the previous row in the table.
 View Next Row button	Click this button to view the next row in the table.
 View Last Row button	Click this button to view the last row in the table.
 View Previous Field button	Click this button to view the previous field in the table.
 Save Changes button	Click this button to save changes to the Custom fields configuration window.
 Cancel Changes button	Click this button to undo changes to the Custom fields configuration window.

Table 5-3 Components of the **Custom fields configuration** window (Continued)

Attribute	Description
Filter Definition row	
Display Name field	Defines the label for the first field below the Select Recipe field in the Sampling editor dialog box. For example, this label was defined as Batch Number in Figure 5-2 on page 5-3.
Enabled list field	Enables or disables the display of the field in the Sampling editor dialog box. • True – Causes the field to be displayed in the Sampling editor dialog box with the label defined in the Display Name field. • False – Prevents the field from being displayed in the Sampling editor dialog box.
Editable list field	Enables or disables editing of the field in the Sampling editor dialog box. • True – Causes the field to be edited by the user in the Sampling editor dialog box. • False – Prevents the field from being edited by the user in the Sampling editor dialog box.
DefaultEntries	Defines values for a list box that will display for the field. Use a comma-delimited string of values.

>> To define custom fields for the Sampling editor dialog box:

1 Open the **Sampling Editor** window as described on page 5-1.

2 Click  the **Configuration** button at the bottom of the **Sampling Editor** window.

The **Custom fields configuration** window appears (see **Figure 5-3** on page 5-6).

3 In the **Custom fields configuration** window, define the display label for the field:

- a.** Click in the **Display Name** column for the field you want to work with.
- b.** Type the text to use as the name of the field.

4 In the **Custom fields configuration** window, set True/False for whether to display the field in the **Sampling Editor** window:

- a.** Click in the **Enable** column for the field you want to work with.
- b.** From the list box, select:
 - **True** – Causes the field to be displayed in the **Sampling editor** dialog box with the label defined in the **Display Name** field.

—OR—

- **False** – Prevents the field from being displayed in the **Sampling editor** dialog box.

- 5 In the **Custom fields configuration** window, set whether the field will allow text data entry:
 - a. Click in the **Editable** column for the field you want to work with.
 - b. From the list box, select:
 - **True** – Causes the field to be edited by the user in the **Sampling editor** dialog box.
 - **False** – Prevents the field from being edited by the user in the **Sampling editor** dialog box.
- 6 When the value in the **Editable** column for the field has been set to **False** in the **Custom fields configuration** window, you can define the names for choices to display in a list box for the field.
 - a. Type the first value.
 - b. Type a comma.
 - c. Type the next value.
 - d. And so forth.

The values must be typed as a comma-separated list without a space after a comma.
- 7 Click  the **Save Changes** button in the **Custom fields configuration** window to save your changes.
- 8 Click  the **Close Window** button to close the **Custom fields configuration** window.
- 9 The custom fields will display the next time the **Sampling editor** dialog box is opened.

Chapter 6

Sensors

The sensor windows are used to view real-time information for each sampling point of instruments connected to the system:

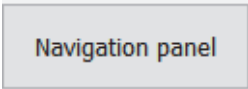
- Particle sensors (see **Particle Sensors Values** on page 6-3)
- Biological sensors (see **Biocapt Values** on page 6-6)
- Analog sensors (see **Analog Inputs** on page 6-9)

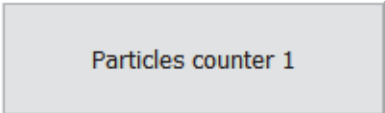
The Navigation panel button allows access to the above sensor pages. These pages are created by default during the installation process. Using the SMART Editor application, pages and area maps can be added.

Particle counter Window

The **Particle counter** window is for display only, and shows summary information for all sensors of every module in the system.

>> **To open the Particle counter window:**

1 Click the **Navigation panel**  button at the bottom of the **Main Page** window.

2 From the pop-up menu, click the **Particle counter**  button.

3 The **Particle counter** window appears (see **Figure 6-1**).

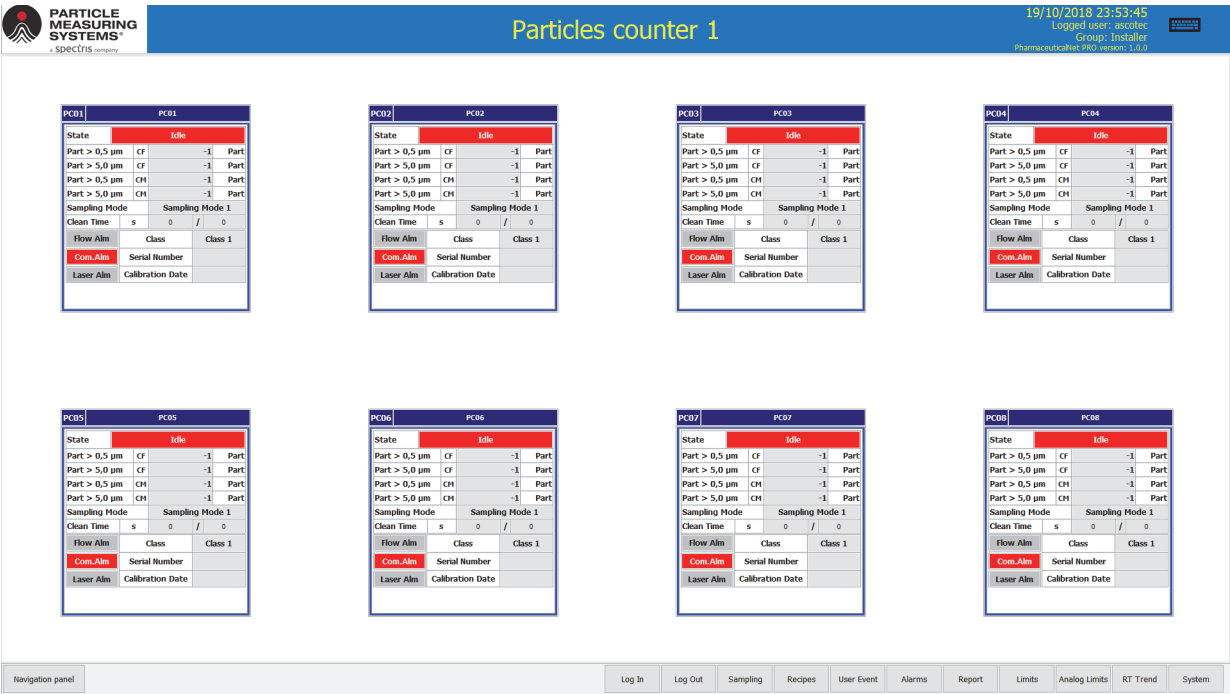


Figure 6-1 Pharmaceutical Net Pro Particle counter window

Particle Sensors Values

The components of the **sampling point information** view in the **Particle Sensors** window are described in **Table 6-1**.

Table 6-1 Particle Sensors window **EBI Channel** view components

Component	Description
Module Sub-Title	Displays a brief description of the current state of the modules, for example: Idle.

Table 6-1 Particle Sensors window EBI Channel view components (Continued)

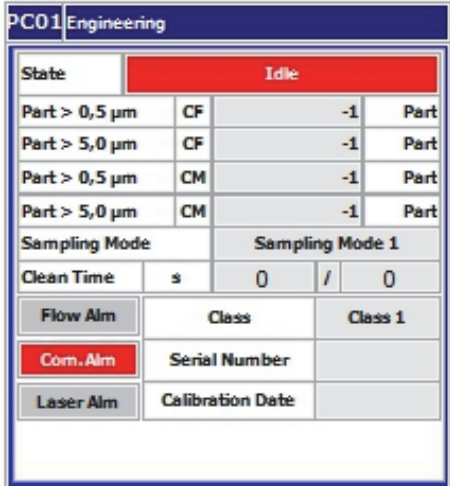
Component	Description
Particle Counter sampling point information 	• Title bar – Displays a short description (S01,S02,...): Long description. • Custom ID – User-specified (see Sensors Editor on page 12-5) sensor ID • Description – User-specified (see Sensors Editor on page 12-5) description of sensor • State – Displays the status of the sampling point: • Idle – Point ready • Starting – Point at the start • Sampling – Point in sampling • Stopping – Point at the end of sampling • Waiting – The system has processed the command and it is waiting the response from the sensor • Fault – System alarm for the sensor (flow, communication, laser, etc). • Cleaning – (only for particle counting) Sensor ON, but without acquisition of data and generation of alarms. • Particles > 0,5µm/CF (cubic foot) – Displays the counts of particles with size > 0.5 µm in a ft³. • Particles > 5,0µm/CF (cubic foot) – Displays the counts of particles with size > 5.0 µm in a ft³. • Particles > 0,5µm/CM (cubic meter) – Displays the counts of particles with size > 0.5µm in a m³. • Particles > 5,0µm/CM (cubic meter) – Displays the counts of particles with size > 5.0 µm in a m³. • Sampling mode – (Volume, Time, or Continuous) – Displays the sampling mode in effect • Volume – Current volume of sampling / target volume (when sampling mode is Volume) • Time – Current time of sampling / target time (when sampling mode is Time) • Clean Time – Current clean time (counter is not sampling but pump is active) / target clean time in seconds • Flow Alm – Solid red indicates the presence of a flow alarm, which means a fault in the vacuum line to the sensor. • Com Alm – Solid red indicates a communication alarm, which means the sensor is not connected and/or not responding. • Laser Alm – Solid red indicates the presence of a laser alarm, which means the laser inside the sensor has faulted.

Figure 6-2 Sampling point – Particle sensors

Table 6-1 Particle Sensors window **EBI Channel** view components (Continued)

Component	Description
Particle Counter sampling point information , cont'd.	- Manual switch ON/OFF button – Switch ON button – Click this button to turn the particle counter ON. The Electronic Signature window will appear. Switch OFF button – Click this button to turn the particle counter OFF. The Electronic Signature window will appear. Class – Displays the particle counter's class Serial Number – Displays the particle counter's serial number Calibration Date – Displays the particle counter's calibration date Predictive Reset button – This button appears when predictive alarm is active. Click this button to reset the alarm status.

Biocapt Window

The **Biocapt** window is for display only.

The **Biocapt** window displays of summary information for all sensors of every module in the system.

>> **To open the Biocapt window:**

- Click **Biocapt 1** the **Biocapt** button at the bottom of the **Main Page** window.
The **Biocapt** window appears (see **Figure 6-3**).

Biocapt Values



Figure 6-3 Biocapt window

The components of the **sampling point information** view in the **Biocapt** window are described in **Table 6-2** on page 6-7.

Table 6-2 Biological Sensors window B0 Remote Channel view components

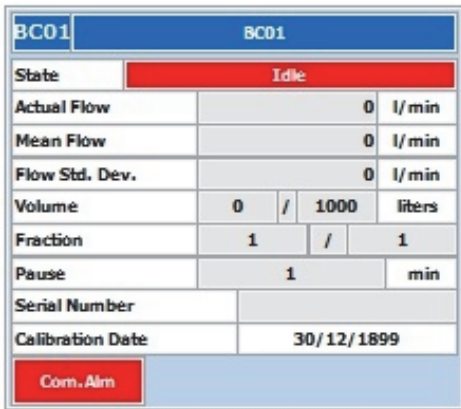
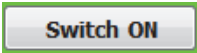
Component	Description
Sensor State	Displays a brief description of the current state of the sensor, for example: Idle.
Biological Sensor sampling point information 	• Title bar – Displays a short description (B01,B02,...): Long description. • State – Displays the status of the sampling point: • Idle – Point ready • Starting – Point at the start • Sampling – Point in sampling • Stopping – Point at the end of sampling • Waiting – The system has processed the command and it is waiting the response from the sensor. • Fault – System alarm for the sensor (flow, communication, laser, etc) • Pause – Sampling paused • Actual Flow – Displays the line's flow rate in l/m. • Mean Flow – Displays the average of flow rate in l/m. • Flow Std. Dev. – Displays the standard deviation of flow rate. • Volume – Current volume of sampling / target volume (when sampling mode is Volume) • Time – Current time of sampling / target time (when sampling mode is Time) • Fraction ... / – Displays the number of fractions set / fractions to reach • Pause – Displays the number of minutes of delay between two fractions • Serial Number – Displays the biological sensor's serial number • Flow Calc Alarm – Solid red indicates the presence of a flow alarm, which means a fault in the vacuum line to the sensor. • Com Alarm – Solid red indicates a communication alarm, which means the sensor is not connected and/or not responding. • Serial Number – Displays the biological sensor's serial number • Calibration Date – Displays the particle counter's calibration date

Figure 6-4 Sampling point – Biological

Table 6-2 Biological Sensors window **B0 Remote Channel** view components (Continued)

Component	Description
	- Manual switch ON/OFF button –  Switch ON button – Click this button to turn the sensor ON. The Electronic Signature window will appear.  Switch OFF button – Click this button to turn the sensor OFF. The Electronic Signature window will appear.

Analog Inputs Window

The **Analog Inputs** window is for display only.

The **Analog Inputs** window displays of summary information for all sensors of every module in the system.

>> **To open the Analog Inputs window:**

- Click **Analog input 1** the **Analog Inputs** button at the bottom of the **Main Page** window.

The **Analog Inputs** window appears (see **Figure 6-3**).

Analog Inputs



Figure 6-5 Analog Inputs window

The components of the **sampling point information** view in the **Analog Inputs** window are described in **Table 6-3** on page 6-10.

WARNING

In order to avoid the overgrowth of the database, it is strongly recommended with ENODE to the set deadband in the SBPC interface to 0.7% of acquisition range for each channel (in terms of samples this means 28 samples).

Table 6-3 Analog Inputs window view components

Component	Description
 Channel box	Displays the input's identification name
 Description box	Displays a description of the input type
 Alarm box	Displays any analog sensor alarms (ex. Com. Alm)
 Value box	Displays the input's value
 Unit box	Displays the units for the input's value

NOTE: The virtual analog channels will default to a higher channel number. Each virtual analog channel value is calculated as a subtraction of two physical analog inputs.

NOTE: Alert and alarm limits for analog channels are now configured in the Analog Limits Editor. See **Analog Inputs Limits** on page 12-20.

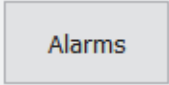
Chapter 7

Alarms

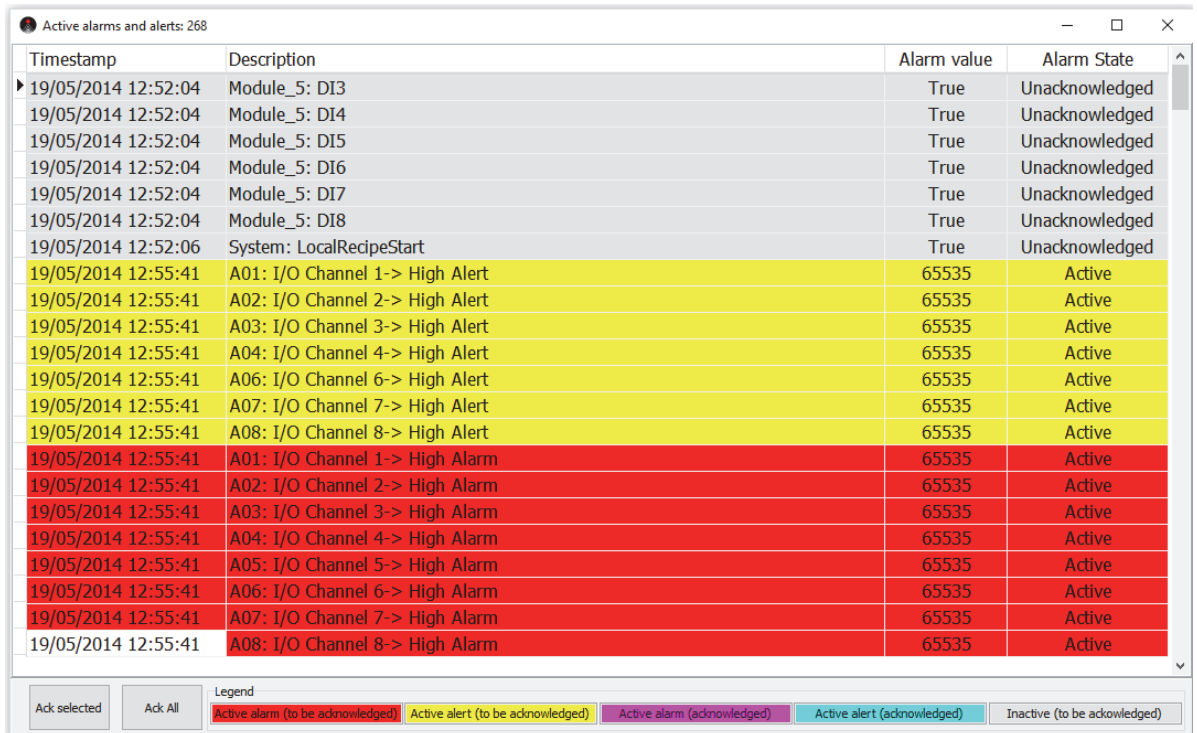
The **Alarms** window (see **Figure 7-1**) is used to acknowledge an alarm.

Alarms Window

>> To open the Alarms window:

- Click  the **Alarms** button at the bottom of the Pharmaceutical Net Pro screen.

The **Alarms** window appears (see **Figure 7-1**).



Active alarms and alerts: 268

Timestamp	Description	Alarm value	Alarm State
19/05/2014 12:52:04	Module_5: DI3	True	Unacknowledged
19/05/2014 12:52:04	Module_5: DI4	True	Unacknowledged
19/05/2014 12:52:04	Module_5: DI5	True	Unacknowledged
19/05/2014 12:52:04	Module_5: DI6	True	Unacknowledged
19/05/2014 12:52:04	Module_5: DI7	True	Unacknowledged
19/05/2014 12:52:04	Module_5: DI8	True	Unacknowledged
19/05/2014 12:52:06	System: LocalRecipeStart	True	Unacknowledged
19/05/2014 12:55:41	A01: I/O Channel 1-> High Alert	65535	Active
19/05/2014 12:55:41	A02: I/O Channel 2-> High Alert	65535	Active
19/05/2014 12:55:41	A03: I/O Channel 3-> High Alert	65535	Active
19/05/2014 12:55:41	A04: I/O Channel 4-> High Alert	65535	Active
19/05/2014 12:55:41	A06: I/O Channel 6-> High Alert	65535	Active
19/05/2014 12:55:41	A07: I/O Channel 7-> High Alert	65535	Active
19/05/2014 12:55:41	A08: I/O Channel 8-> High Alert	65535	Active
19/05/2014 12:55:41	A01: I/O Channel 1-> High Alarm	65535	Active
19/05/2014 12:55:41	A02: I/O Channel 2-> High Alarm	65535	Active
19/05/2014 12:55:41	A03: I/O Channel 3-> High Alarm	65535	Active
19/05/2014 12:55:41	A04: I/O Channel 4-> High Alarm	65535	Active
19/05/2014 12:55:41	A05: I/O Channel 5-> High Alarm	65535	Active
19/05/2014 12:55:41	A06: I/O Channel 6-> High Alarm	65535	Active
19/05/2014 12:55:41	A07: I/O Channel 7-> High Alarm	65535	Active
19/05/2014 12:55:41	A08: I/O Channel 8-> High Alarm	65535	Active

Legend

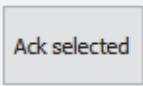
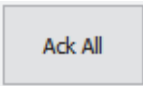
Ack selected Ack All

Active alarm (to be acknowledged) Active alert (to be acknowledged) Active alarm (acknowledged) Active alert (acknowledged) Inactive (to be acknowledged)

Figure 7-1 Alarms window

The components of the **Alarms** window are described in **Table 7-1** on page 7-2.

Table 7-1 Alarms window components

Component	Description	
Alarm data table		
Change time column	Date and Time at which the alarm is triggered.	
Description column	Short description of what triggered the alarm.	
Alarm value column	Value of the component when it triggered the alarm.	
Alarm State column	State of the alarm: • Active (not acknowledged) • Active (acknowledged) • Inactive (not acknowledged)	
 Ack selected button	Click this button to acknowledge the selected alarm (after electronic signature).	
 Ack All button	Click this button to acknowledge all the alarms of the list (after electronic signature).	
Legend section	Active Alarm (to be acknowledged) – red	
	Active Alert (to be acknowledged) – yellow	
	Active Alarm (acknowledged) – purple	
	Active Alert (acknowledged) – cyan	
	Inactive (to be acknowledged) - gray	

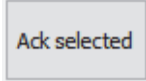
Acknowledge an Alarm

Pharmaceutical Net Pro provides at least the following system alarms:

- Connection with particle sensors
- Connection between management software (SCADA) and FacilityPro®
- Flow alarm for particle sensor or BioCapt®
- Laser alarm for particle sensor
- System communication alarm (internal FacilityPro communication)

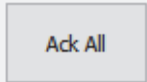
Acknowledging an alarm requires entry of a password in the **Electronic Signature** window.

>> To acknowledge an alarm:

- 1 Open the **Alarms** window as described on page 7-1.
- 2 In the **Alarms** window (see **Figure 7-1** on page 7-1), locate the alarm you want to acknowledge.
Use the vertical scroll bar or the down/up arrow keys on the keyboard to move up and down through the list.
- 3 Select the alarm and either double-click on it, or click  the **Ack Selected** button at the bottom of the window.
The **Electronic Signature** window appears.
- 4 Enter your electronic signature (see **Electronic Signature** on page 1-5).

After you enter the electronic signature, the system performs the acknowledge of the selected alarm.

>> To acknowledge multiple alarms at once:

- 1 Open the **Alarms** window as described on page 7-1.
- 2 Click  the **Ack All** button at the bottom of the window, or hold down the **SHIFT** key on your keyboard to select multiple rows.
The **Electronic Signature** window appears.
- 3 Enter your electronic signature (see **Electronic Signature** on page 1-5).

This page is intentionally left blank.

Chapter 8

Reports

Pharmaceutical Net Pro software generates reports via:

- the **Report** window (see Report Window on page 8-2)
- the **Sampling Report** window (see Sampling Report on page 8-48)

Overview

Reports available through the Report Generator

The general report types that can be generated from the **Report** window are:

- **Audit Report**
Log-ins, Alarms and Alarms Acknowledge, Electronic Signature events.
See example on page 8-38.
- **Trend Report**
Sampling data from the sensors in graphical format.
See example in page 8-46.
- **Data Report**
Sampling data from the sensors in tabular format.
See example on page 8-42.

A **Report** window report can be filtered for one of the following parameters:

- **Select By Group**
Selection is made on a pre-defined group of modules.
See Create a report “by group” on page 8-21.
- **Select By Sampling**
Extraction of data is filtered for recipe.
See Create a report “by sampling” on page 8-28.

Generation of reports via the **Report** window is described in Report Window beginning on page 8-2.

Reports available through the Sampling Report window

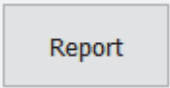
The **Sampling Report** is generated from the **Sampling Report** window (see **Figure 8-30** on page 8-48).

A **Sampling Report** can be generated for each recipe that has been started during a specific time period (see Sampling Report on page 8-48).

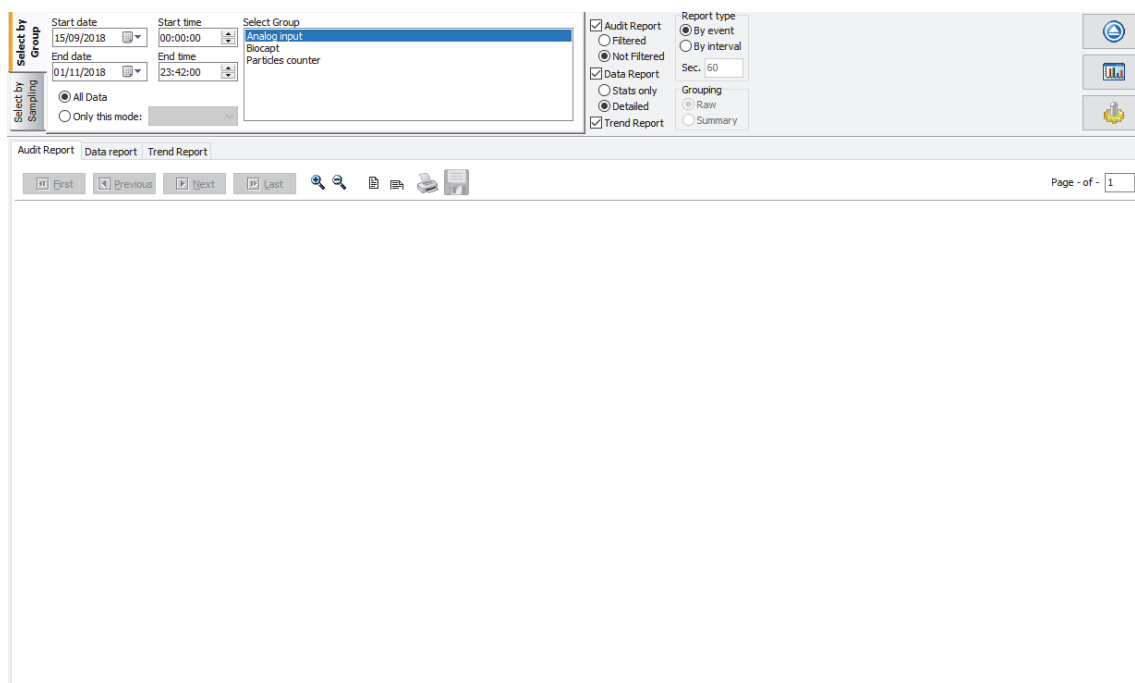
Report Window

Access to the Report Generator is through the **Report** window. The **Report** window has configuration fields across the top of the window and separate tabs for displaying each type of report in a text box (see **Figure 8-1**).

>> To open the Report window:

- 1 Click  the **Report** button at the bottom of the Pharmaceutical Net Pro screen.

The **Report** window appears (see **Figure 8-1**).



The screenshot shows the 'Report' window with the 'Select by Group' tab selected. The interface includes several configuration fields and checkboxes. On the left, there are dropdowns for 'Start date' (15/09/2018) and 'End date' (01/11/2018), and time fields for 'Start time' (00:00:00) and 'End time' (23:42:00). A 'Select Group' dropdown is set to 'Analysis input'. Below these are radio buttons for 'All Data' (selected) and 'Only this mode:'. To the right, there are checkboxes for 'Audit Report' (checked), 'Filtered' (unchecked), 'Not Filtered' (checked), 'Data Report' (checked), 'Stats only' (unchecked), 'Detailed' (checked), and 'Trend Report' (checked). Further right, there are radio buttons for 'Report type' (By event selected, By interval unselected) and a 'Sec.' field set to 60. Below these are radio buttons for 'Grouping' (Raw selected, Summary unselected). At the bottom, there are tabs for 'Audit Report', 'Data report', and 'Trend Report'. A toolbar with icons for first, previous, next, last, search, and print is visible. The main area is a large empty text box. The bottom right corner shows 'Page - of - 1'.

Figure 8-1 Report window, **Select by Group** tab fields

The buttons in the top right corner of the **Report** window are described in **Table 8-1**.

Table 8-1 Report window buttons

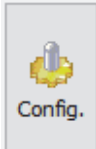
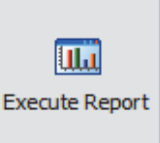
Component	Description
 Config. button	Click this button to open the Tags catalog window where the parameters displayed in the Report window can be configured.

Table 8-1 Report window buttons (Continued)

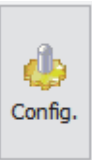
Component	Description
 Exit Page button	Click this button to close the session with the Report window. Any reports generated during the session will close.
 Execute Report button	Click this button to open the “add comment” window and generate the report with the parameters set up in the Report window. See page 8-27 or page 8-33

Configuration of Tags Catalog

The **Tags catalog** window lets an Administrator associate sensors with groups and create and modify recipes.

>> To open the Tags catalog window:

- 1 Log in as a user with Administrator privileges.
- 2 Open the **Report** window as described on page 8-2.

- 3 Click  the **Config.** button.

The **Tags catalog** window appears.

Tags Catalog tabs

Sensor Type

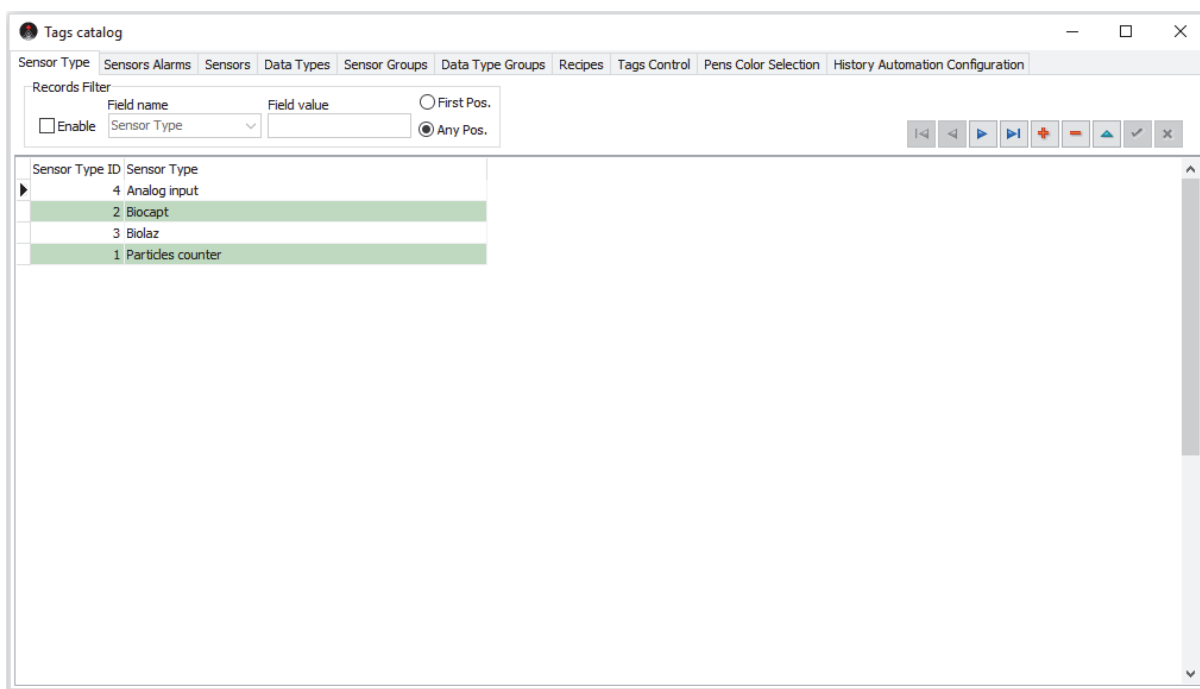


Figure 8-2 Tags catalog window, **Sensor Type** tab

Use this tab to add to, edit, delete, and sort through the **Sensor Type** list (i.e. Analog, Particles, RVU).

Sensors Alarms

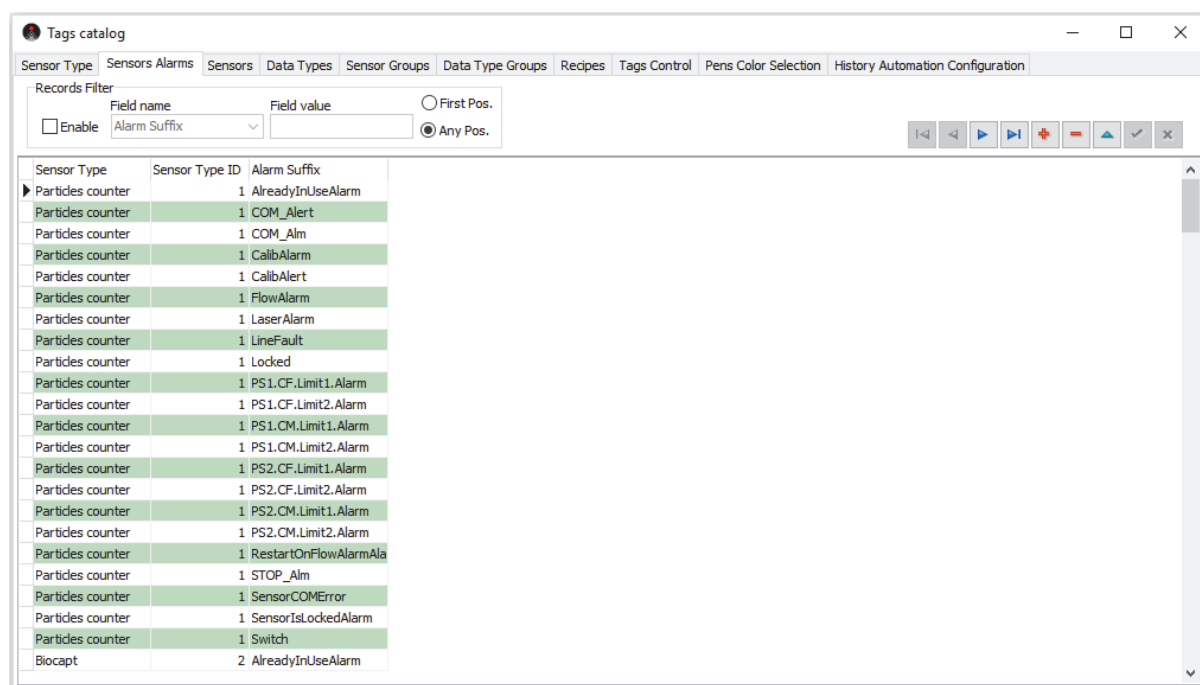


Figure 8-3 Tags catalog window, **Sensors Alarms** tab

Use the **Sensors Alarms** tab to add, edit, delete, and sort through sensor alarms. Use **Field name** and **Field value** to filter using the categories of **Sensor Type** (i.e. Analog, Particles, Biological), **Sensor Type ID** (a number assigned to a type of **Alarm Suffix**), and **Alarm Suffix** (specified by its associated alarm tag).

Sensors

Tags catalog

Sensor Type | Sensors Alarms | **Sensors** | Data Types | Sensor Groups | Data Type Groups | Recipes | Tags Control | Pens Color Selection | History Automation Configuration

Records Filter: ☐ Enable Field name: FacilityPro number Field value: ☐ First Pos. ☒ Any Pos. Duplicate selected row

Sensor Type	Sensor Type ID	FacilityPro number	Module	Sensor number	Sensor Name	Tag Description
Partides counter	1	1	1	1	PC01: PC01	PC01: PC01
Partides counter	1	1	1	2	PC02: PC02	PC02: PC02
Partides counter	1	1	1	3	PC03: PC03	PC03: PC03
Partides counter	1	1	1	4	PC04: PC04	PC04: PC04
Partides counter	1	1	1	5	PC05: PC05	PC05: PC05
Partides counter	1	1	1	6	PC06: PC06	PC06: PC06
Partides counter	1	1	1	7	PC07: PC07	PC07: PC07
Partides counter	1	1	1	8	PC08: PC08	PC08: PC08
Biocapt	2	1	2	1	BC01: BC01	BC01: BC01
Biocapt	2	1	2	2	BC02: BC02	BC02: BC02
Biocapt	2	1	2	3	BC03: BC03	BC03: BC03
Biocapt	2	1	2	4	BC04: BC04	BC04: BC04
Biocapt	2	1	2	5	BC05: BC05	BC05: BC05
Biocapt	2	1	2	6	BC06: BC06	BC06: BC06
Biocapt	2	1	2	7	BC07: BC07	BC07: BC07
Biocapt	2	1	2	8	BC08: BC08	BC08: BC08
Biocapt	2	1	2	9	BC09: BC09	BC09: BC09
Biocapt	2	1	2	10	BC10: BC10	BC10: BC10
Biocapt	2	1	2	11	BC11: BC11	BC11: BC11
Biocapt	2	1	2	12	BC12: BC12	BC12: BC12
Analog input	4	1	5	1	A001: A001	A001: A001
Analog input	4	1	5	2	A002: Test Room Temperature	A002: Test Room Temperature
Analog input	4	1	5	3	A003: A003	A003: A003

Figure 8-4 Tags catalog window, **Sensors** tab

Use the **Sensors** tab to add, edit, delete, and sort through entries based on the following fields: **Sensor Type**, **Sensor Type ID**, **FacilityPro number**, **Module**, **Sensor number**, **Sensor Name** and **Tag Description**. Duplicate a selected row using the associated button adjacent to the **Records Filter**.

All values will have functional defaults in place prior to any editing.

Data Types

Sensor Type	Sensor Type ID	Data Type Name	Tag Suffix	Tag Description	In Data Report	In Trend Rep	Right Axis	Mode Tag Name	Alarm State Tag Name
Analog input	4	Analog Alarm State	AlarmState	Analog Alarm State	☐	☐	☐		
Analog input	4	Analog Value	RealValue	Value	☒	☒	☐		Analog Alarm State
Biocapt	2	Biocapt Flow State	FlowState	Biocapt Flow State	☐	☐	☐		
Biocapt	2	Biocapt Volume	Volume	Volume (l)	☒	☒	☐		
Biocapt	2	Flow	Flow	Flow (l/min)	☒	☒	☐		Biocapt Flow State
Particles count	1	Part 0.5 CF	PS1.CFCount	Part > 0.5 CF	☒	☒	☐	Part sampling mode	Part 0.5 CF Alarm State
Particles count	1	Part 0.5 CF Alarm State	PS1.CF_AlarmState	0.5 CF Alarm State	☐	☐	☐		
Particles count	1	Part 0.5 CM	PS1.CMCount	Part > 0.5 CM	☒	☒	☐	Part sampling mode	Part 0.5 CM Alarm State
Particles count	1	Part 0.5 CM Alarm State	PS1.CM_AlarmState	0.5 CM Alarm State	☐	☐	☐		
Particles count	1	Part 0.5 raw	PS1.RawCount	Part > 0.5 raw	☒	☒	☐	Part sampling mode	
Particles count	1	Part 5.0 CF	PS2.CFCount	Part > 5.0 CF	☒	☒	☐	Part sampling mode	Part 5.0 CF Alarm State
Particles count	1	Part 5.0 CF Alarm State	PS2.CF_AlarmState	5.0 CF Alarm State	☐	☐	☐		
Particles count	1	Part 5.0 CM	PS2.CMCount	Part > 5.0 CM	☒	☒	☐	Part sampling mode	Part 5.0 CM Alarm State
Particles count	1	Part 5.0 CM Alarm State	PS2.CM_AlarmState	5.0 CM Alarm State	☐	☐	☐		
Particles count	1	Part 5.0 raw	PS2.RawCount	Part > 5.0 raw	☒	☒	☐	Part sampling mode	
Particles count	1	Part Volume	Volume	Volume (l)	☒	☒	☐	Part sampling mode	
Particles count	1	Part sampling mode	SamplingMode	Sampling mode	☐	☐	☐		

Figure 8-5 Tags catalog window, Data Types tab

Use the **Data Types** tab to add, edit, delete, and sort through entries based on all possible criteria (use the bottom slider to view all columns).

All values will have functional defaults in place prior to any editing.

Sensor Groups

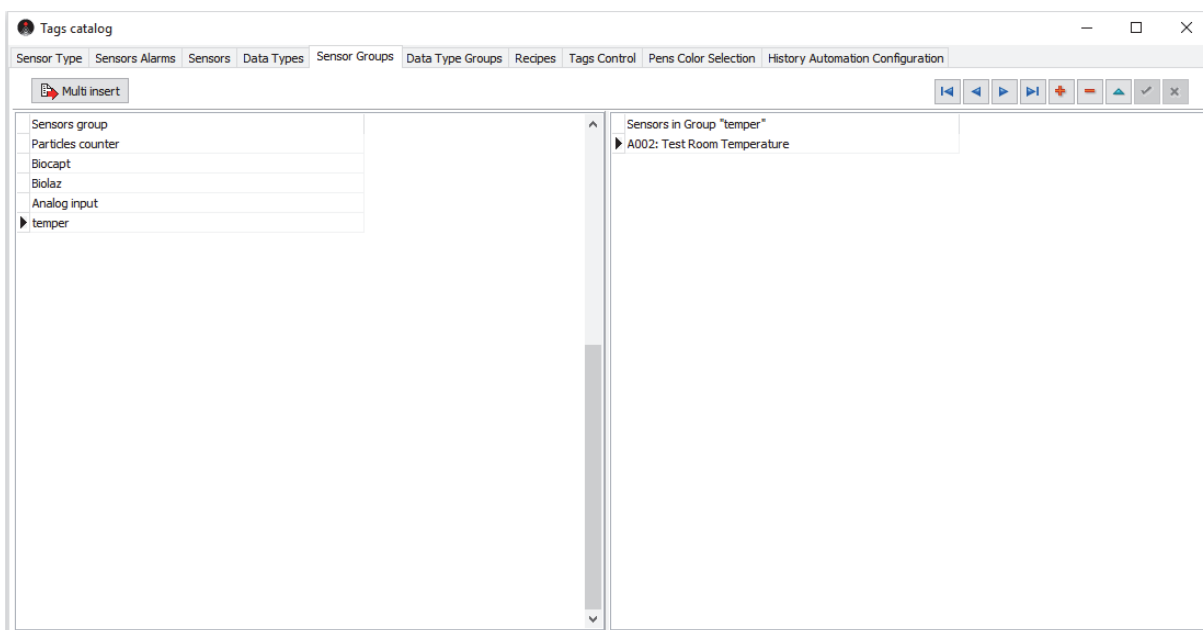


Figure 8-6 Tags catalog window, **Sensor Groups** tab

Use the **Sensor Groups** tab to associate sensors with groups. See Configure Sensor Group Tags on page 8-16.

Data Type Groups

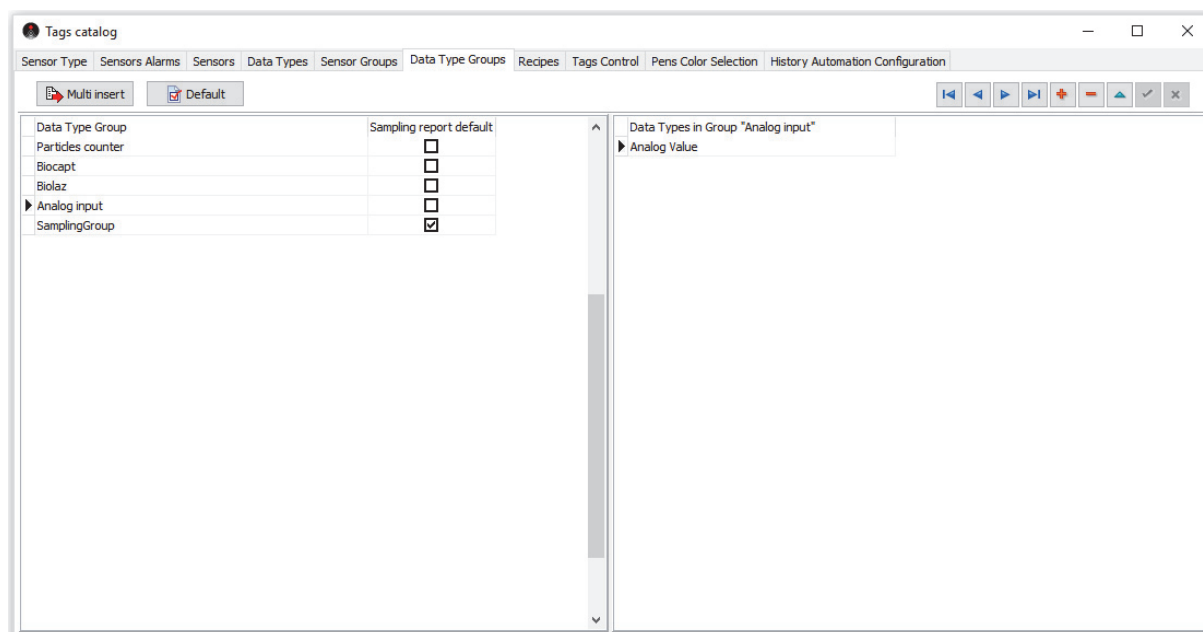
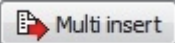
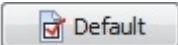


Figure 8-7 Tags catalog window, **Data Type Groups** tab

Use the **Data Type Groups** tab to associate **Data Types** (i.e. MiniCapt Flow) with **Data Type Groups** (i.e. MiniCapt). Default **Data Type Groups** to be displayed for **Sampling Reports** can be checkmarked beneath **Sampling report default**.

Use the **Multi insert** button  to select multiple **Data Types** to associate with a specific **Data Type Group**.

Default  restores the selected entry's default values.

Recipes

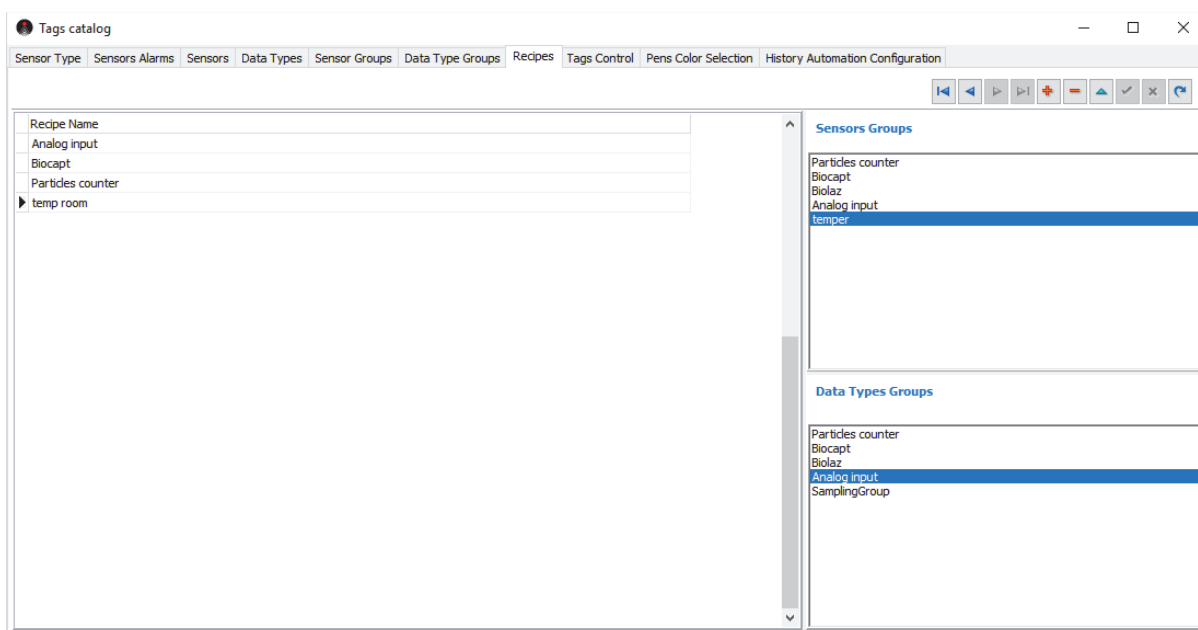


Figure 8-8 Tags catalog window, Recipes tab

Use the **Recipes** tab to assign **Sensors Groups** and **Data Types Groups** to specific recipes (located under Recipe Name, i.e. MiniCapt, Particles, RVU). Entries under **Recipe Name** can be added or deleted. See Configure Recipe Tags on page 8-18 for more details.

Use the **Undo** arrow  to undo the last change made.

Tags Control

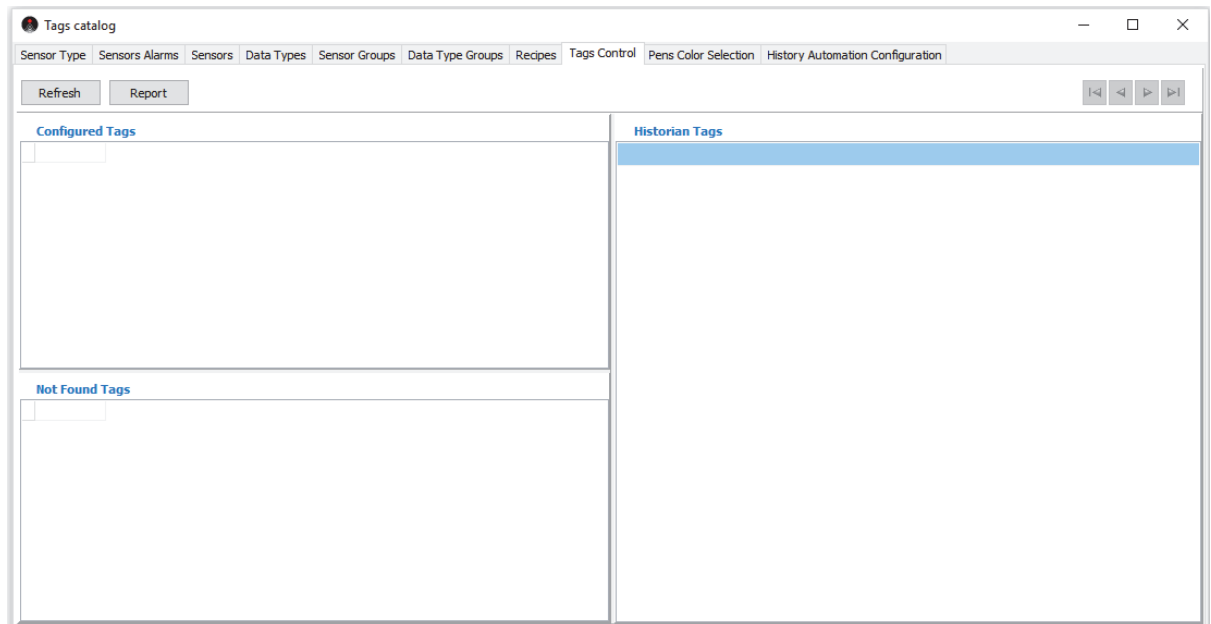


Figure 8-9 Tags catalog window, **Tags Control** tab

Use the **Tags Control** tab to **Report** new tags, navigate through **Configured Tags**, **Not Found Tags** and **Historian Tags**, and **Refresh** tag listings.

Pens Color Selection

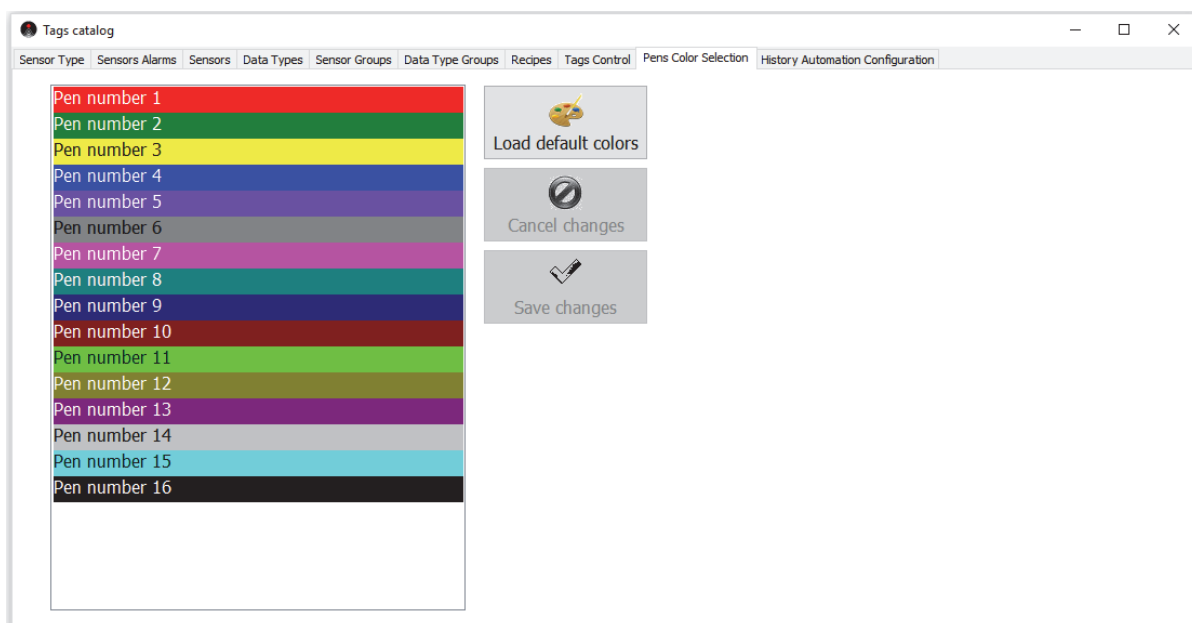


Figure 8-10 Tags catalog window, Pens Color Selection tab

Use the **Pens Color Selection** tab to specify colors of sensor data in a **Trend Report**. The order of use begins with **Pen number 1**.

History Automation Configuration

Tags catalog

Sensor Type | Sensors Alarms | Sensors | Data Types | Sensor Groups | Data Type Groups | Recipes | Tags Control | Pens Color Selection | History Automation Configuration

History automation configuration parameters

☒ Export CSV data files ☒ Export statistical data Execution time: 10:00:00

Recipe: temp room

Target directory: D:\Test_anto Select

Timestamp format

Short Date Format: dd/MM/yyyy

Long Time Format: hh:mm:ss

Get local configuration

Samplings automatic PDF export parameters

FacilityPro recipe	FacilityPro	Interval	Audit	Data	Trend	Summary	Summary Type	Summary Intervals
* 1	0	☒	☒	☒	☐	Min	2	
* 2	0	☒	☒	☒	☐	Sum	2	

PDF files export directory: D:\PDF Select

Save changes Cancel changes

Figure 8-11 Tags catalog window, **History Automation Config**tab

Use the **History Automation Configuration** tab to automatically export CSV files, statistical data and PDF reports associated with a specific FacilityPro recipe at a set time and to a set directory.

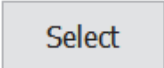
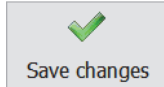
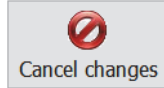
Table 8-2 Components of the **History Automation Config** tab

Component	Description
History automation configuration parameters	
Export CSV data files	Checkmark this box to export CSV data files containing all data for each sensor in the selected Recipe group for the past 24-hour period.
Export statistical data	Checkmark this box to export statistical data containing all statistics for each sensor in the selected Recipe group for the past 24-hour period.
Recipe	List of available recipe Groups
Execution time 10:00:00	Time when automatic export will occur in the same format as the OS. Automatic export occurs daily.
Execution time	
Target directory	Filepath of where the CSV and statistical data will save
Select	Opens Windows explorer to allow user to choose the desired filepath

Table 8-2 Components of the **History Automation Config** tab

Component	Description
Timestamp format	
Short Date Format	How the date will be displayed as part of the exported data files
Long Time Format	How the time will be displayed as part of the exported data files
Get local configuration Get local configuration	Changes the Timestamp format to the same as the current FacilityPro settings, as seen on the Main Page.
Samplings automatic PDF export parameters	
 View First Row button	Click this button to view the first row in the table.
 View Previous Row button	Click this button to view the previous row in the table.
 View Next Row button	Click this button to view the next row in the table.
 View Last Row button	Click this button to view the last row in the table.
 Plus button	Click this button to open the list of FacilityPro recipes for the purpose of adding another row.
 Minus button	Click this button to delete the selected row.
 Save Changes button	Click this button to save changes to the Samplings automatic PDF export parameters area.
 Cancel Changes button	Click this button to undo changes to the Samplings automatic PDF export parameters area.
FacilityPro recipe	Displays the name of the recipe(s) to be included in the sampling export in alphabetical order. The elipsis button  can be used to change the chosen recipe. Choosing the asterisk option (*) will extract all recipes configured with the selected FacilityPro processor.
FacilityPro	Displays the FacilityPro processor number associated with the recipe. The ellipsis button  can be used to change the chosen recipe.
Interval	This field defines an “every nth” value (in seconds) for which subset of sample data to include in the report. For example, if this value is ‘30’, the data included in the report will only be those timestamped every 30 seconds from the start of sampling. If this field is 0, only event data will be included.
Audit	Checkmark this box to include the audit report as part of the recipe sampling export.

Table 8-2 Components of the **History Automation Config** tab

Component	Description
Data	Checkmark this box to include the data report as part of the recipe sampling export.
Trend	Checkmark this box to include the trend report as part of the recipe sampling export.
Summary	Checkmark to enable the Summary Type and Summary Intervals fields.
Summary Type	Calculating the trend can be done with any of the following options: • Minimum option (Min) – Select this option to use the minimum value of the samples in the sub interval. • Maximum option (Max) – Select this option to use the maximum value of the samples in the sub interval. • Average option (Avg) – Select this option to use the average of the samples in the sub interval. • Sum option – Select this option to use the sum of the value of each sample of the interval.
Summary Intervals	This field applies a Historical Trend function which uses fewer samples than the samples stored in the database for the requested time range (between Start date and End date) to create a historical trend view (at least 1 sample every minute). You can set a sub-interval (number of samples) to display collapsed into one sample. A minimum value of 2 is required.
PDF files export directory	Filepath where the PDF reports will be stored for access by the user.
 Select	Opens Windows explorer to allow the user to choose the desired filepath.
 Save changes	Saves adjustments made to any parameters in the History Automation Configuration tab.
 Cancel changes	Cancels any adjustments made to any of the parameters in the History Automation Configuration tab.

Configure Sensor Group Tags

An Administrator can associate sensors with groups using the **Sensor Groups** tab in the **Tags catalog** window.

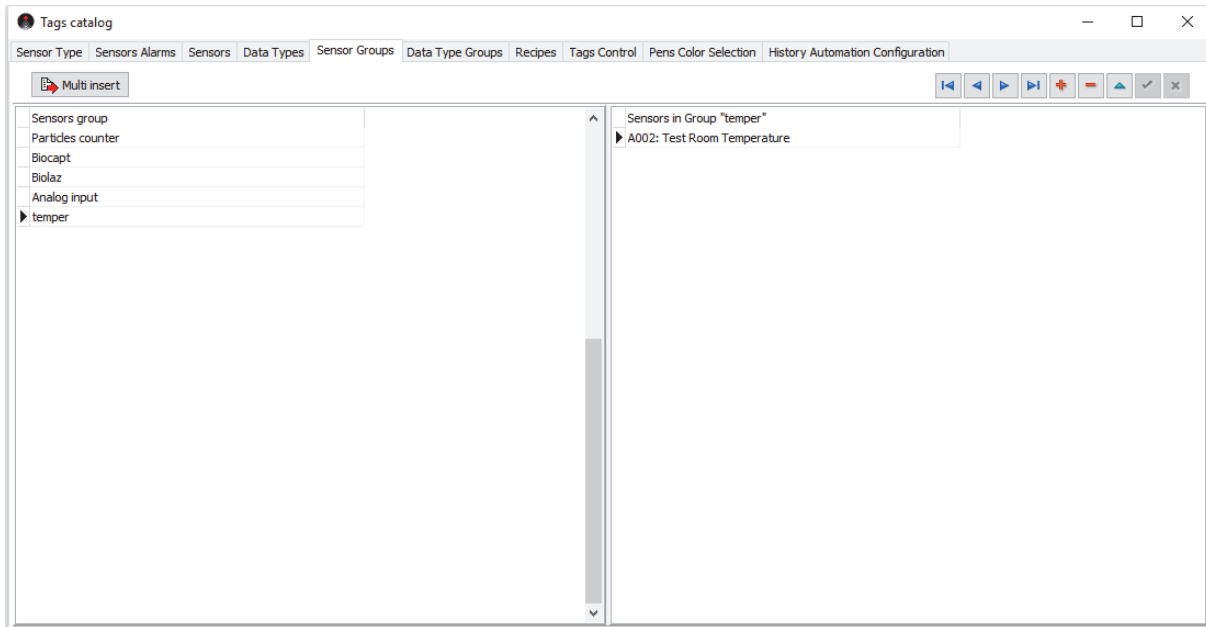
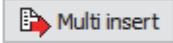


Figure 8-12 Tags catalog window, **Sensor Groups** tab

>> To add a new Sensor group:

- 1 Open the **Tags catalog** window as described on page 8-3.
- 2 Click the **Sensors Groups** tab in the **Tags catalog** window.
- 3 In the **Sensor Groups** tab of the **Tags catalog** window, select one of the existing groups in the **Sensors Groups** list.
- 4 Click  the **Plus** button.
A field for the name of the new Sensor group appears in the **Sensors Groups** column of the **Sensors Groups** tab in the **Tags catalog** window.
- 5 Type the name of the new Sensor group.
- 6 Click the  **Accept** button.
- 7 Click  the **Multi insert** button.
The **Select sensors in group** window appears (see **Figure 8-13**).

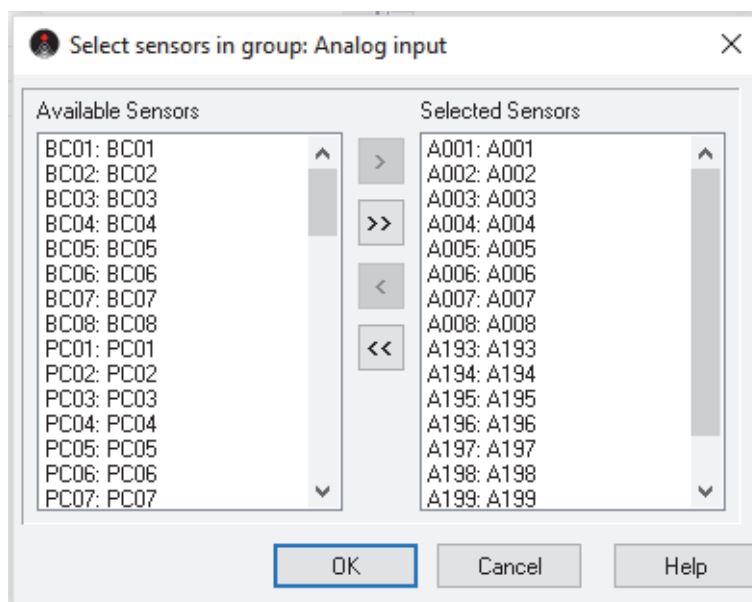
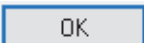


Figure 8-13 Select sensors in group window

- 8** In the **Available Sensors** list of the **Select sensors in group** window, select all sensors that you want to add to the new group.
- 9** Click  the > button to add the selected sensors to the **Selected Sensors** list.
- 10** Click  the **OK** button in the **Select sensors in group** window to close it.
- 11** Click  the **X** button to close the **Tags catalog** window.

Configure Recipe Tags

An Administrator can create and modify recipes using the **Recipes** tab in the **Tags catalog** window.

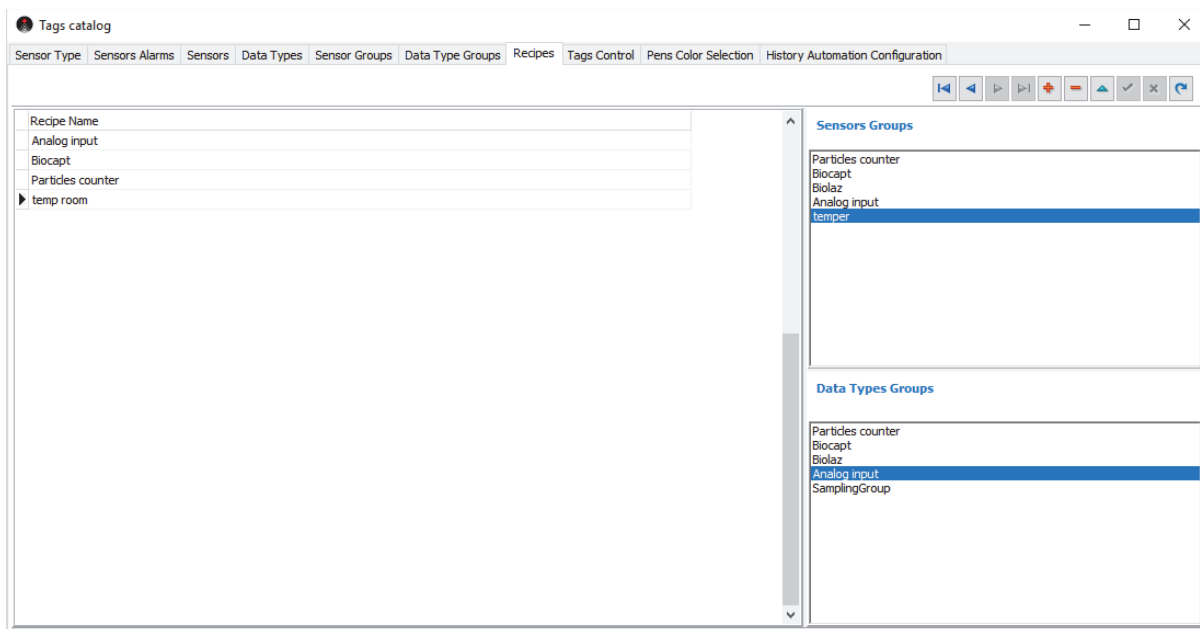


Figure 8-14 Tags catalog window, Recipes tab

Recipes are collections of data to be displayed in **Report Generator**, obtained as a combination of Sensors Groups and Data types.

>> To add new recipe:

- 1 Open the **Tags catalog** window as described on page 8-3.
- 2 Click the **Recipes** tab in the **Tags catalog** window.
- 3 In the **Recipes** tab of the **Tags catalog** window, click  the **Plus** button.
A field for the name of the new recipe appears in the **Recipe Name** column of the **Recipes** tab in the **Tags catalog** window.
- 4 Type the name of the new recipe.
- 5 Select the desired Sensor group from the **Sensors Groups** list.
- 6 Select the desired Data Type group from the **Data Types Groups** list.
- 7 Click the  **Accept** button.
- 8 Click  the **X** button to close the **Tags catalog** window.

Configure Data Type Group Tags

An Administrator can associate Data types with groups.

>> To add a new Data Type Group:

- 1 Open the **Tags catalog** window as described on page 8-3.
- 2 Click the **Data Type Groups** tab in the **Tags catalog** window.

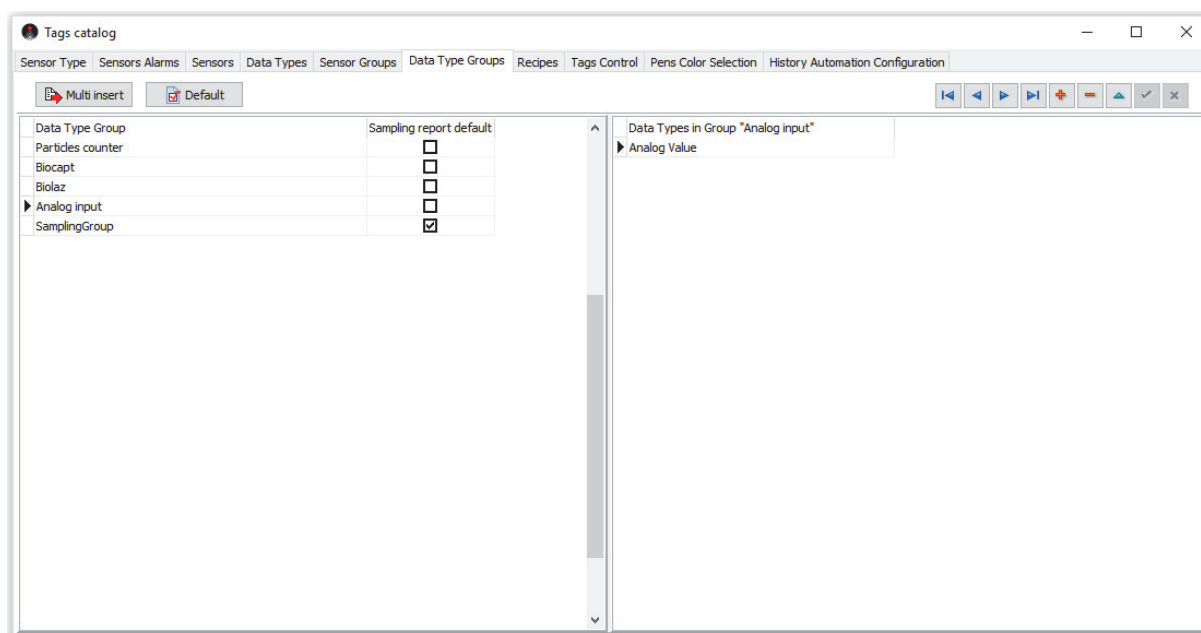
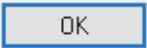


Figure 8-15 Tags catalog window, Data Type Groups tab

- 3 In the **Recipes** tab of the **Tags catalog** window, click  the **Plus** button.
A field for the name of the new Group appears in the **Data Type Group** column of the **Data Type Groups** tab in the **Tags catalog** window.
- 4 Type the name of the new Group.
- 5 Click the  **Accept** button.
- 6 Click  the **Multi insert** button.
The **Select data types in group** window appears.
- 7 Select the desired Data Type group from the **Data Types Groups** list.
- 8 Click  the > button to add the selected sensors to the **Data types in selected group** list.
- 9 Click  the **OK** button in the **Select data types in group** window to close it.
- 10 Click  the **X** button to close the **Tags catalog** window.

Creating Reports through the Report Window

Creating any report through the **Report** window involves the following general steps:

1 Choose between these tabs, then select parameters for the report:

- **Select by Group**

See Create a report “by group” on page 8-21.

- **Select by Sampling**

See Create a report “by sampling” on page 8-28.

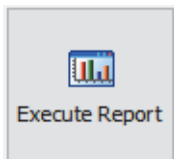
2 Select the report you want to generate.

See Select report(s) to generate on page 8-25.

3 Set a date/time range for the report.

See Set a date and time range for a report on page 8-30.

4 Click



the **Execute Report** button to type note text you want to include

in the header of the report and generate the report.

See Execute the report on page 8-27.

The system automatically generates the **Audit / Data / Trend** report and displays it in the text box of the **Report** window.

After a report has been generated and displayed in the **Report** window, you can:

- Review the report on screen.

See Change the View in the Report Displayed in the Report Window on page 8-35.

- Print the report.

See Print a Report from the Report Window on page 8-37.

- Save a copy of the report as a **PDF** file or a **CSV** (comma-separated values) file.

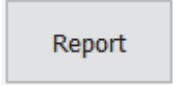
See Export a Report from the Report Window on page 8-37.

Create a report “by group”

You can generate a report for a pre-defined group of modules. For example, groups of modules might include:

- **All Analogs** – all Analog sensors installed in the system.
- **All BIO** – all Biological samplers installed in the system.
- **All PC** – all Particle sensors installed in the system.

>> **To display the parameters for generating a report based on “by group”:**

- 1 Click  the **Report** button at the bottom of the Pharmaceutical Net Pro screen.

The **Report** window appears (see **Figure 8-1** on page 8-2).

- 2 Click  the **Select by Group** tab.

The parameters available display along the top of the **Report** window (see **Figure 8-16**)



Figure 8-16 Parameter fields in the **Report** window for **Select by Group** reports

The parameters available for reports created “by group” in the **Report** window are described in **Table 8-3**.

Table 8-3 **Select by Group** tab **Report** window components

Component	Description
Start date field and list field	Defines the day, month, and year for the beginning of the time range to be included in the report.
Start time field and spinner buttons	Defines the hour, minute, and second for the beginning of the time range to be included in the report, in the same format as the OS.
End date field and list field	Defines the day, month, and year for the end of the time range to be included in the report.
End time field and spinner buttons	Defines the hour, minute, and second for the end of the time range to be included in the report, in the same format as the OS.
 All Data option	Select this option to include in the report all event data collected.

Table 8-3 Select by Group tab Report window components (Continued)

Component	Description
☒ Only this mode option	Select this option to select the mode for the report from the field: • Normal Operation • At rest • In operation • Custom
Select Group list	Only one of the pre-defined groups can be selected as the basis for the an Audit Report report(s) during a session if it is being filtered with the ☒ Filtered option.
☒ Audit Report check box and options See an example on page 8-38	• ☒ When checked, an Audit Report will be generated and displayed in the Audit Report tab when the Execute Report button is clicked. • ☐ When not checked, no Audit Report will be generated when the Execute Report button is clicked.
☒ Filtered option	☒ When selected, only the data available for the group selected in the Select Group list will be included when the Audit report is generated.
☒ Not Filtered option	☒ When selected, data for all groups will be included when the Audit report is generated.
☒ Data Report check box See an example on page 8-42	• ☒ When checked, a Data Report will be generated and displayed in the Data Report tab when the Execute Report button is clicked. • ☐ When not checked, no Data Report will be generated when the Execute Report button is clicked.
☒ Detailed option	☒ When selected, all the data available for the group selected in the Select Group list will be included when the Data report is generated.
☒ Stats Only option	☒ When selected, only the Statistical data available for the group selected in the Select Group list will be included when the Data report is generated.
☒ Trend Report check box See an example on page 8-46	• ☒ When checked, a Trend Report will be generated and displayed in the Trend Report tab when the Execute Report button is clicked. • ☐ When not checked, no Trend Report will be generated when the Execute Report button is clicked.

Table 8-3 Select by Group tab Report window components (Continued)

Component	Description
Intervals field	This field defines an “every nth” value for which subset of samples to include in the report. If this field is null, all the data from the sampling will be included in the report.
Report Type options ☒ Raw option ☒ Summary option	☒ Select this option to hide the Summarize pane. ☒ Select this option to enable the Intervals field and Calculate options in the Summarize pane.
Summarize field and options Number of Intervals field Calculate options	This field applies a Historical Trend function which uses fewer samples than the samples stored in the database for the requested time range to create a historical trend view (at least 1 sample every minute). You can set a sub-interval (number of samples) to display collapsed into one sample. The sub-interval’s value to use for calculating the trend can be one of the following options: • ☒ Minimum option – Select this option to use the minimum value of the samples in the sub interval. • ☒ Maximum option – Select this option to use the maximum value of the samples in the sub interval. • ☒ Average option – Select this option to use the average of the samples in the sub interval. • ☒ Sum option – Select this option to use the sum of the value of each sample of the interval.

Set a date and time range for a report

Every report requires a start date and time, and an end date and time.

>> To set the start date for the report:

- 1 In the **Start Date** field, click  the list box button.
A **Calendar** popup appears (see **Figure 8-17**).
- 2 Use the  left and  right arrows in the **Calendar** popup to select the month and year for the start date of the report.
- 3 Select the day of the month.

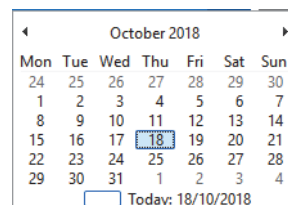


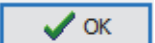
Figure 8-17 Calendar popup

- 4 Click in the **Start date** field to close the **Calendar** popup.
The date you selected in the **Calendar** popup displays in the **Start Date** field.

>> To set the end date for the report:

- 1 Repeat the steps described above in To set the start date for the report: using the **End Date** field instead of the **Start Date** field.

>> To set the start time for the report:

- 1 In the **Start Time** field, edit the hour, minute, second using the format **hh:mm:ss**,
–OR–
On SCADA Client PCs, click  the spinner buttons in the **Start Time** field.
- 2 Click  the **OK** button in the **Time Set** popup to close it.
The time you selected appears in the **Start Time** field.

>> To set the end time for the report:

- 1 Repeat the steps described in To set the start time for the report: on page 8-31 using the **End Time** field instead of the **End Time** field.

Select a data subset

- 1 To include all data in the report, select the  **All Data** option.
- 2 To include a data subset in the report, select the  **Only this mode** option:
 - a. From the **Only this mode** list box, select the mode to apply to the report.
Examples of modes include:
 - **Normal Operation**
 - **At rest**
 - **Custom**

Select a group of modules to include in the report

- 1 In the **Select Group** list, select a group of modules to use as the basis for the report.
Examples of groups are:
 - **All Analogs** – all Analog sensors installed in the system.
 - **All BIO** – all Biological samplers installed in the system.
 - **All PC** – all Particle sensors installed in the system.

Select report(s) to generate

One or more of the following reports can be generated at one time:

- **Audit Report** – Lists the following events:
 - Log-ins
 - Electronic Signatures
 - Alarms
 - Alarm Acknowledges

See an illustration in Audit Report examples – Select by Group – Audit Report on page 8-38.

- **Trend Report**

Historical trend of sampling data from the sensors rendered in graphical format.

See an illustration in Trend Report examples – Select by Samplings on page 8-46.

- **Data Report**

Sampling data from the sensors in tabular format.

See an illustration in Data Report examples – Select by Group on page 8-42.

If you choose more than one report to generate, each report will be available in a separate tab in the **Report** window after the reports are generated.

>> To select the report subject(s):

- 1 To select the **Audit Report**, place a checkmark in the ☒ **Audit Report** check box and select one of the following options:
 - ☒ **Filtered** option – Causes the data displayed to be limited to only the data in the group. (Filters are configured in the **Custom fields configuration** window (see Configuring Custom Fields to Apply to a Sampling on page 5-6.)
 - ☒ **Not Filtered** option – Causes no filter to be applied to the report.
- 2 To select the **Data Report**, place a checkmark in the ☒ **Data Report** check box and select one of the following options:
 - ☒ **Detailed** – When selected, all the data available for the group selected in the **Select Group** list will be included when the **Data** report is generated.
 - ☒ **Stats Only** – When selected, only the Statistical data available for the group selected in the **Select Group** list will be included when the **Data** report is generated.
- 3 To select the **Trend Report**, place a checkmark in the ☒ **Trend Report** check box.

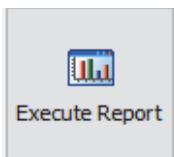
Select an interval, specify the type for the report**>> To select the a certain interval for which data to include and specify “raw” or “summary” data:**

- 1 In the **Interval(s)** field, enter an “every nth” number to limit the number of samples to include in the report.
- 2 In the **Report Type** pane, select one of the following options:
 - ☒ **Raw** – Select this option to use raw data in the report.
 - ☒ **Summary** – Select this option to display the **Summarize** pane with the **Number of intervals** field and the **Calculate** options.

For a Trend Report, set calculation parameters

- 1 In the **Number of Intervals** field, enter the number of samples to display collapsed into one sample.
- 2 From the **Calculate** pane, select which sub-interval’s value to use for calculating the trend:
 - ☒ **Minimum** option – The minimum value of the samples in the sub interval.
 - ☒ **Maximum** option – The maximum value of the samples in the sub interval.
 - ☒ **Average** option – The average of the samples in the sub interval.
 - ☒ **Sum** option – The sum of the value of each sample of the interval.

Execute the report

- 1 Click  the **Execute Report** button.

- 2 The **Input a comment to print on reports** dialog box appears (see **Figure 8-18** on page 8-27).

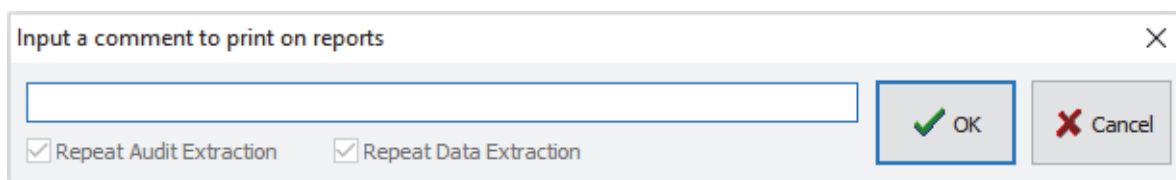
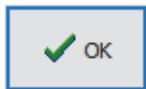


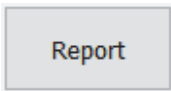
Figure 8-18 **Input a comment to print on reports** dialog box

- 3 In the **Input a comment to print on reports** dialog box, enter note text you want included in the header of the report.
- 4 Click  the **OK** button in the **Input a comment to print on reports** dialog box.
- The system automatically generates **Audit / Data / Trend** report and displays it in the text box of the **Report** window.

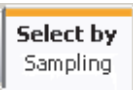
Create a report “by sampling”

The parameters available for reports created “by sampling” are shown in **Figure 8-19** and described in **Table 8-4**.

>> To display the parameters for generating a report based on “by sampling”:

1 Click  the **Report** button at the bottom of the Pharmaceutical Net Pro

screen. The **Report** window appears (see **Figure 8-1** on page 8-2).

2 Click  the **Select by Sampling** button.

The parameters available display along the top of the **Report** window (see **Figure 8-19**)

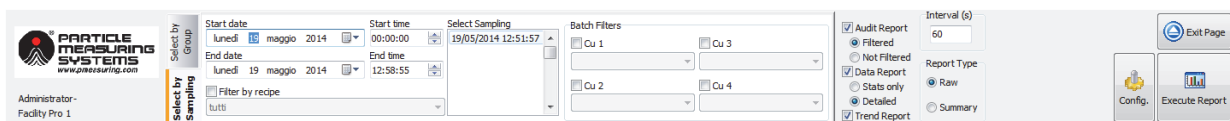


Figure 8-19 Parameter fields in the **Report** window for **Select by Sampling** reports

The components of the **Report** window for the **Select by Sampling** mode are described in **Table 8-4**.

Table 8-4 **Select by Sampling** mode **Report** window components

Component	Description
Start date field and list field	Defines the day, month, and year for the beginning of the time range to be included in the report.
Start time field and spinner buttons	Defines the hour, minute, and second for the beginning of the time range to be included in the report, in the same format as the OS.
End date field and list field	Defines the day, month, and year for the end of the time range to be included in the report.
End time field and spinner buttons	Defines the hour, minute, and second for the end of the time range to be included in the report, in the same format as the OS.
☒ Filter by recipe check box and list field	☒ When checked, you can select recipe from the list to apply as a filter for the report.
Select Sampling list	One currently scheduled sampling can be selected from this list.

Table 8-4 Select by Sampling mode Report window components (Continued)

Component	Description
Batch Filters check boxes and list fields	☒ Custom Field 1 – When checked, you can select the filter to apply to the report. ☒ Custom Field 2 – When checked, you can select the filter to apply to the report. ☒ Custom Field 3 – When checked, you can select the filter to apply to the report. ☒ Custom Field 4 – When checked, you can select the filter to apply to the report. See Custom fields configuration window on page 5-6.
☒ Audit Report check box and options ☒ Filtered option ☒ Not Filtered option	☒ When checked, an Audit Report will be generated and displayed in the Audit Report tab when the Execute Report button is clicked. ☐ When not checked, no Audit Report will be generated when the Execute Report button is clicked. ☒ When selected, only the data available for the custom filter selected in the Batch Filters list will be included when the Audit report is generated. ☒ When selected, all data for the selected sampling for all groups will be included when the Audit report is generated.
☒ Data Report check box See an example on Data Report examples – Select by Group on page 8-42 ☒ Detailed option ☒ Status Only option	☒ When checked, a Data Report will be generated and displayed in the Data Report tab when the Execute Report button is clicked. ☐ When not checked, no Data Report will be generated when the Execute Report button is clicked. ☒ When selected, all the data available for the sampling selected in the Select Sampling list will be included when the Data report is generated. ☒ When selected, only the Statistical data available for the sampling selected in the Select Sampling list will be included when the Data report is generated.
☒ Trend Report check box	☒ When checked, a Trend Report will be generated and displayed in the Trend Report tab when the Execute Report button is clicked. ☐ When not checked, no Trend Report will be generated when the Execute Report button is clicked.

Table 8-4 Select by Sampling mode Report window components (Continued)

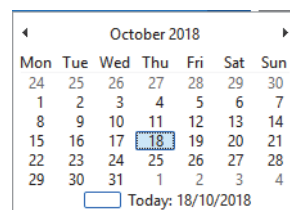
Component	Description
Intervals field	This field defines an “every nth” value for which subset of samples to include in the report. If this field is null, all the data from the sampling will be included in the report.
Report Type options ☒ Raw option ☒ Summary option	☒ When selected, hides Summarize pane. ☒ When selected, enables the Intervals field and Calculate options in the Summarize pane.
Summarize field and options Number of Intervals field Calculate options	This field applies a Historical Trend function which uses fewer samples than the samples stored in the database for the requested time range to create a historical trend view (at least 1 sample every minute). You can set a sub-interval (number of samples) to display collapsed into one sample. The sub-interval’s value to use for calculating the trend can be one of the following options: ☒ Minimum option – The minimum value of the samples in the sub interval. ☒ Maximum option – The maximum value of the samples in the sub interval. ☒ Average option – The average of the samples in the sub interval. ☒ Sum option – The sum of the value of each sample of the interval.

Set a date and time range for a report

Every report requires a start date and time, and an end date and time.

>> To set the start date for the report:

- 1 In the **Start Date** field, click  the list box button. A **Calendar** popup appears (see **Figure 8-20**).
- 2 Use the  left and  right arrows in the **Calendar** popup to select the month and year for the start date of the report.
- 3 Select the day of the month.

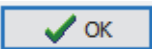
**Figure 8-20 Calendar popup**

- 4 Click in the **Start date** field to close the **Calendar** popup. The date you selected in the **Calendar** popup displays in the **Start Date** field.

>> To set the end date for the report:

- 1 Repeat the steps described above in To set the start date for the report: using the **End Date** field instead of the **Start Date** field.

>> To set the start time for the report:

- 1 In the **Start Time** field, edit the hour, minute, second using the same format as the operating system (OS),
-OR-
On SCADA Client PCs, click  the spinner buttons in the **Start Time** field.
- 2 Click  the **OK** button in the **Time Set** popup to close it.
The time you selected appears in the **Start Time** field.

>> To set the end time for the report:

- 1 Repeat the steps described in To set the start time for the report: on page 8-31 using the **End Time** field instead of the **End Time** field.

Select a recipe, a sampling, and filters

- 1 To limit the report to events for a single recipe:
 - a. Place a ☒ checkmark in the **Filter by recipe** check box.
 - b. From the field below the **Filter by recipe** check box, select the Recipe you want to use for the report.
- 2 From the **Select Sampling** field, select the sampling to use in the report.
- 3 To apply filters to the report in the **Batch Filters** pane:
 - a. Place a ☒ checkmark in the **Custom Field 1** check box to activate the filter.
 - b. Place a ☒ checkmark in the **Custom Field 2** check box to activate the filter.
 - c. Place a ☒ checkmark in the **Custom Field 3** check box to activate the filter.
 - d. Place a ☒ checkmark in the **Custom Field 4** check box to activate the filter.

Select a report to generate

One or more of the following reports can be generated at one time:

- **Audit Report** – Lists the following events:

- Log-ins
- Electronic Signatures
- Alarms
- Alarm Acknowledges

See an illustration in Audit Report examples – Select by Group – Audit Report on page 8-38.

- **Trend Report**

Sampling data from the sensors in graphical format.

See an illustration in Trend Report examples – Select by Samplings on page 8-46.

- **Data Report**

Sampling data from the sensors in tabular format.

See an illustration in Data Report examples – Select by Group on page 8-42.

The **Report** window has separate tabs for type of report.

>> To select the report subject(s):

1 To select the **Audit Report**, place a checkmark in the ☒ **Audit Report** check box and select one of the following ☐ options:

- ☒ **Filtered** option – Causes the data displayed to be limited to only the data in the group selected in the **Select Group** field. (Filters are configured in the **Custom fields configuration** window (see Configuring Custom Fields to Apply to a Sampling on page 5-6.)
- ☐ **Not Filtered** option – Causes no filter to be applied to the report.

2 To select the **Data Report**, place a checkmark in the ☒ **Data Report** check box, and select one of the following ☐ options:

- ☒ **Detailed** option – When selected, all the data available for the sampling selected in the **Select Sampling** list will be included when the **Data** report is generated.
- ☐ **Stats Only** option – When selected, only the Statistical data available for the sampling selected in the **Select Sampling** list will be included when the **Data** report is generated

3 To select the **Trend Report**, place a checkmark in the ☒ **Trend Report** check box.

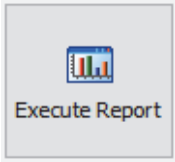
Select an interval, specify the type for the report

- 1 In the **Interval(s)** field, enter an “every nth” number to limit the number of samples to include in the report.
- 2 In the **Report Type** pane, select one of the following  options:
 -  **Raw** – Select this option to use raw data in the report.
 -  **Summary** – Select this option to display the **Summarize** pane with the **Number of intervals** field and the **Calculate** options.

For a Trend Report, set calculation parameters

- 3 In the **Number of Intervals** field, enter the number of samples to display collapsed into one sample.
- 4 From the **Calculate** pane, select which sub-interval’s value to use for calculating the trend:
 -  **Minimum** option – The minimum value of the samples in the sub interval.
 -  **Maximum** option – The maximum value of the samples in the sub interval.
 -  **Average** option – The average of the samples in the sub interval.
 -  **Sum** option – The sum of the value of each sample of the interval.

Execute the report

- 1 Click  the **Execute Report** button.

- 2 The **Input a comment to print on reports** dialog box appears (see **Figure 8-21** on page 8-33).

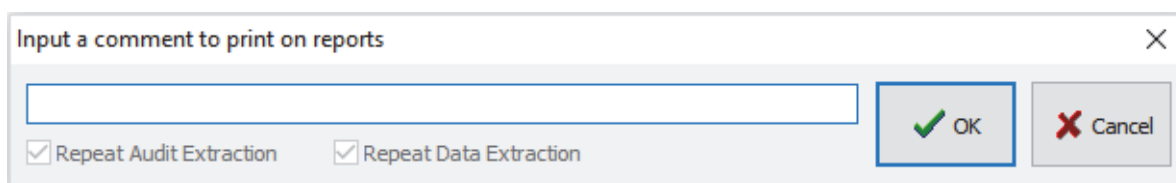
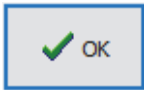


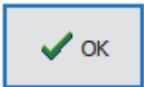
Figure 8-21 Input a comment to print on reports dialog box

- 3 In the **Input a comment to print on reports** dialog box, enter note text you want included in the header of the report.
- 4 Click  the **OK** button in the **Input a comment to print on reports** dialog box to close it.

- 5** If **Data or Trend** check boxes are selected, you will be able to select between PDF and CSV data types.

The data types contained in the data type group you selected are listed in the **Data types in selected group** pane.

- e.** In the **Data types in selected group** pane, select a data type.

- f.** Click  the **OK** button in the **Select data types group** dialog box to close it.

- 6** The system automatically generates the **Audit / Data / Trend** report and displays it in the text box of the **Report** window.

Change the View in the Report Displayed in the Report Window

To view a report you generated, click its tab in the **Report** window.

>> To change the view of the report displayed in the Report window:

Use the **Report** window controls shown in **Figure 8-22** and described in **Table 8-5**. General and report-specific controls in the Report window to change the view of a report.



Figure 8-22 Audit Report window controls for changing the view of a generated report

The general controls for changing the view of a report in the **Report** window are described in **Table 8-5**.

Table 8-5 General and report-specific controls in the **Report** window

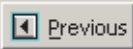
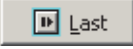
General Controls	Description
 First button	Click this button to view the first page of the report.
 Previous button	Click this button to view the previous page of the report.
 Next button	Click this button to view the next page of the report.
 Last button	Click this button to view the last page of the report.
 Zoom In button	Click this button to magnify the report.
 Zoom Out button	Click this button to reduce the magnification of the report.
 Fit to page button	Click this button to fit the view to show the entire page of the report.
 Fit to width button	Click this button to fit the view to show the entire width of the report.
 Print report button	Click this button to open the Select pages to print dialog box so you can select the pages of the report to send to a printer. See Select pages to print dialog box on page 8-37.
 Save report button	Click this button to open the Choose the file format dialog box so you can save a PDF file or a CSV file of the report.
Page n of m current page field	Type the number of the page you want to view, then press the Enter key on the keyboard to jump to that page.

Table 8-5 General and report-specific controls in the **Report** window (Continued)

Audit Report Controls	Description
☒ All records option	Select this option to view all event types generated for the Audit Report . (See example on page 8-38.)
☒ Audit records only option	Select this option to view only audit events.
☒ Alarms only option	Select this option to view only alarm events.
☒ Alarms and Acknowledge option	Select this option to view alarms and acknowledgements.
☒ Select record ID option	Select this option to open the Insert alarm ID dialog box where you can enter an Alarm ID so you can generate a report for that record. Figure 8-23 Insert alarm ID dialog box
☒ Event Log only option	Select this option to view events log created by the users.
Trend Report Controls	Description
☒ All Pages option	Select this option to apply the viewing options to all pages in the Trend Report display. (See example on page 8-46.)
☒ Current Page option	Select this option to apply the viewing options to the page currently being viewed in the Trend Report display.
Slider	Drag the slider button to move the view left or right.
Values per page field	Sets the number of intervals to display on the page.
☒ Allow hiding check box	☒ When checked, limits the data displayed to the selected sensor.
☒ 3D style check box	☒ When checked, applies the 3-dimensional graphing mode to the display. ☐ When not checked, applies the 2-dimensional graphing mode to the display.
☒ Show Alarm Labels check box	☒ When checked, alarms a labels on the graph.
☒ Left Axis check box	☒ When checked, data are shown in logarithmic scale on the left axis, if it has been set in the Tag Catalog.
☒ Right Axis check box	☒ When checked, data are showed in logarithmic scale on the right axis, if it has been set in the Tag Catalog.
☒ Show < 1 Values check box	☒ When checked, together with Left Axis or Right Axis, includes < 1 values in the logarithmic graph.

Table 8-5 General and report-specific controls in the **Report** window (Continued)

Legend	Lists the main parameters used to create the report and the colors that represent an instrument or parameter.
---------------	---

Print a Report from the Report Window

>> To print a report from the Report window:

- 1 Generate the report.
The report displays in the text box of the **Report** window.
- 2 Click  the **Print report** icon in the toolbar at the top of the report in the **Report** window.
A **Select Pages To Print** window appears (see **Figure 8-24**).

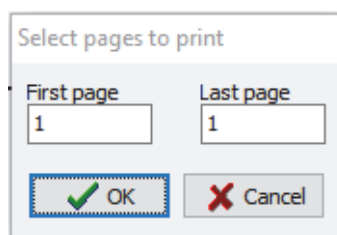


Figure 8-24 Select pages to print dialog box

- 3 Enter the pages to print.
- 4 Click  the **OK** button in the **Select pages to print** dialog box

Export a Report from the Report Window

Reports generated through the **Report** window can be saved as the following file types:

- PDF – Portable Document Format for viewing in Adobe Acrobat
- CSV – Comma-separated values for use in a spreadsheet application

>> To export a report from the Report window:

- 1 Generate the report. The report displays in the text box of the **Report** window.
- 2 Click  the **Save report** icon in the toolbar at the top of the report in the **Report** window to choose between PDF and CSV file type options. Choose **Cancel** to exit.
- 3 Select the file format in the **Choose the file format** dialog box. The **Choose the file format** dialog box closes.
 - Pharmaceutical Net Pro exports a PDF of the report to a folder in the general path **c:\Export \PDF** and a CSV file in the general path **c:\Export \CSV**.
 - The full path displays a message with the full name of the exported report file.

Report Examples

Examples of the reports that can be generated from the **Report** window are described in the following sections:

Audit Report examples – Select by Group – Audit Report, this page

Data Report examples – Select by Group on page 8-42

Trend Report examples – Select by Samplings on page 8-46

Audit Report examples – Select by Group – Audit Report

An example of an **Audit Report – Select by Group** displayed in the **Report** window is shown in **Figure 8-25**.

Particle Measuring Systems
spectris company

Audit report
Comment:
Printed by - ascotec

From: 17/10/2018 00:00:00 To: 20/10/2018 00:16:48
All records

ID	Date and time	Source	Description	Value	Status	Performed by	Comment
193	18/10/2018 17:30:01	-	A193: A193-> Sensor COM Alarm	True	ALM		
194	18/10/2018 17:30:01	-	A193: A193-> Sensor COM Alarm	2	ALM		
195	18/10/2018 17:30:01	-	A194: A194-> Sensor COM Alarm	True	ALM		
196	18/10/2018 17:30:01	-	A194: A194-> Sensor COM Alarm	2	ALM		
197	18/10/2018 17:30:01	-	A195: A195-> Sensor COM Alarm	True	ALM		
198	18/10/2018 17:30:01	-	A195: A195-> Sensor COM Alarm	2	ALM		
199	18/10/2018 17:30:01	-	A196: A196-> Sensor COM Alarm	True	ALM		
200	18/10/2018 17:30:01	-	A196: A196-> Sensor COM Alarm	2	ALM		
201	18/10/2018 17:30:01	-	A197: A197-> Sensor COM Alarm	True	ALM		
202	18/10/2018 17:30:01	-	A197: A197-> Sensor COM Alarm	2	ALM		
203	18/10/2018 17:30:01	-	A198: A198-> Sensor COM Alarm	True	ALM		
204	18/10/2018 17:30:01	-	A198: A198-> Sensor COM Alarm	2	ALM		
205	18/10/2018 17:30:01	-	A199: A199-> Sensor COM Alarm	True	ALM		
206	18/10/2018 17:30:01	-	A199: A199-> Sensor COM Alarm	2	ALM		
207	18/10/2018 17:30:01	-	A200: A200-> Sensor COM Alarm	True	ALM		
208	18/10/2018 17:30:01	-	A200: A200-> Sensor COM Alarm	2	ALM		
2	18/10/2018 17:30:05	-	PC01: PC01-> Pneumatic Fault Alarm	True	ALM		
5	18/10/2018 17:30:05	-	PC10: PC10-> Pneumatic Fault Alarm	True	ALM		
8	18/10/2018 17:30:05	-	PC11: PC11-> Pneumatic Fault Alarm	True	ALM		
11	18/10/2018 17:30:05	-	PC12: PC12-> Pneumatic Fault Alarm	True	ALM		

Figure 8-25 Example of an **Audit Report Select by Group** displayed in the **Report** window

Header information – Audit Report – Select by Group

The header at the top of the first page of an **Audit Report – Select by Group** contains:

- Company logo
- Report name
- Comment text included when the report was created
- Report range – The date/time span covered in the report

- Data filtering applied
 - **All records**
 - **Audit records only**
 - **Alarm only**
 - **Alarm and AcknowledgeTime interval required: From (date/time) to (date/time)**
- Printed By – Name of the user logged in

Report body – Audit Report – Select by Group

The body of the **Audit Report – Select by Group** lists information for each event recorded in the audit log.

- Column headings:
 - **ID** – The event identification number
 - **Date and Time** – The event's date and time
 - **Node** – The iFix data source
 - **Description** – A description of the event
 - **Value** – For alarms, the value for the TAG
 - **Status** – Current status of the event reported:
 - **CFN** – Active alarm (Change From Normal)
 - **HIHI** – A status indicating the high threshold alarm exceeded
 - **HI** – A status indicating the high alert threshold exceeded
 - **LOLO** – A status indicating the low threshold alarm exceeded
 - **LO** – A status indicating the low alert threshold exceeded
 - **Performed by** – The user who performed the electronic signature
 - **Comment** – Notes inserted by the operator

Footer information – Audit Report – Select by Group

The footer of each report contains:

- Printing date and time – In the same format as the OS
- **FacilityPro <station name room name>**
- Page n of m – Number of the current page and total number of pages in the report

Last page – Audit Report – Select by Group

The last page of a report includes some blank lines for the operator to fill in on the printed report:

Performed by _____ Date _____ Reviewed and Approved by _____ Date _____

Audit Report examples – Select by Sampling Audit Report

An example of an **Audit Report – Select by Sampling** displayed in the **Report** window is shown in **Figure 8-26**.

PARTICLE MEASURING SYSTEMS
spectris company

ascotec
Facility Pro 1

Start date: 19 October 2018
End date: 20 October 2018
Start time: 00:00:00
End time: 00:16:48
Select Sampling: 19/10/2018 00:41:20
Batch Filters: CustomField1, CustomField2, CustomField3, CustomField4
Audit Report: ☒ Filtered, ☐ Not Filtered
Data Report: ☐ Data only, ☒ Detailed
Trend Report: ☐ Trend Report, ☐ Summary
Report type: ☐ By event, ☒ By interval
Sec: 5
Grouping: ☒ None, ☐ Summary

Aventino - Facility Pro 1
Recipe: B1A0221 (All records)
Mode: Sampling Mode 1
Started at: 19/10/2018 00:41:03
CustomField1:
CustomField2:
CustomField3:
CustomField4:
Sampling notes:
Comment:
Group: Installer
User: ascotec

ID	Date and time	Source	Description	Value	Status	Performed by	Comment
Performed by: _____ Date: _____ Reviewed and approved by: _____ Date: _____							

Figure 8-26 Example of an **Audit Report – Select by Sampling** displayed in the **Report** window

Header information – Audit Report – Select by Sampling

The header at the top of the first page of a **Audit Report – Select by Sampling** contains:

- Company logo
- **FacilityPro <station name room name>**
- Recipe name – The name of the recipe selected for the report.
- Time the recipe started running – **Started at mm/dd/yyyy or dd/mm/yyyy hh:mm:ss AM or PM**
- Mode – The mode of the sampling selected for the report:
 - **Normal Operation**
 - **At Rest**
 - **Custom**
- Custom field 1, if any specified
- Custom field 2, if any specified
- Custom field 3, if any specified
- Custom field 4, if any specified
- Sampling notes – User defined

- Comment text included when the report was created
- Group – name of the login group the logged-in user belongs to
- User – name of the user logged in when the report was generated

Report body – Audit Report – Select by Sampling

The body of the **Audit Report – Select by Sampling** lists information for each event recorded in the audit log.

- Column headings:
 - **ID** – The event identification number
 - **Date and Time** – The event's date and time
 - **Node** – The iFix data source
 - **Description** – A description of the event
 - **Value** – For alarms, the value for the TAG
 - **Status** – Current status of the event reported:
 - **CFN** – Active alarm (Change From Normal)
 - **HIHI** – A status indicating the high threshold alarm exceeding
 - **HI** – A status indicating the high alert threshold exceeding
 - **LoLo** – A status indicating the low threshold alarm exceeding
 - **Lo** – A status indicating the low alert threshold exceeding
 - **Performed by** – The user who performed the electronic signature
 - **Comment** – Notes inserted by the operator

Footer information – Audit Report – Select by Sampling

The footer of each report contains:

- Printing date and time – In the same format as the OS
- **FacilityPro <station name room name>**
- Page n of m – Number of the current page and total number of pages in the report

Last page – Audit Report – Select by Sampling

The last page of a report includes some blank lines for the operator to fill in on the printed report:

Performed by _____ Date _____ Reviewed and Approved by _____ Date _____

Data Report examples – Select by Group

The **Data Report** creates a table of sampling data from the sensors and calculated values for minimums, maximums, averages, and standard deviations.

An example of an **Data Report – Select by Group** displayed in the **Report** window is shown in **Figure 8-27**.

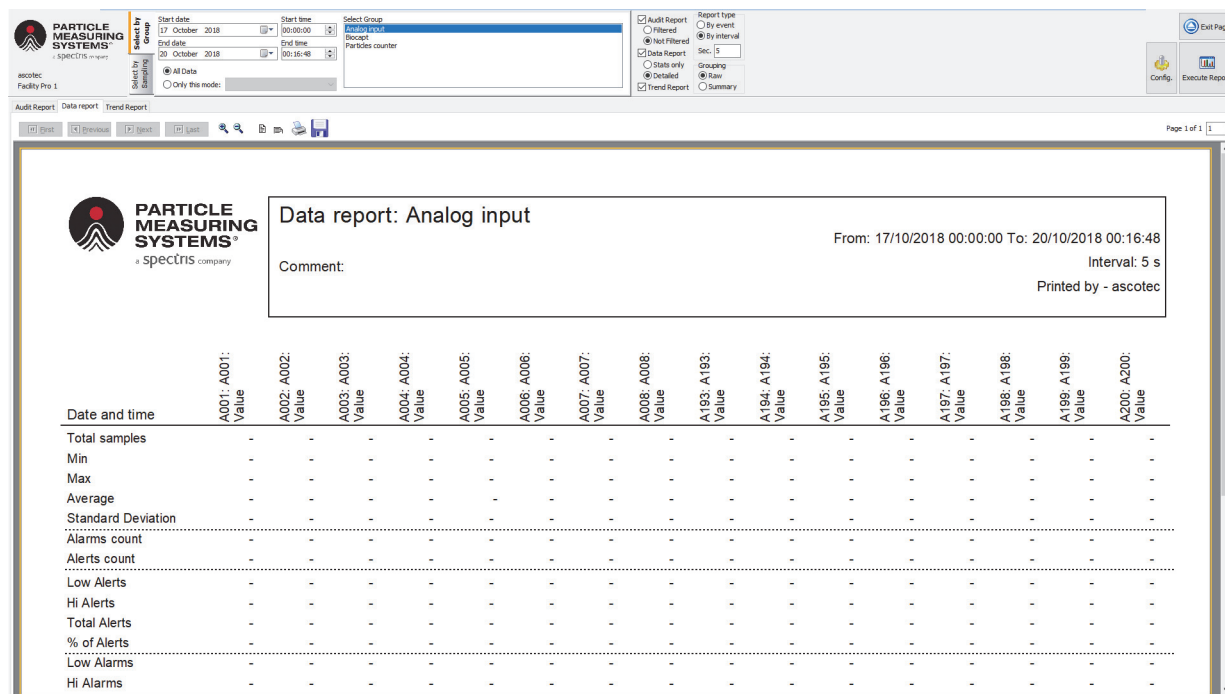


Figure 8-27 Example **Data Report – Select by Group** displayed in the **Report** window

Header information – Data Report – Select by Group

The header at the top of each page of a report contains:

- Company logo
- Report name
- Report range – The date/time span covered in the report
- Data interval
- Comment text included when the report was created
- Printed by – name of the user logged in for printing

Report body – Data Report – Select by Group

- Column headings:
 - **Date and Time** – The event's date and time

- Measurements – Values of the samples, and with an enhanced font, indicates if the value exceeded the threshold of alert or alarm.
For PC reports, for each sampler, column headings include:

- **S0n: Flow(l/min)** – The number of liters sampled per minute, where **n** is the number of the sampler
- **S0n – Part>0.5um/CF**
- **S0n – Part>0.5um/CM**
- **S0n – Part>5.0um/CF**
- **S0n – Part>5.0um/CM**
- **S0n – Volume(lt)**

Report summary – Data Report – Select by Group

Appears on the last page of the report.

- **Total samples** – total number of samples in the column;
- **Min** – the minimum value reported in the column;
- **Max** – the maximum value reported in the column;
- **Average** – The average value of samples in the column;
- **Standard deviation** – With reference to the average value of samples in the column
- **Low Alerts** – The number of low alerts;
- **Hi Alerts** – The number of high alerts;
- **% of Alerts**– The percentage of alerts to total samples;
- **Low Alarms**– The number of low alarms;
- **Hi Alarms**– The number of high alarms;
- **% of Alarms**– The percentage of alarms to total samples.

Footer information – Data Report – Select by Group

The footer of each report contains:

- Printing date and time
- **FacilityPro <station name room name>**
- Page n of m – Number of the current page and total number of pages in the report

Last page – Data Report – Select by Group

The last page of a report includes some blank lines for the operator to fill in on the printed report:

Performed by _____ Date _____ Reviewed and Approved by _____ Date _____

Data Report examples – Select by Samplings

The **Data Report** creates a table of sampling data from the sensors and calculated values for minimums, maximums, averages, and standard deviations.

An example of an **Data Report – Select by Samplings** displayed in the **Report** window is shown in **Figure 8-27**.

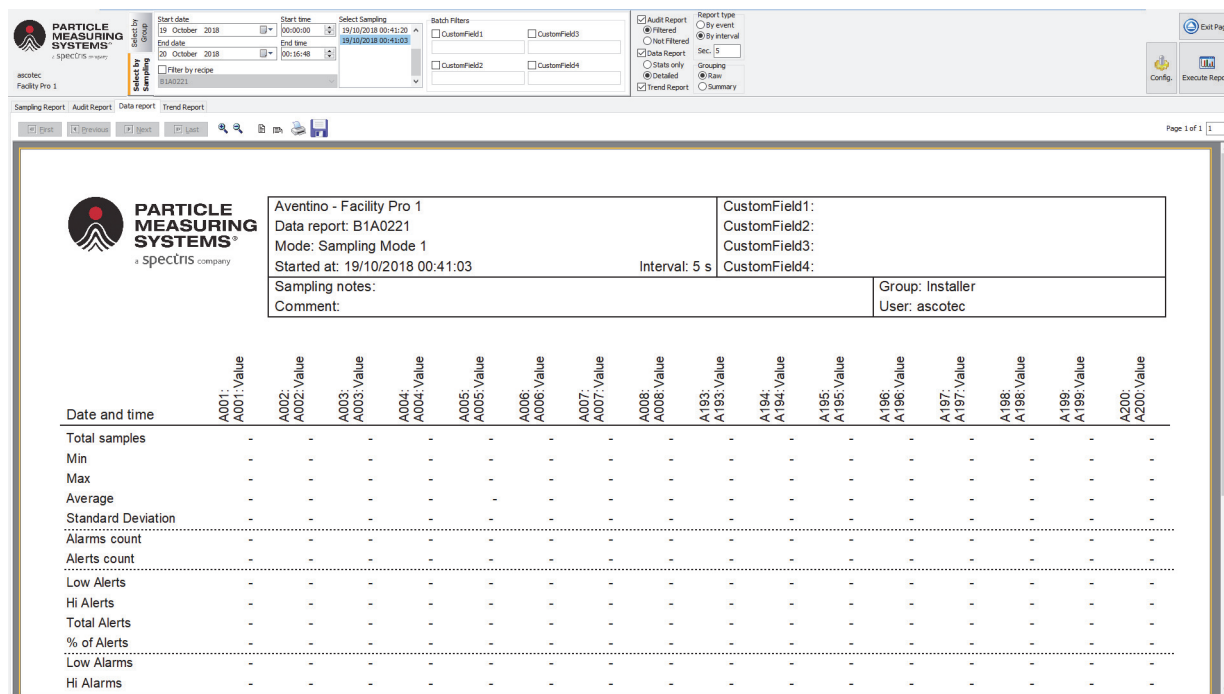


Figure 8-28 Example **Data Report – Select by Samplings** displayed in the **Report** window

Header information – Data Report – Select by Samplings

The header at the top of each page of a report contains:

- Company logo
- FacilityPro <station name room name>Report name
- Recipe name - The name of the recipe selected for the report.
- Time the recipe started running - **Started at mm/dd/yyyy or dd/mm/yyyy hh:mm:ss AM or PM**
- Mode - The mode of the sampling selected for the report:
 - **Normal Operation**
 - **At Rest**
 - **Custom**
- Data Interval
- Custom field 1, if any specified
- Custom field 2, if any specified
- Custom field 3, if any specified
- Custom field 4, if any specified

- Sampling notes - User defined
- Comment text included when the report was created
- Group - name of the login group the logged-in user belongs to
- User - name of the user logged in when the report was generated

Report body – Data Report – Select by Samplings

- Column headings:
 - **Date and Time** – The event's date and time
 - **Measurements** – Values of the samples, and with an enhanced font, indicates if the value exceeded the threshold of alert or alarm.
For PC reports, for each sampler, column headings include:
 - **S0n: Flow(l/min)** – The number of liters sampled per minute, where **n** is the number of the sampler
 - **S0n – Part>0.5um/CF**
 - **S0n – Part>0.5um/CM**
 - **S0n – Part>5.0um/CF**
 - **S0n – Part>5.0um/CM**
 - **S0n – Volume(lt)**

Report summary – Data Report – Select by Samplings

Appears on the last page of the report.

- **Total samples** – total number of samples in the column;
- **Min** – the minimum value reported in the column;
- **Max** – the maximum value reported in the column;
- **Average** – The average value of samples in the column;
- **Standard deviation** – With reference to the average value of samples in the column
- **Low Alerts** – The number of low alerts;
- **Hi Alerts** – The number of high alerts;
- **% of Alerts**– The percentage of alerts to total samples;
- **Low Alarms**– The number of low alarms;
- **Hi Alarms**– The number of high alarms;
- **% of Alarms**– The percentage of alarms to total samples.

Footer information – Data Report – Select by Samplings

The footer of each report contains:

- Printing date and time
- **FacilityPro <station name room name>**
- Page n of m – Number of the current page and total number of pages in the report

Last page – Data Report – Select by Samplings

The last page of a report includes some blank lines for the operator to fill in on the printed report:

Performed by _____ Date _____ Reviewed and Approved by _____ Date _____

Trend Report examples – Select by Samplings

The **Trend Report** creates a Historical trend graph of sampling data from the sensors. An example of a **Trend Report** displayed in the **Report** window is shown in **Figure 8-29**.

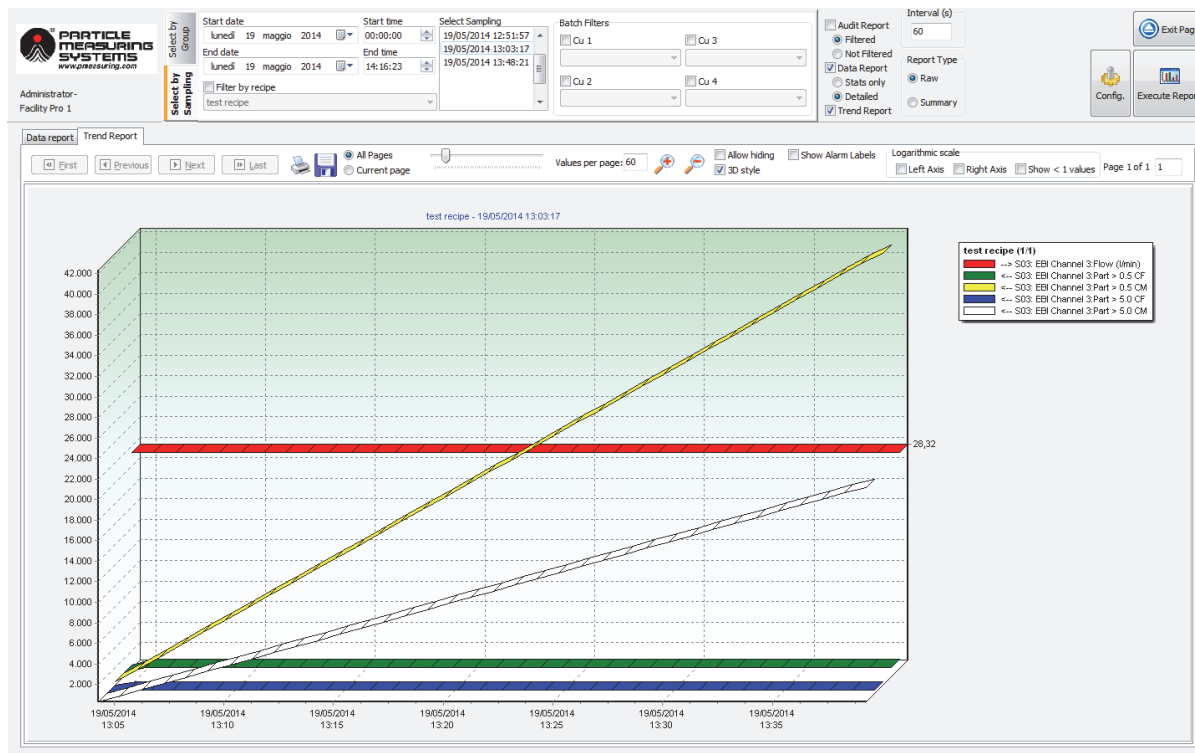


Figure 8-29 Example **Trend Report** displayed in the **Report** window

In the graphical display of the **Trend Report** you can choose:

- 3D/2D mode
- Display point value
- Zoom the time axis (number of samples per page for trend)
- Enable or disable hiding of filterable data
- Show or hide alarm labels
- Set a logarithmic scale on left or right axis

The general viewing controls and controls specific to the **Trend Report** are described in **Table 8-5** General and report-specific controls in the Report window on page 8-35.

Header information – Trend Report – Select by Samplings

The header at the top of each page of a report contains:

- Company logo
- **FacilityPro** <station name room name>
- Recipe name –

- Time the recipe started running – **Started at mm/dd/yyyy or dd/mm/yyyy hh:mm:ss AM or PM**
- Custom field 1, if any specified
- Custom field 2, if any specified
- Custom field 3, if any specified
- Custom field 4, if any specified
- Comment text included when the report was created
- Sampling notes, if inserted at the start of the sampling
- Group – name of the login group the logged-in user belongs to
- User – name of the user logged in when the report was generated

Header information - Trend Report Exported PDF - Select by Group

The header at the top of each page of a report contains:

- Company logo
- Report name
- Report range - The date/time span covered in the report
- Data interval
- Comment text included when the report was created
- Printed by - name of the user logged in for printing

Footer information - Trend Report Exported PDF

The footer of each report contains:

- Printing date and time
- **FacilityPro <station name room name>**
- Page n of m – Number of the current page and total number of pages in the report

Last page - Trend Report

The last page of a report includes some blank lines for the operator to fill in on the printed report:

Performed by _____ Date _____ Reviewed and Approved by _____ Date _____

Sampling Report

The **Sampling Report** is used to provide reports for each recipe that has been started in a required time period. Generating a **Sampling Report** retrieves data from the database and arranges the following information in the report:

- Recipe name
- Start date/Start time for each sensor of the recipe
- End date/End time
- Errors and alarms found during the sampling

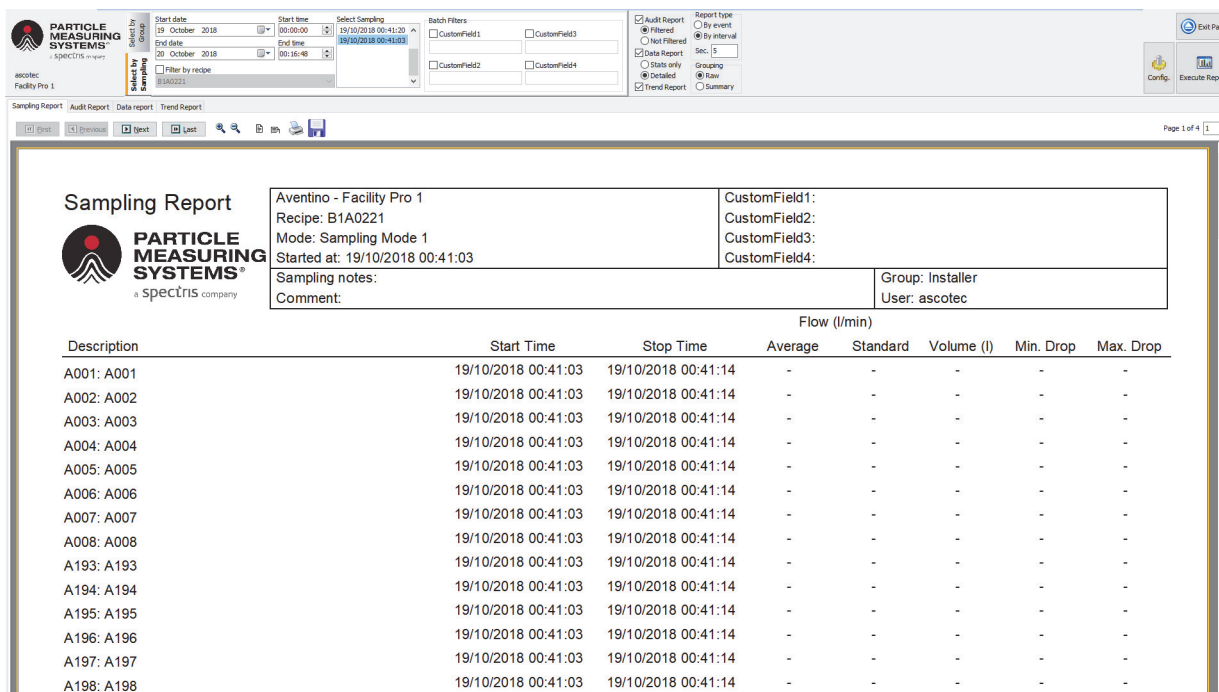
The objective of this function is to represent significant historical data present in the database for the sampling to selected recipe events (alarms, start / stop point) of the system (Audit) recorded in the DB events and alarms for the selected sampling recipe.

WARNING

It is strongly recommended to perform Database maintenance in order to keep dimension under 5GB. Bigger database size can lead to Report extraction slowdown.

Sampling Report tab

A sampling report will be generated in the **Sampling Report** tab. Configurable settings are found in the top menu.



Sampling Report

Particulate - Facility Pro 1
Recipe: B1A0221
Mode: Sampling Mode 1
Started at: 19/10/2018 00:41:03

CustomField1:
CustomField2:
CustomField3:
CustomField4:

Sampling notes:
Comment:

Group: Installer
User: ascotec

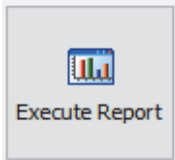
Description	Start Time	Stop Time	Flow (l/min)				
			Average	Standard	Volume (l)	Min. Drop	Max. Drop
A001: A001	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A002: A002	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A003: A003	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A004: A004	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A005: A005	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A006: A006	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A007: A007	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A008: A008	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A193: A193	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A194: A194	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A195: A195	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A196: A196	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A197: A197	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-
A198: A198	19/10/2018 00:41:03	19/10/2018 00:41:14	-	-	-	-	-

Figure 8-30 Sampling Report window

Create a Sampling Report

Creating any sampling report involves the following general steps:

- 1 Set a date/time range for the report
See Set a date and time range for a report on page 8-30.
- 2 From the **Select Sampling** field, select the sampling you want to use in the report.
- 3 Apply filters, if necessary.

- 4 Click  the **Execute Report** button to type note text you want to

include in the header of the report and generate the report.

See Execute the report on page 8-27.

The system automatically generates the **Sampling Report** and displays it in the **Sampling Report** tab.

After a report has been generated and displayed in the **Sampling Report** tab, you can:

- Review the report on screen.
See Change the view in report displayed in the Sampling Report window on page 8-49.
- Print the report.
See Print a Sampling Report on page 8-49.
- Save a copy of the report as a **PDF** file.
See Export a Sampling Report from the Sampling Report Window on page 8-50.

Change the view in report displayed in the Sampling Report window

The controls for the **Sampling Report** display are the same as the general display controls described in **Table 8-5** General and report-specific controls in the Report window on page 8-35.

Print a Sampling Report

>> To print a Sampling Report:

- 1 Generate the report (see Create a Sampling Report on page 8-49).
The report displays in the text box of the **Sampling Report** tab.
- 2 Click  the **Printer** icon in the toolbar at the top of the report in the **Sampling Report** window.
A **Print** window appears.
- 3 Select which pages to print.

Export a Sampling Report from the Sampling Report Window

A **Sampling Report** generated through the **Sampling Report** tab can be saved as a PDF – Portable Document Format – for viewing in Adobe Acrobat

>> To export a Sampling Report from the Sampling Report tab:

1 Generate the report.

The report displays in the text box of the **Sampling Report** tab.

2 Click the **Save report** icon in the toolbar at the top of the report in the **Sampling Report** window.

Pharmaceutical Net Pro exports a PDF of the report to a folder in the general path **c:\Export \PDF**.

- The full path displays in a message with the full name of the exported report file.

Sampling Report components

An example of a **Sampling Report** displayed in the **Sampling Report** tab is shown in **Figure 8-30** on page 8-48.

Header information – Sampling Report

The header at the top of the first page of a **Sampling Report** contains:

- Report name – **Sampling Report**
- Company logo
- **FacilityPro <station name room name>**
- Recipe name –
- Time the recipe started running – **Started at mm/dd/yyyy or dd/mm/yyyy hh:mm:ss AM or PM**
- Mode – The mode of the sampling selected for the report:
 - **Normal Operation**
 - **At Rest**
 - **Custom**
- Batch Number – User defined
- Process type:
 - **Custom**
- Sampling notes – User defined
- Comment text included when the report was created
- Group – name of the login group the logged-in user belongs to
- User – name of the user logged in when the report was generated

Report body – First section – Sampling Report

- Column headings:
 - **Channel** – The name of the channel described
 - **Description** – A description defined for the channel
 - **Start Time** – The start time for the sampling
 - **Stop Time** – The end time for the sampling
 - **Flow (L/min)** – Heading for the following columns:
 - **Average** – Average flow in liters/minute in the format **nn.nn**
 - **Std. Dev.** – Standard deviation for the flow in the format **nn.nn**
 - **Volume (l)** – Number of liters measured in the format **nn.nn**
 - **Min. Drop** – Minimum line pressure drop in the format **nn.nn**
 - **Max. Drop** – Maximum line pressure drop in the format **nn.nn**

Report body – Second section – Sampling Alarms – Sampling Report

- Column headings:
 - **Alarm date and time** – The date and time the alarm occurred
 - **Channel** – The name of the channel described
 - **Description** – A description defined for the channel
 - **Alarm** – Device ID: Channel name and qualifier **Alert** or **Alarm**
 - **State** – Description of the status of the event (current value that generated the alarm)

Footer information – Sampling Report

The footer of each report contains:

- Print time – PDF creation date and time
- Printed by – Name of the user logged in
- Page n of m – Number of the current page and total number of pages in the report

Last page

The last page of a report includes some blank lines for the operator to fill in on the printed report:

Performed by _____ Date _____ Reviewed and Approved by _____ Date _____

This page is intentionally left blank.

Chapter 9

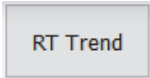
Real Time Trend

The **Real Time Trend** window allows the user to view a continuous stream of data provided by one of the available recipe sampling groups and its associated sampling channels.

Data is saved when a change event occurs (when the value of the data changes) to save memory space. In the event maximum assets are used (all 16 pens of the trend graph changing once per second), the maximum duration of a trend report is 2 days, 23 hours and 59 minutes.

Real Time Trend Window

>> To open the Real Time Trend window:

- 1 Click  the **Real Time Trend** button at the bottom of the Pharmaceutical Net Pro screen.
The **Real Time Trend** window appears (see **Figure 9-1**).

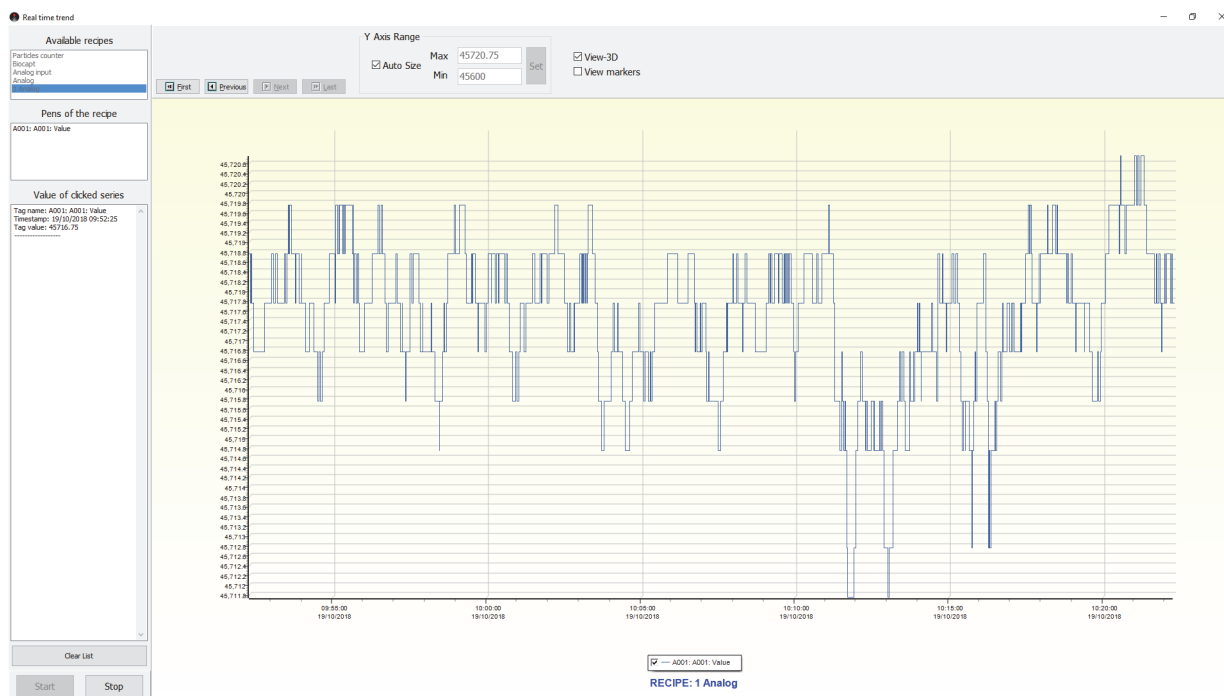


Figure 9-1 Real Time Trend window

Table 9-1 contains all components displayed on the **RT Trend** window and their usage.

Table 9-1 Components of the Real Time Trend window

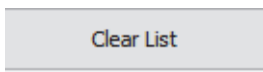
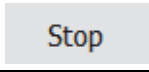
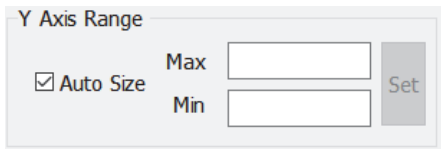
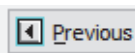
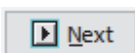
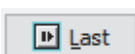
Component	Description
Available recipes	Lists all recipe groups configured with the processor.
Pens of the recipe	Lists all channels in the recipe group selected in Available recipes .
Value of clicked series	Lists the current values of the individual channels of the recipe group, including Tag name , Timestamp and Tag value .
 Clear List	Clears Value of clicked series field.
 Start	Starts real time trending.
 Stop	Stops real time trending.
Debug Information	This area contains tag information pertaining to debugging the tool. Here, the First and Last timestamp in series can be viewed.

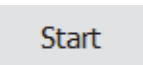
Table 9-1 Components of the **Real Time Trend** window

Component	Description
Y Axis Range	Enter a Max and Min value to limit the vertical view of data plots to desired limits. To enable the Max and Min range, remove the checkmark from Auto Size, enter the desired range and click Set. To automatically adjust the view based on the highest and lowest values, checkmark Auto Size. 
View-3D	Checkmark View-3D to change the real time view from the default 2D to 3D.
View markers	Checkmark View markers to attach markers listed in the legend to the real time trend lines in addition to their associated values.
Legend area	Lists the markers of each channel including the label and line color. To hide a channel from view, click the adjacent checkmark box so that it is blank.
 First	Navigates timeplot to first data point in series.
 Previous	Navigates timeplot to previous data point in series.
 Next	Navigates timeplot to next data point in series.
 Last	Navigates timeplot to last data point in series.

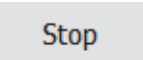
Starting and Stopping a Real Time Trend

The **Real Time Trend** window displays overtop the main screen, allowing the user to minimize it to use other functions of the software while it is running, and also open another **Real Time Trend** window. No more than four **Real Time Trend** windows can be open at once.

>> To begin a trend:

- 1 Click the desired recipe group listed in **Available recipes**.
Pens of the recipe automatically populates once a recipe group is chosen.
- 2 Click the **Start**  button. Real time trending begins.

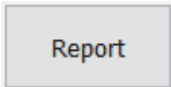
>> To end a trend:

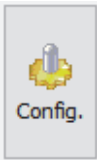
- 1 Click the **Stop**  button.
- 2 Perform the **Electronic Signature**. This does not pause the display of data. Trending cannot be started again at the last timestamp in the data series, but will begin again at 00:00:00. Current trend data will be lost.

Editing Pens of the Recipe

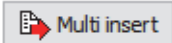
Real time trending is only allowed for a recipe with a lower limit of 1 pen and an upper limit of 16 pens.

>> To edit the number of pens:

- 1 Click the **Reports**  button.

- 2 Click the **Config.**  button to open the **Tags Catalog**.

- 3 Adjustment of the listing of pens in the **Real Time Trend** window is done using the **Sensor Groups** tab and **Data Type Groups** tab. Use the **Multi insert**

 button to edit each listing.

The number of pens is equal to the number of sensors listed in the **Sensors in Group** column (**Sensor Groups** tab) multiplied by the number of **Data Types in Group** entries (**Data Type Groups** tab) for the same, applicable **Sensors group/Data Type Group**.

For example, if a **Sensors Group** includes both particle and analog sensors, certain **Data Types** will only apply to certain sensors (i.e., Analog Value is only applicable to an analog sensor).

- 4 Colors of the individual sensor trend lines can be adjusted with the **Pens Color Selection** tab.
- 5 Once adjustments have been made, exit the **Reports** window and click the **RT Trend** button. The **Pens of the recipe** listing is now adjusted in the new window (old windows are unaffected).

Chapter 10

Run Time Logic

The purpose of the Run Time Logic Editor is to provide a graphical interface for the editing of digital output activation. With this tool, you can create any logic with “Available tags” to drive physical digital output to activate lamps, sirens, or any device with a switch.

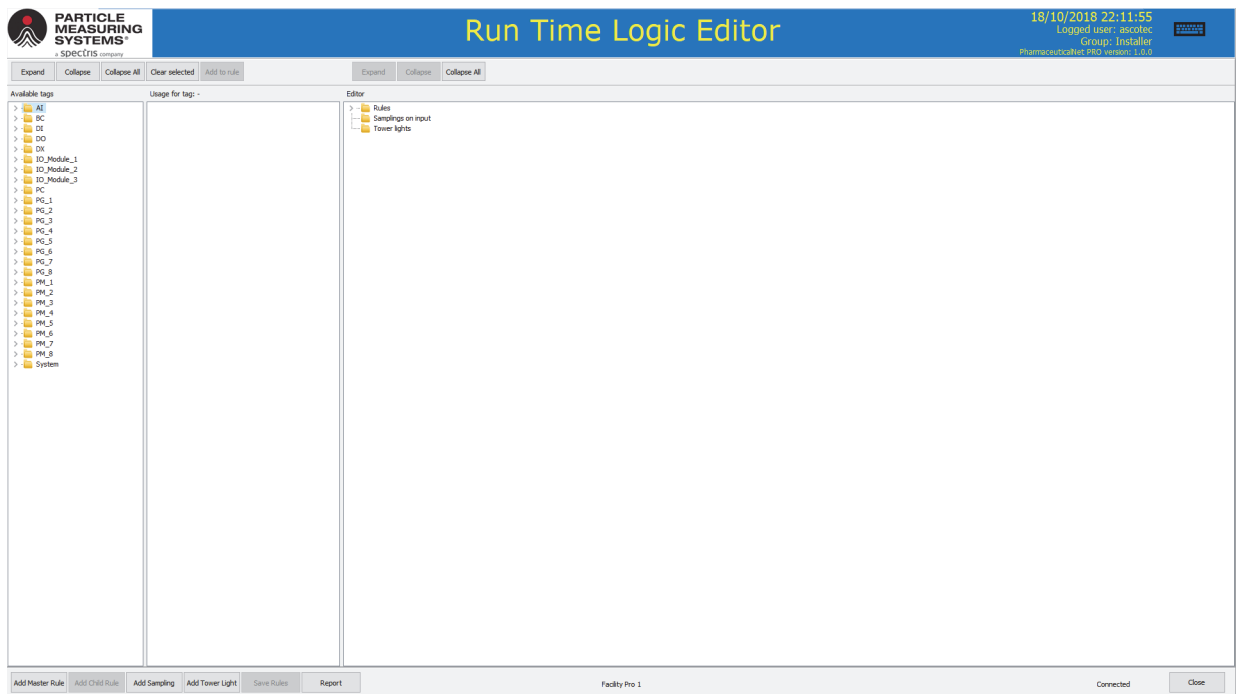


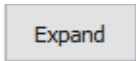
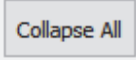
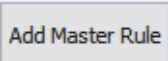
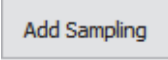
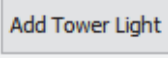
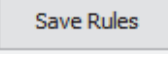
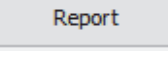
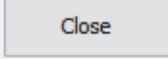
Figure 10-1 Run Time Logic Editor window

There are three panes of the Run Time Logic Editor. From left to right:

- **Available Tags pane**
- **Usage for Tag pane** on page 10-11
- **Editor pane** on page 10-11

Table 10-1 contains all buttons displayed on the Run Time Logic Editor screen and their usage.

Table 10-1 Buttons of the **Run Time Logic Editor** screen

Button	Description
 Expand	Expands the current selection. A node can be expanded if a  arrow is adjacent to the node title.
 Collapse	Collapses the current node selection. A node can be collapsed if a  arrow is adjacent to the node title.
 Collapse All	Collapses all nodes currently expanded.
 Add Master Rule	Opens the Rule Editor window to create a new rule.
 Add Child Rule	Opens the Rule Editor window to create a new child rule.
 Add Sampling	Opens the Sampling on input window to create a new sampling rule.
 Add Tower Light	Opens the TowerLightEditForm window to create a new tower light rule.
 Save Rules	Use this button to save the created rules.
 Report	Use this button to save/print a PDF copy detailing the configured rules. The display is the same as shown in the Editor pane.
 Close	Returns the user to the System window.

Available Tags pane

This section outlines the definitions and usage of each available tag contained in the Run Time Logic Editor.

Only a single usage of a unique output is allowed. Any input can be used an unlimited amount of times.

Each alarm has an alarm out. Alarm out is triggered to 1 at the same time of the alarm but changes to 0 after acknowledgement (DO_Ack = 1). For example, with the use of COM_AlarmOut instead of COM_Alarm, you can silence the siren for a sensor sent to calibration, acknowledging the communication alarm from SCADA.

Analog Input Folder

The number of analog (including virtual) inputs is dependent on system configuration.

Table 10-2 Analog Input Folder

Tags	Description	Usage
CalibAlarm	Alarm triggered from calibration expiration.	Input
CalibAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
CalibAlert	Alert triggered from calibration expiration.	Input
CalibAlertOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
COM_Alm	Alarm triggered when the sensor/pneumatic module is disconnected.	Input
COM_AlmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
DO_Ack	Used as an input to set all system alarms equal to 0 (off). Used as an output for confirmation if acknowledgement from SCADA occurs for any system alarm.	Input, Output
HiAlarm	Alert triggered when the analog value exceeds the Hi alert limit (after set delay).	Input
HiAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
HiHiAlarm	Alarm triggered when the analog value exceeds the Hi alarm Limit (after set delay).	Input
HiHiAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
Installed	Confirms if the sensor is installed.	Input
LOAlarm	Alert triggered when the analog value is lower than the low alert limit (after set delay).	Input
LOAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
LogData	Confirms when the data is logged to the database.	Input
LOLOAlarm	Alarm triggered when the analog value is lower than the low alarm limit.	Input
LOLOAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
RangeError	Alarm triggered when the analog value is 1000 samples lower than the lowest sample calibration point.	Input
RangeErrorOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input

BioCapt (BC) Folder

The number of BC folders is dependent on the number of configured biological sensors.

Table 10-3 BioCapt (BC) Folder

Tags	Description	Usage
AlreadyInUseAlarm	Alarm triggered from calibration expiration.	Input
AlreadyInUseAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
CalibAlarm	Alarm triggered from calibration expiration.	Input
CalibAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
CalibAlert	Alert triggered from calibration expiration.	Input
CalibAlertOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
COM_Alm	Alarm triggered when the sensor/pneumatic module is disconnected.	Input, Output
COM_AlmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
DelayedStart	Alarm triggered when the sensor has a Delayed Start status.	Input
DO_Ack	Used as an input to set all system alarms equal to 0 (off). Used as an output for confirmation if acknowledgement from SCADA occurs for any system alarm.	Input, Output
DropAlarm	Alarm triggered when the line pressure drop exceeds set limits during sampling.	Input
DropAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
DropHiAlarmEnabled	Alarm triggered when drop alarm high is enabled.	Input
DropLowAlarmEnabled	Alarm triggered when drop alarm low is enabled.	Input
EnableSwitch	Alarm triggered when it is not possible to use the sensor switch.	Input
FlowAlarm	Alarm triggered when the flow alarm occurs.	Input
FlowAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
FlowCalculationAlarm	Alarm triggered when the analog value is 1000 samples lower than the lowest sample calibration point.	Input
FlowCalculationAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
Installed	Confirms if the sensor is installed.	Input
LineFault	Alarm triggered when the sampling is interrupted.	Input

Table 10-3 BioCapt (BC) Folder

Tags	Description	Usage
LineFaultOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
Locked	Used as an output to disable the sensor from sampling. Used as an input for confirmation of when the sensor is locked.	Input, Output
LogData	Confirms when the data is logged in your database.	Input
SensorIsLockedAlarm	Alarm triggered when the sensor is locked.	Input
SensorIsLockedAlarmOut	Triggered to 1 at the time of the alarm but switches to 0 after acknowledgement or DO_Ack=1.	Input
SensorIsSampling	Alarm triggered when the sensor is in use.	Input
STOP_Alm	Alarm triggered when time or volume sampling is interrupted by a user.	Input
STOP_AlmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
Switch	Used as an input for confirmation of the sensor switch status. Used as an output to start sensor sampling.	Input, Output

Digital Input (DI) Folder

Digital Input (DI), numbered from 1 to as many as 64 (dependent on system configuration), refers to physical I/O boards installed. Each I/O board has 8 digital inputs. The I/O Board 1 has input from 1 to 8, I/O Board 2 has input from 9 to 16 and so on.

Table 10-4 Digital Input (DI) Folder

Tags	Description	Usage
Di0XX (01-XX)	Digital input from the I/O Boards	Input

Digital Output (DO) Folder

Digital Output (DO) refers to the number of physical I/O boards installed. Each I/O board has 16 digital outputs.

Table 10-5 Digital Output (DO) Folder

Tags	Description	Usage
Do0XX (1-32)	Digital output from the I/O Boards.	Input, Output

Virtual Digital Input/Output (DX) Folder

DX can be used as a virtual DI/DO provided by the OPC interface. DX can be activated with buttons in SCADA's graphical interface or in any OPC client connected to the FacilityPro processor module. Each FacilityPro has 64 DX that can be used from SCADA or as intermediate variables for other rules.

Table 10-6 Digital Output (DO) Folder

Tags	Description	Usage
Dx0XX (1-64)	Virtual digital input or output	Input, Output

I/O Module Folder

Each I/O module (displayed as IO_Module_X) has a maximum of 3 connected ADAM boards.

Table 10-7 I/O Module Folder

Tags	Description	Usage
DO_Ack	Used as an input to set all system alarms equal to 0 (off). Used as an output for confirmation if acknowledgement from SCADA occurs for any system alarm.	Input, Output
ModuleComAlarm	Alarm triggered when the I/O Board is disconnected.	Input
ModuleComAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
TestMode	Alarm triggered when installer accesses the Hardware Test page and enables the manual control of the board output.	Input

Particle Counter Folder

The Particle Counter folder contains every available tag that can be used with each connected particle counter.

Table 10-8 Particle Counter (PC) Folder

Tags	Description	Usage
AlreadyInUseAlarm	Alarm triggered when the sensor is sampling and another sampling is requested.	Input
AlreadyInUseAlarmOut	Alarm triggered when the I/O Board is disconnected.	Input
CalibAlarm	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input

Table 10-8 Particle Counter (PC) Folder

Tags	Description	Usage
CalibAlarmOut	Alarm triggered when installer accesses the Hardware Test page and enables the manual control of the board output.	Input
CalibAlert	Alert triggered from calibration expiration.	Input
CalibAlertOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
COM_Alm	Alarm triggered when the sensor/pneumatic module is disconnected.	Input
COM_AlmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement or DO_Ack=1.	Input
DelayedStart	Alarm triggered when the sensor has a Delayed Start status.	Input
DO_Ack	Used as an input to set all system alarms equal to 0 (off). Used as an output for confirmation if acknowledgement from SCADA occurs for any system alarm.	Input, Output
EnableSwitch	Alarm triggered when it is not possible to use the sensor switch.	Input
FlowAlarm	Alarm triggered when flow alarm occurs.	Input
FlowAlarmOut	Triggered to 1 at the time of the alarm.	Input
Installed	Confirms if the sensor is installed.	Input
LineFault	Alarm triggered when the sampling is interrupted.	Input
LineFaultOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
Locked	Used as an output to disable the sensor from sampling. Used as an input for confirmation of when the sensor is locked.	Input, Output
LogData	Confirms when the data is logged in your database.	Input
PneumaticModuleAlarm	Alarm triggered when the pneumatic module has an alarm.	Input
PneumaticModuleAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
Alarm <i>PS1>CF>Limit1 or PS1>CM>Limit1</i>	Alarm triggered when the cubic foot concentration of particles exceeds limit settings.	Input
AlarmOut <i>PS1>CF>Limit1 or PS1>CM>Limit1</i>	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
CF_OK <i>Navigate: PS1>CF</i>	Alarm triggered when cubic foot concentration is reached.	Input
CM_OK <i>Navigate: PS1>CM</i>	Alarm triggered when cubic meter concentration is reached.	Input
SensorIsLockedAlarm	Alarm triggered when the sensor is locked.	Input

Table 10-8 Particle Counter (PC) Folder

Tags	Description	Usage
SensorIsLockedAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
SensorIsSampling	Alarm triggered when the sensor is in use.	Input
STOP_Alm	Alarm triggered when time or volume sampling is interrupted by a user.	Input
STOP_AlmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
Switch	Used as an input for confirmation of the sensor switch status. Used as an output to start sensor sampling.	Input, Output

Pumps Group Folder

The Pumps Group folder contains every available tag that can be used with each connected pump.

Table 10-9 Pumps Group Folder

Tags	Description	Usage
Alarm1	Alarm triggers when the maximum pneumatic pressure between modules in the same pump group limit 1 is exceeded.	Input
Alarm2	Alarm triggers when the maximum pneumatic pressure between modules in the same pump group limit 2 is exceeded.	Input
HardwareTestMode	Alarm triggered when installer accesses the Hardware Test page and enables the manual control of a module in the same pump group.	Input
Pump1Test	Alarm triggered when Pump 1 is on.	Input
Pump2Test	Alarm triggered when Pump 2 is on.	Input
Pump3Test	Alarm triggered when Pump 3 is on.	Input
Pump4Test	Alarm triggered when Pump 4 is on.	Input

Pneumatic Module Folder

The number of pneumatic modules listed is dependent on system configuration. Each sensor has 8 channels (8 channel valves and 8 flow valves).

Table 10-10 Pumps Group Folder

Tags	Description	Usage
CalibAlarm	Alarm triggered from calibration expiration.	Input
CalibAlarmOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
CalibAlert	Alert triggered from calibration expiration.	Input
CalibAlertOut	Triggered to 1 at the time of the alarm, but switches to 0 after acknowledgement of DO_Ack=1.	Input
ChanValve <i>Channel_X</i>	Alarm is triggered when the channel valve is on.	Input
FlowValve <i>Channel_X</i>	Alarm triggered when the flow valve is on.	Input
COM_Alm	Alarm triggered when the sensor/pneumatic module is disconnected.	Input
DO_Ack	Used as an input to set all system alarms equal to 0 (off). Used as an output for confirmation if acknowledgement from SCADA occurs for any system alarm.	Input, Output
PressureTestFault	Alarm triggered on the initial pressure test fault of the sensor/pneumatic module.	Input
RBValve	Alarm triggered when the barometric pressure valve is on.	Input

System Folder

System tags refer to SCADA for communication and information exchange (SCADA Alarm).

Table 10-11 System Folder

Tags	Description	Usage
AICalibAlarmLimit	Status (Enabled/Disabled) of the Analog Sensor Calibration Alarm	Input
AICalibAlertLimit	Status (Enabled/Disabled) of the Analog Sensor Calibration Alarm	Input
BufferAlarm1	Alarm triggered when disconnection time between SCADA A and the FacilityPro exceed the set buffer time.	Input
BufferAlarm1Out	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input

Table 10-11 System Folder

Tags	Description	Usage
BufferAlarm2	Alarm triggered when disconnection time between SCADA B and the FacilityPro exceed the set buffer time.	Input
BufferAlarm2Out	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
ComAlarm1	FacilityPro communication alarm with Node A.	Input
ComAlarm1Out	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
ComAlarm2	FacilityPro communication alarm with Node B.	Input
ComAlarm2Out	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
DO_Ack	Used as an input to set all system alarms equal to 0 (off). Used as an output for confirmation if acknowledgement from SCADA occurs for any system alarm.	Input, Output
LifeBeat1_Alarm	SCADA A communication alarm from OPC Link.	Input
LifeBeat1_AlarmOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
LifeBeat2_Alarm	SCADA B communication alarm from OPC Link.	Input
LifeBeat2_AlarmOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
PCCalibAlarmLimitEnabled	Status (Enabled/Disabled) of the particle calibration alarm.	Input
PCCalibAlertLimitEnabled	Status (Enabled/Disabled) of the particle sensor calibration alert.	Input
PMCalibAlarmLimitEnabled	Status (Enabled/Disabled) of the pneumatic module calibration alarm.	Input
PMCalibAlertLimitEnabled	Status (Enabled/Disabled) of the pneumatic module calibration alert.	Input
Scada1Alarm	Alarm from SCADA A to FacilityPro processor.	Input
Scada1AlarmOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input
Scada2Alarm	Alarm from SCADA B to FacilityPro processor.	Input
Scada2AlarmOut	Triggered to 1 at the time of the alert, but switches to 0 after acknowledgement or DO_Ack=1.	Input

The logic categories to activate digital output or sensors include:

- **Rule Editor** on page 10-12
- **Sampling on Input Editor** on page 10-14
- **Tower Lights** on page 10-15

Usage for Tag pane

The Usage for Tag pane will display the rule associated with the selected tag.

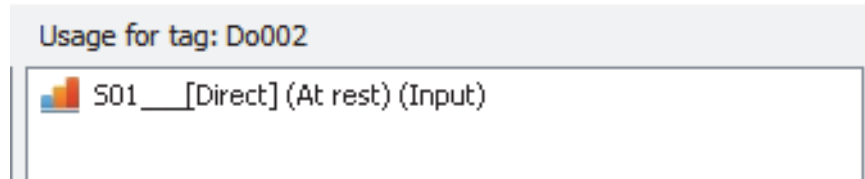


Figure 10-2 Available tag selection displays its usage

In **Figure 10-2**, the Available Tag selected is “Do002”. The usage associated with this tag is as an Input for **Sampling on input** Rule 1 (S01).

Editor pane

The Editor pane contains all rules organized by type (**Rules**, **Sampling on input**, and **Tower lights**)

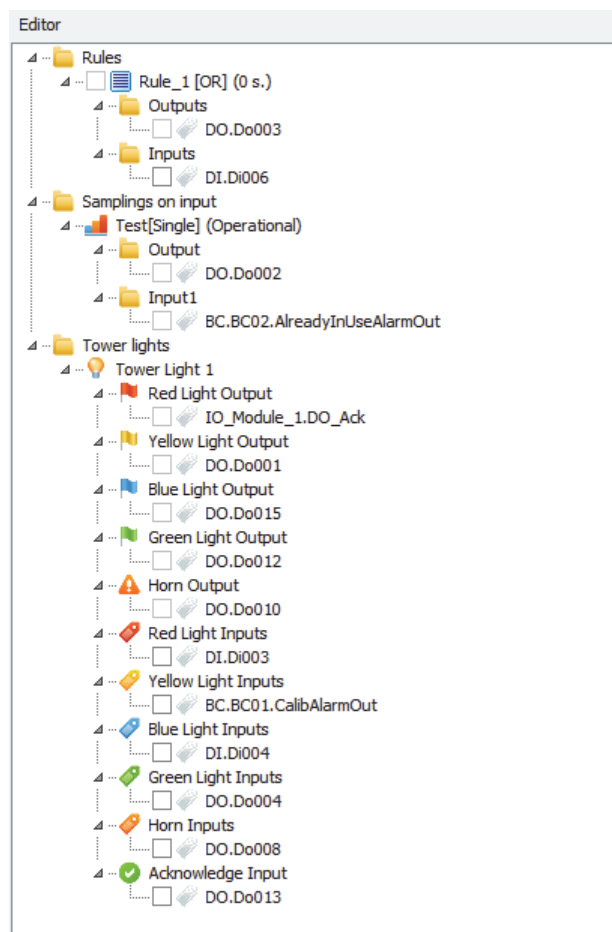


Figure 10-3 Editor pane contains all edited rules organized by type

Rule Editor

Use the Rule Editor to create logic to drive digital output (DO).

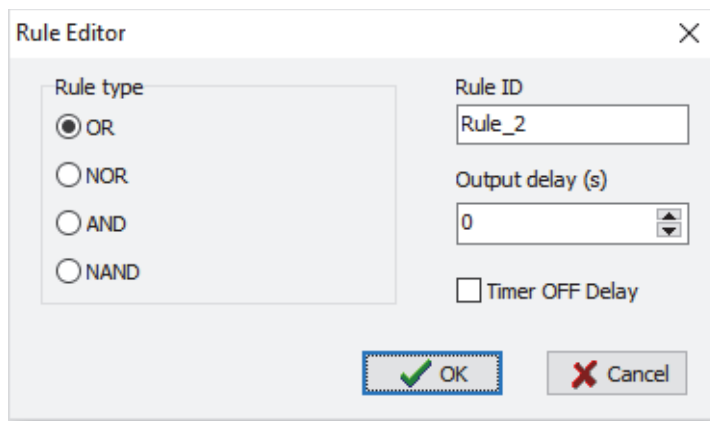


Figure 10-4 Rule Editor window

Rule Type

DO activation is a product of combinatory logic that uses the operators “and”, “or”, “not” and “delay”. The list of Rule types (see Figure 4-4) are derived from “and”, “or” and “not”:

- OR
- NOR
- AND
- NAND

The cause and effect of each logic statement is summarized in the tables below.

Table 10-12 Effect on output of turning all input on/off

		Output	
		ON	OFF
All Input	ON	NOR	OR
	OFF	AND	NAND

Table 10-13 Effect on output of turning any input on/off

		Output	
		ON	OFF
Any Input	ON	NAND	AND
	OFF	OR	NOR

The operator “delay” is used in the following way:

- When **Timer OFF Delay** is checked, the output of the rule will = 0 (become false) after the set **Output delay** time (between 0 and 3600 seconds).
- When **Timer OFF Delay** is left unchecked, the output activation (= 1) of the rule will be delayed after the set **Output delay** time.

Example Rule Editor Rule Creation

Use **Figure 10-4** and **Figure 10-5** as a guide for the following steps:

- 1 To begin, select **Add Master Rule** to open the Rule Editor window.
- 2 Assume the rule requires the output to activate only when all associated inputs are deactivated. For this assumption, under Rule type, select “OR”. Create a unique **Rule ID** to describe the rule.
- 3 Assume an output activation delay of 0 seconds. Set the **Output delay** to be 0 seconds, and leave the **Timer OFF Delay** box unchecked.
- 4 Once finished, select **OK**.
- 5 Upon rule creation, **Outputs** and **Inputs** folders will appear beneath the **Rule ID** details. Drag the desired tags from the Available Tags pane and drop them into the applicable folders. The rule in **Figure 10-5** uses one tag from the Digital Input (DI) folder, and one from the Digital Output (DO) folder: Di006 and Do003.

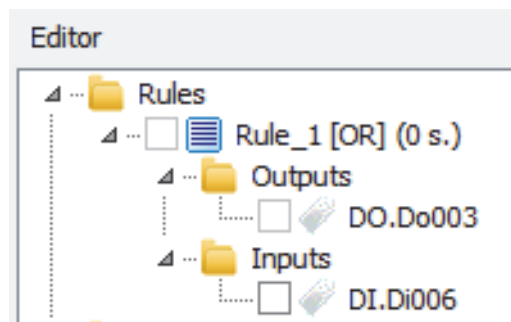


Figure 10-5 Example Rule created with Rule Editor

Creating a Child Rule

Use this procedure to create an additional logic statement within a rule. A child rule can be added to the **Inputs** folder of a rule.

- 1 After selecting the Inputs folder of the rule, select the **Add Child Rule** button.
- 2 The **Rule Editor** window will appear (see **Figure 10-6**). In this window, you will be able to set the **Rule type**, **Output delay** and **Timer OFF Delay** option.

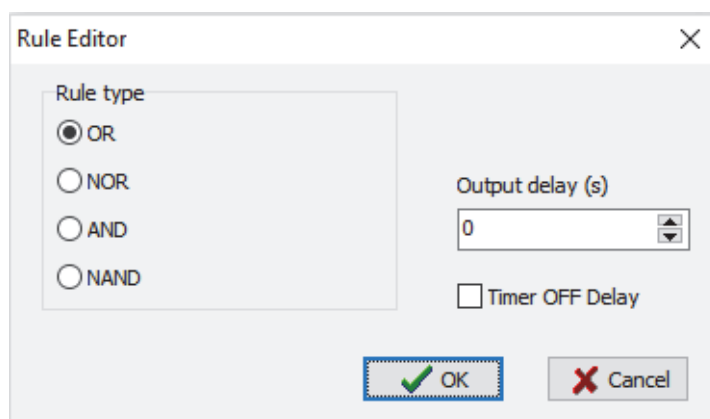


Figure 10-6 Child rule creation in **Rule Editor**

- 3** Assume that our child rule uses **Rule type** “OR”, with the output activating after an **Output delay** of 0 seconds (**Timer OFF Delay** is unchecked).
- 4** Once finished, select **OK**.
- 5** An additional **Inputs** folder is created within the original rule’s input. Both rule statements are used for Output activation/deactivation.
- 6** Save the created rule by selecting the **Save Rules** button from the bottom menu.

Sampling on Input Editor

Use the **Sampling on input editor** to start/stop a pre-configured recipe with digital input. A digital output is available as feedback to notify that the recipe is sampling.

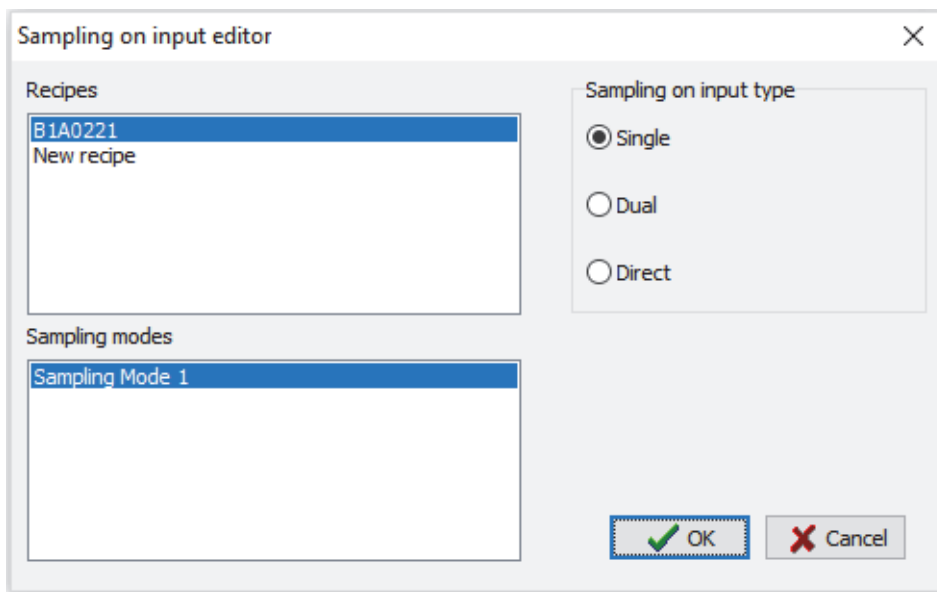


Figure 10-7 Sampling on input editor window

Sampling on input type can be selected as:

- **Single:** Allows a single digital input for the application of pushbutton sampling activation, with a second push deactivating the sampling.
- **Dual:** Allows for two separate manual inputs for the application of a pushbutton to start sampling and a second pushbutton to stop sampling.
- **Direct:** Allows the use of a switch instead of a pushbutton to drive sampling. When the switch is closed, the sampling is started. When open, the sampling is stopped.

Sampling modes define the state of operation applied to the selected recipe. Options include **At rest** and **Operational** conditions.

All associations between DI and recipes are saved together with all rules definitions in the Rules.dat file.

Recipes can be created using SCADA for implementation in this special rule (see **Chapter 4 Recipes**).

Tower Lights

Use the **TowerLightEditForm** to create rules to drive light activation with the use of input/ output tags.

Figure 10-8 TowerLightEditForm window

A Tower Light rule is a special rule that runs separately from all other rules. When a tower light rule is added, the following may be specified:

- 5 digital outputs to drive light activation
- 5 digital inputs to drive the logic
- One additional digital input (**Acknowledge Input**) for siren silencing (with a **Horn off delay** option)
- A timer for siren reactivation if the alarm should persist after silencing (**Ack. reset delay**).

Table 10-14 Tower Light Outputs and Inputs

Icon	Description
	User-specified tower light rule name. Primary node for a tower light rule.
	Red Light Output Activates when Red Light Input is one. Green and Yellow Light Output turns off when activated.
	Yellow Light Output Activates when Yellow Light Input is on. Green Light Output turns off when activated.
	Blue Light Output Activates when Blue Light Input is on. Independent operation from other lights.
	Green Light Input Activates when Green Light Input is on, indicated the associated channel has a sampling status. All other lights are off when activated.
	Horn Output Activates when Red Light Input is activated.
	Red Light Input Inputs required to be true to cause Red Light activation.
	Yellow Light Input Inputs required to be true to cause Yellow Light activation.
	Blue Light Input Inputs required to be true to cause Blue Light activation.
	Green Light Input Inputs required to be true to cause Green Light activation.
	Horn Input Inputs required to be true to cause Horn activation (in addition to Red Light Inputs).
	Acknowledge Input Silences the horn (deactivates Horn Output). Use the Ack. reset delay to set the timer in seconds (between 0 and 3600) before horn reactivation if the alarm persists after silencing.

The **Horn off delay** is the time in seconds (between 0 and 3600) before the **Horn Output** becomes false (=0) after the horn is activated.

The **Ack. reset delay** is the amount of lapsed time before the horn is reactivated after Acknowledgement Input is activated. The variable can be set to be between 0 to 3600 seconds.

Example Tower Light Rule Creation

Use Figure 4-10, Figure 4-11 and Figure 4-12 as a guide for the following steps:

- 1** To begin, select **Add Tower Light** to open the TowerLightEditForm window.
- 2** Create a unique **Tower Light ID** for the rule, or use the default “Tower Light X” format.
- 3** In the **Horn off delay** field, specify the number of seconds before the horn is silenced.
- 4** In the **Ack. reset delay** field, specify the number of seconds before the horn will reactivate after Acknowledgement Input is activated.
- 5** Once finished, select **OK**.
- 6** A listing of Tower Light Outputs and Inputs will appear. **Figure 10-9** displays an example Tower Light rule.

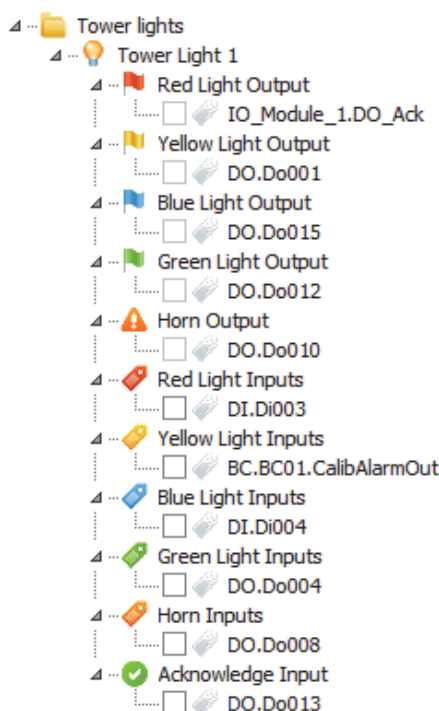


Figure 10-9 Example Tower Light rule

Testing Output Activation

Use the Run Time Logic Editor to manually activate inputs in order to test output activation for specific rules.

Checkmark  the desired inputs in order to test output activation within a specific rule. When an output is activated, its adjacent box will turn green  , in addition to the box adjacent to the rule's description. A white box  indicates inactivation.

Chapter 11

User Management

Default Users

After protocol execution, the FacilityPro Environmental Monitoring System will have the users listed in **Table 11-1**.

Table 11-1 Default users

Username	Password	Group
INSTALLER	-	Installer
NOBODY	-	-

CAUTION

DO NOT delete or change the following users in the Pharmaceutical Net Pro User Manager:

“Nobody” user

The user **Nobody** is logged in when all others are logged out. The user **Nobody** has access to the following FacilityPro windows:

- View Main Page

User Groups

There are five default user groups:

- User (Operator)
- Supervisor
- Poweruser
- Administrator
- Emergency

Group functions

The functions of the default user groups are itemized in **Table 11-2**.

Table 11-2 Default User Groups' functions

Function	OPERATORS Group	POWERUSERS Group	SUPERVISORS Group	ADMINISTRATORS Group
Main Page	View	View	View	View
Recipe	View	View	View Create Modify Delete	View Create Modify Delete
Sampling	View Create Abort	View Create Abort	View Create Edit Delete Abort	View Create Edit Delete Abort Edit Custom Fields
All Sensors Values	View	View	View	View
Sampling Channels	—	—	Switch ON/OFF	Switch ON/OFF
Alarm	View	View Acknowledge	View Acknowledge	View Acknowledge
SamplingReport	View Print Export	View Print Export	View Print Export	View Print Export
Report Generator	View Print Export	View Print Export	View Print Export	View Print Export Configure
System	—	—	—	View
Sensor Editor	View	View	View	View Configure
Limits Editor	View	View	View	View Configure
User Manager	—	—	—	View Create Modify Delete
Backup & Restore	—	—	—	View Edit
Close (to Operating System)	—	—	—	View
Sampling channel	—	—	Switch ON/OFF	Switch ON/OFF
Change password	Modify	Modify	Modify	Modify

Emergency User

An Emergency user can be created for disaster recovery access. This user is saved locally and can be used even in case of domain configuration. Emergency group functions are identical to Administrator group functions, except no additional Emergency users can be created.

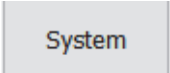
Managing Users

Management of users includes:

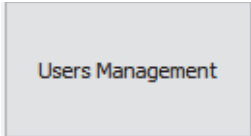
- Create a New User on page 11-5
- Modify a User definition on page 11-6
- Delete a User on page 11-6

User Management window

>> To open the User Management window:

- 1 Click  the **System** button at the bottom of the Pharmaceutical Net Pro screen.

The **System management** window appears (see **Figure 12-1** on page 12-2).

- 2 Click  the **Users Management** button in the **System management** window.

The **Users Management** window appears (see **Figure 11-1**).

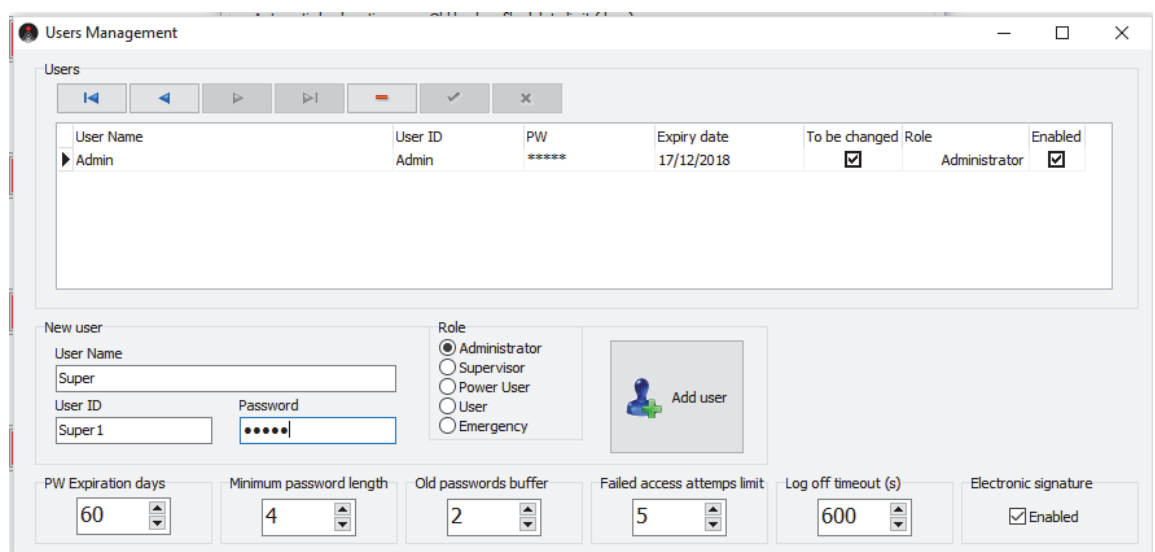


Figure 11-1 Pharmaceutical Net Pro **Users Management** window

The components in the **Users Management** window are described in **Table 11-1** on page 11-4.

Table 11-1 User Management window components

Component	Description
Users list	A list of the current configured users, showing for each: • UserName • User ID • Role • Administrator • Supervisor • Power user • User • Expiry date • To be changed field
New user section	Section for creating new users. It is possible to define UserName, user ID, password and role.
User security options section	Section for defining user security options, either for new or existing users. It is possible to set: • number of days for the password to expire • minimum password length • number of old password that the system remembers and that is not possible to set again • number of seconds until the auto-log off • enable or disable electronic signature.

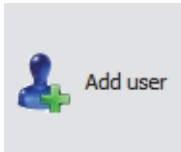
NOTE: When resetting your password after expiration, it will not be possible to use any of the five previous passwords associated with that user.

NOTE: Administrators are able to reset the password of any user. When a user changes their password, the operation will be traced in the audit report.

Create a New User

>> **To create a new user:**

- 1 Open the **Users Management** window as described on page 11-3.
- 2 Fill in all the information about the new user, including **User ID**, **User Name** and **Password**.
- 3 Define security policies including password expiration, number of passwords and so forth.

- 4 Click  the **Add user** button.

The **Electronic Signature** window appears.

- 5 Perform the electronic signature (see Electronic Signature on page 1-5) to save the user definition.
- 6 To close the **Users Management** window, click  the **X** button at the top right corner of the window.

Modify a User definition

>> To modify a user's definition:

- 1 Open the **Users Management** window as described on page 11-3.
- 2 In the **Users** list near the top of the **Users Management** window, select the user to modify.
- 3 Modify the user's settings.
- 4 Click  the green checkmark button.
The **Electronic Signature** window appears.
- 5 Perform the electronic signature (see Electronic Signature on page 1-5) to save the user definition.
- 6 To close the **Users Management** window, click  the **X** button at the top right corner of the window.

Delete a User

>> To delete a user:

- 1 Open the **Users Management** window as described on page 11-3.
- 2 In the **Users** list near the top of the **Users Management** window, select the user to delete.
- 3 Click  the **Delete** button at the top of the **Users Management** window.
The **Electronic Signature** window appears.
- 4 Perform the electronic signature (see Electronic Signature on page 1-5) to save the changes to the user definition.
- 5 To close the **Users Management** window, click  the **X** button at the top right corner of the window.

Chapter 12

System Tasks

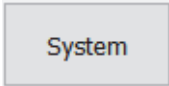
System tasks in Pharmaceutical Net Pro include the following:

- Access the user management interface (see **Chapter 11** User Management, Managing Users on page 11-3)
- Edit sensor configuration (see Sensors Editor on page 12-5)
- Backup and restore (see Backup and Restore on page 12-8)
- Shutdown and restart the system (see page 12-10)
- Editing of digital output activation (see **Chapter 10** Run Time Logic)
- Edit particle and analog limits (see Limits Editor on page 12-13 and Analog Inputs Limits on page 12-20)
- Reset the buffer (see Reset the Buffer on page 12-12)
- Translating entire system in other languages (see Change the Display Language on page 12-11)

System management Window

Many system task windows are accessible through the **System management** window.

>> To open the System management window:

- 1 Click  the **System** button at the bottom of the FacilityPro screen.

The **System management** window appears (see **Figure 12-1**).

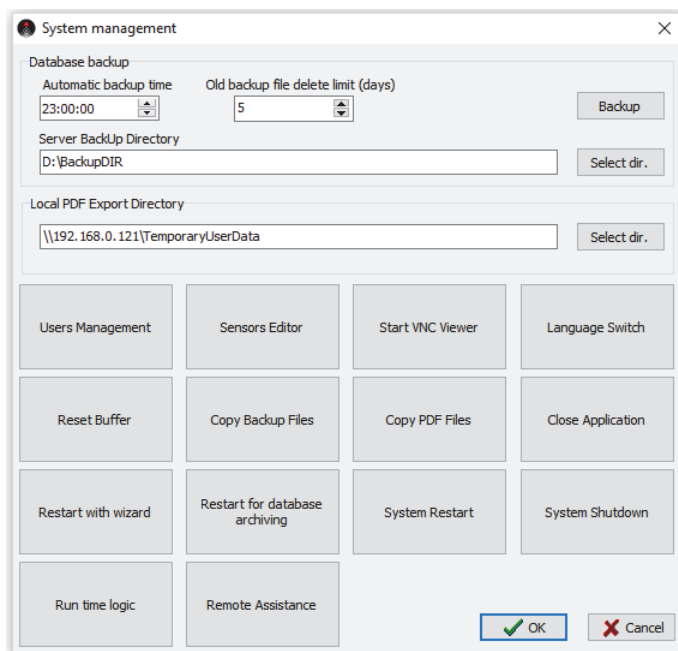


Figure 12-1 FacilityPro **System management** window

The buttons in the **System management** window are described in **Table 12-1**.

Table 12-1 **System management** window components

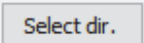
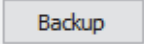
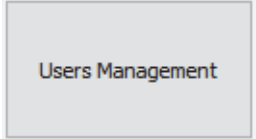
Component	Description
Automatic backup time	Defines the hour, minute, and seconds for initiating an automatic backup of data. Use the up and down arrows to set the time for the backup.
Old backup file delete limit (days)	Defines the number of days before backup files are automatically deleted from the Server BackUp Directory .
Server BackUp Directory	Type the desired backup filepath into this field or click the adjacent Select dir. button.
Local PDF Export Directory	Type the desired PDF export filepath into this field or click the adjacent Select dir. button.
 Select dir. button	Click this button to open Windows Explorer to choose the destination folder for the backup.
 Backup button	Click this button to manually perform a database backup (see Manual backup on page 12-9).
 Users Management button	An Administrator can click this button to access the user management interface (described in Chapter 11 User Management).

Table 12-1 System management window components (Continued)

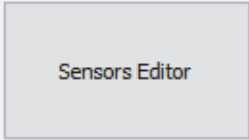
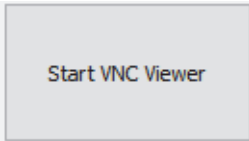
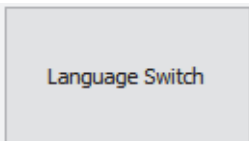
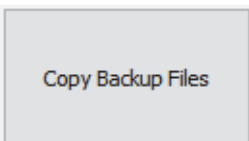
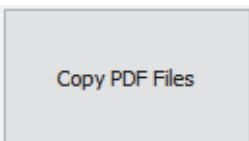
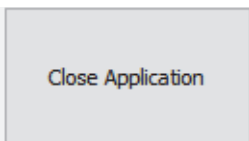
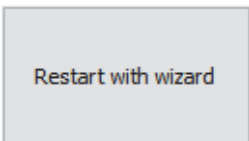
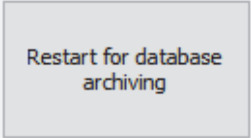
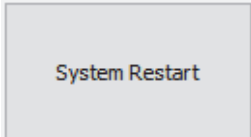
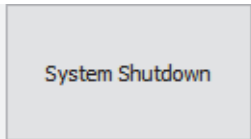
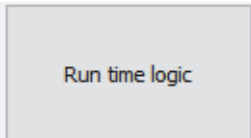
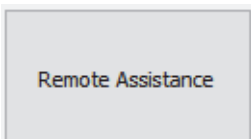
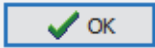
Component	Description
 Sensors Editor button	Click this button to edit sensor configuration (see Sensors Editor on page 12-5).
 Start VNC Viewer button	Click this button to start VNC Viewer connection with the Processor Module.
 Language Switch button	An Administrator can click this button to access the Translate window where the language for the displays can be selected (see Change the Display Language on page 12-11)
 Reset Buffer button	Click this button during the Debuffering to reset the data buffer (see Reset the Buffer on page 12-12).
 Copy Backup Files button	An Administrator can click this button to copy the backup files to an external hard drive.
 Copy PDF Files button	An Administrator can click this button to copy the PDF files to an external hard drive.
 Close Application button	Click this button to close the Pharmaceutical Net Pro software application.
 Restart with wizard button	Click this button to restart the machine and open the Aventino Setup Wizard (see Chapter 13 Aventino Setup Options).

Table 12-1 System management window components (Continued)

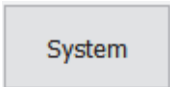
Component	Description
 Restart for database archiving	Click this button to restart the machine, archive the current database and open a new database initialized with the latest data and alarms. Once the machine restarts and the archiving option is available, click Archive . When the database reaches a preconfigured size, an alarm message will display notifying the user that it is time to archive.
 System Restart button	Click this button to restart the system (see Restart the System on page 12-9).
 System Shutdown button	Click this button to shut down the system (see Shutdown the System on page 12-10).
 Run time logic button	Opens the Run Time Logic Editor screen. See Chapter 10 Run Time Logic for more details.
 Remote Assistance button	If connected to the internet, clicking this button opens TeamViewer with the ID and password displayed for usage by outside support. If not connected to the internet, clicking this button opens Windows Explorer to the C:\WINDOWS\system32 folder.
 OK button	Click to accept changes and close the System management window.
 Cancel button	Click to close the System management window.

Sensors Editor

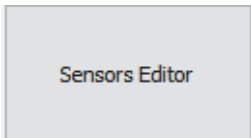
The **Sensors Editor** window is used to view (Operator users) and change (Installer user only) the descriptions of individual sampling points that populate labels for data entry fields and display-only fields in other windows of the Pharmaceutical Net Pro software application.

Sensors Editor window

>> To open the Sensors Editor window:

1 Click  the **System** button at the bottom of the Pharmaceutical Net Pro screen.

The **System management** window appears (see **Figure 12-1** on page 12-2).

2 Click  the **Sensors Editor** button in the **System management** window.

The **Sensors Editor** window appears (see **Figure 12-2**).

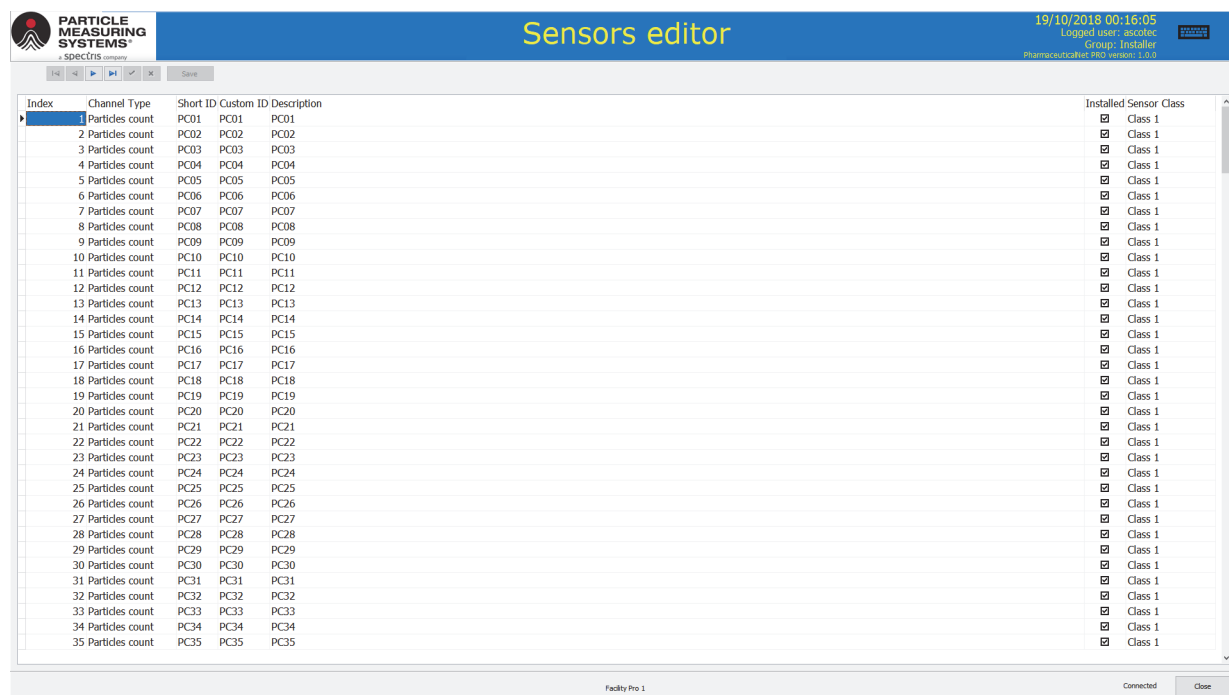
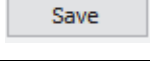


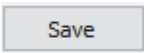
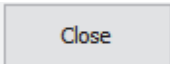
Figure 12-2 Sensors Editor window

The components of the **Sensors Editor** window are described in **Table 12-2**.

Table 12-2 Sensors Editor window components

Component	Description
Toolbar	The buttons on toolbar at the top of the Sensors Editor window are used to navigate to the record in the table you want to work with and to “post” and save your edits.
 First Record button	Click this button to highlight the first row in the table.
 Previous Record button	Click this button to highlight the previous row in the table.
 Next Record button	Click this button to highlight the next row in the table.
 Last Record button	Click this button to highlight the last row in the table.
 Post Edit button	Click this button to “accept” your changes. The Electronic Signature window will appear (see Electronic Signature on page 1-5).
 Undo Edit button	Click this button to “undo” your most recent edit.
 Save button	Click this button to save changes to the Sensors Editor window.
Table components	
ID column	The ID of a sensor.
Module Type column	The sensor’s type.
Module column	The sensor’s number.
Channel column	A channel number.
Description column	A description of the sensor.
 Close button	Click this button to close the Sensors Editor window.

>> To edit the definitions in the Sensor Editor window:

- 1** Open the **Sensors Editor** window as described on page 12-5.
- 2** Click on the text you want to change.
- 3** Type the new text.
- 4** Click  the **Post Edit** toolbar button.
 - The **Save** button becomes enabled.
 - The **Electronic Signature** window appears.
- 5** Perform the electronic signature (see Electronic Signature on page 1-5) to save the changes to the **Sensors Editor** window.
- 6** Click  the **Save** button.
- 7** Click  the **Close** button to close the **Sensors Editor** window.

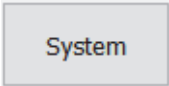
NOTE: Only the Installer can change the Sensor's name.

Backup and Restore

The **System management** window (see **Figure 12-1**) is used to administer database backups and restore a database from a backup.

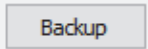
Daily backup configuration

>> To configure the daily backup:

- 1 Click  the **System** button at the bottom of the Pharmaceutical Net Pro screen.
The **System management** window appears (see **Figure 12-1**).
- 2 In the **Automatic backup time** field, use the up and down arrows to set the time of day for the backup.
- 3 In the **Old backup delete limit (days)** field, adjust the number of days before the old backup files are purged from the directory.
- 4 In the **Server BackUp Directory** field, set the filepath where the backup file will be saved.
- 5 Click  the **OK** button in the **System management** window.

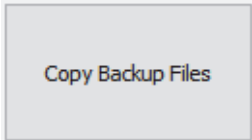
Manual backup

>> To perform a manual backup:

- 1 Open the **System management** window as described on page 12-9.
- 2 Click  the **Backup** button in the **System management** window.
The backup files will be saved in the filepath specified in the **Server BackUp Directory** field.

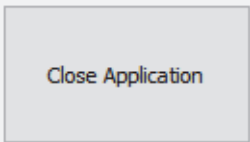
Copy backup files

>> To copy the backup files:

- 1 Plug a USB drive into the server module.
- 2 Click  the **Copy Backup Files** button in the **System management** window.
- 3 From there, the file to be copied and the destination external drive can be selected.

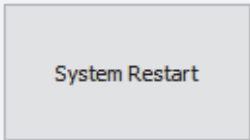
Close the Pharmaceutical Net Pro software application

An Administrator user can close the Pharmaceutical Net Pro software, and access the

Operating System by clicking  the **Close Application** button in the **System management** window.

Restart the System

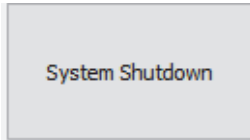
An Administrator user can restart the entire system (Server + Clients) by clicking

 the **System Restart** button in the **System management** window.

This will restart the Server machine and Pharmaceutical Net Pro software application on the Client machines.

Shutdown the System

An administrator user can shut down the entire system (Server + Clients) by clicking



the **System Shutdown** button in the **System management**

window.

This will result in a complete power-off for the server machine, and a shut down of the Pharmaceutical Net Pro software application on the client machines.

Shutdown a Disconnected Client

When a Pharmaceutical Net Pro client is open but disconnected from the server, the only option to close it is to press the combination keys **Alt + F4** on the keyboard.

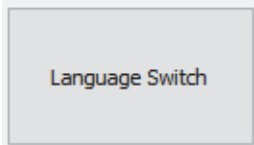
This will result in a complete power-off for the client machine.

Change the Display Language

An Administrator user can select the language for the Pharmaceutical Net Pro software displays.

>> **To change the display language:**

1 Open the **System management** window as described on page 12-9.

2 Click  the **Language Switch** button in the **System management** window.

The **FacilityPro Tools Language Switch** window appears (see **Figure 12-3**).

The **FacilityPro Tools Language Switch** window has options for the languages supported for the Pharmaceutical Net Pro software displays.

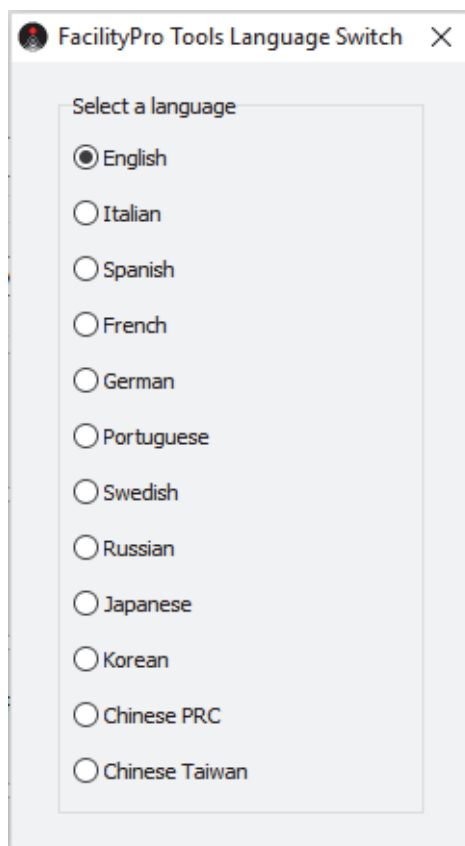
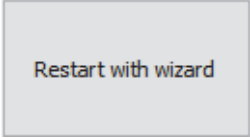


Figure 12-3 FacilityPro Tools Language Switch window

3 In the **FacilityPro Tools Language Switch** window, select  the option for the desired language.

4 Click  the X button to close the **FacilityPro Tools Language Switch** window.

- 5 Click  the **Restart with wizard** button in the **System**

management window.

The wizard window appears.

- 6 Use the wizard to restart the Pharmaceutical Net Pro software application.

Reset the Buffer

An Administrator user can reset the data buffer while the system is debuffering.

NOTE: After 8 hours of buffering (collecting data with FacilityPro disconnected), the system goes in a Buffer Overrun state, and stops collecting data. In this case, once reconnected to the Pharmaceutical Net Pro server machine, in order to re-activate the connection the **Reset Buffer** button has to be pressed.

>> To reset the buffer:

- 1 Open the **System management** window as described on page 12-9.

- 2 Click  the **Reset Buffer** button in the **System management**

window.

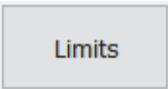
Limits Editor

The **Limits Editor** window can be used to view and edit the particle limits used by the system to handle warnings and alarms.

Only Administrator users have sufficient privileges to make changes to this list.

Limits Editor window

>> To open the **Limits Editor** window:

- 1 Click  the **Limits** button at the bottom of the Pharmaceutical Net Pro screen.
- The **Limits Editor** window appears (see **Figure 12-4**).

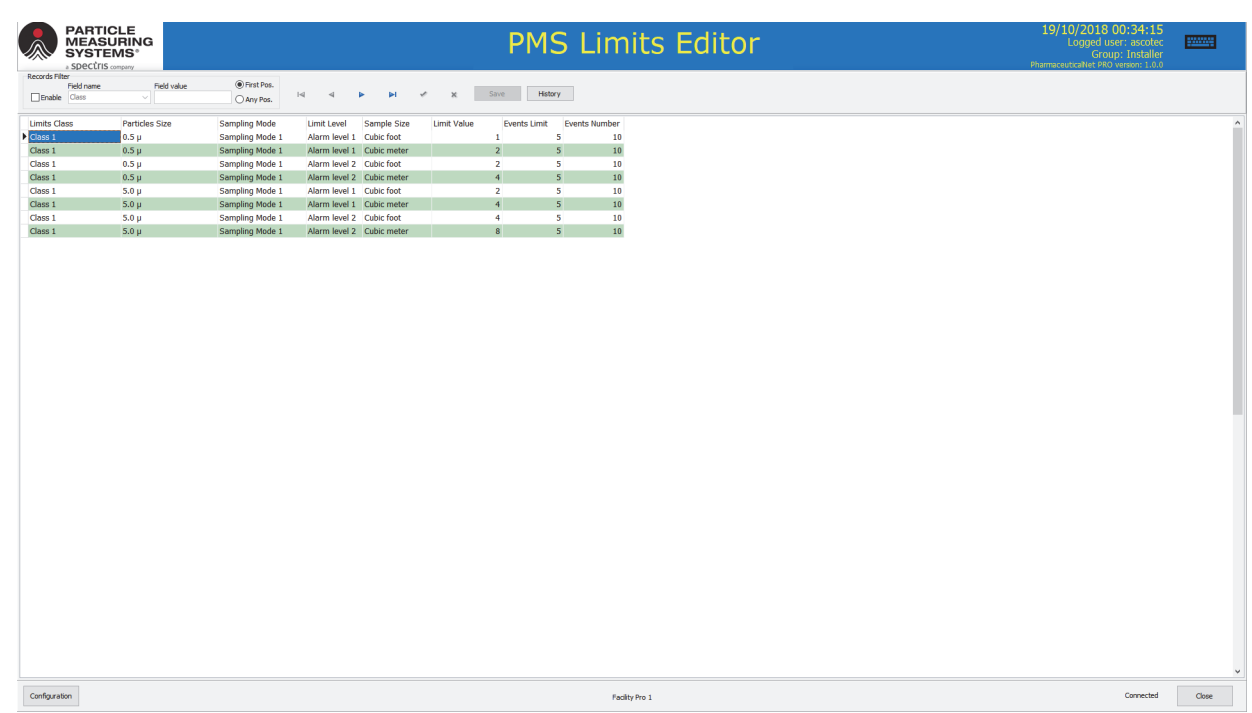


Figure 12-4 Limits Editor window

The components of the **Limits Editor** window are described in **Table 12-3** on page 12-14.

Table 12-3 Components of the **Limits Editor** window

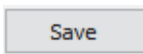
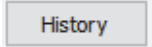
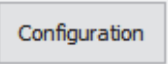
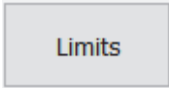
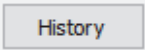
Components	Description
Records Filter area	
☒ Enable check box	☒ When checked, enables the two fields to the right: Field name and Field value. ☐ When not checked disables the two fields to the right: Field name and Field value.
Field name field	When the ☒ Enable check box is checked, use this list box to select the name of the following fields to use with a Field value to focus the values in the table to just those selections: Limit type Particles size Sampling mode Limits class
Field value field	When the ☒ Enable check box is checked and a field name is selected in the Field name field, enter the value for limiting the scope of the values displayed in the table.
☒ First Pos. option	☒ When selected, positions  the row pointer at the first row in the table.
☒ Any Pos. option	☒ When selected, positions  the row pointer wherever you click on a row.
Navigation buttons	Use the arrow buttons at the top of the Limits Editor window to view a different record of the table.
 First Record button	Click this button to view the beginning of the list.
 Previous Record button	Click this button to view the previous record of the list.
 Next Record button	Click this button to view the next record of the list.
 Last Record button	Click this button to view the last record of the list.
 Save button	Click this button to save changes made in the Limits Editor window.

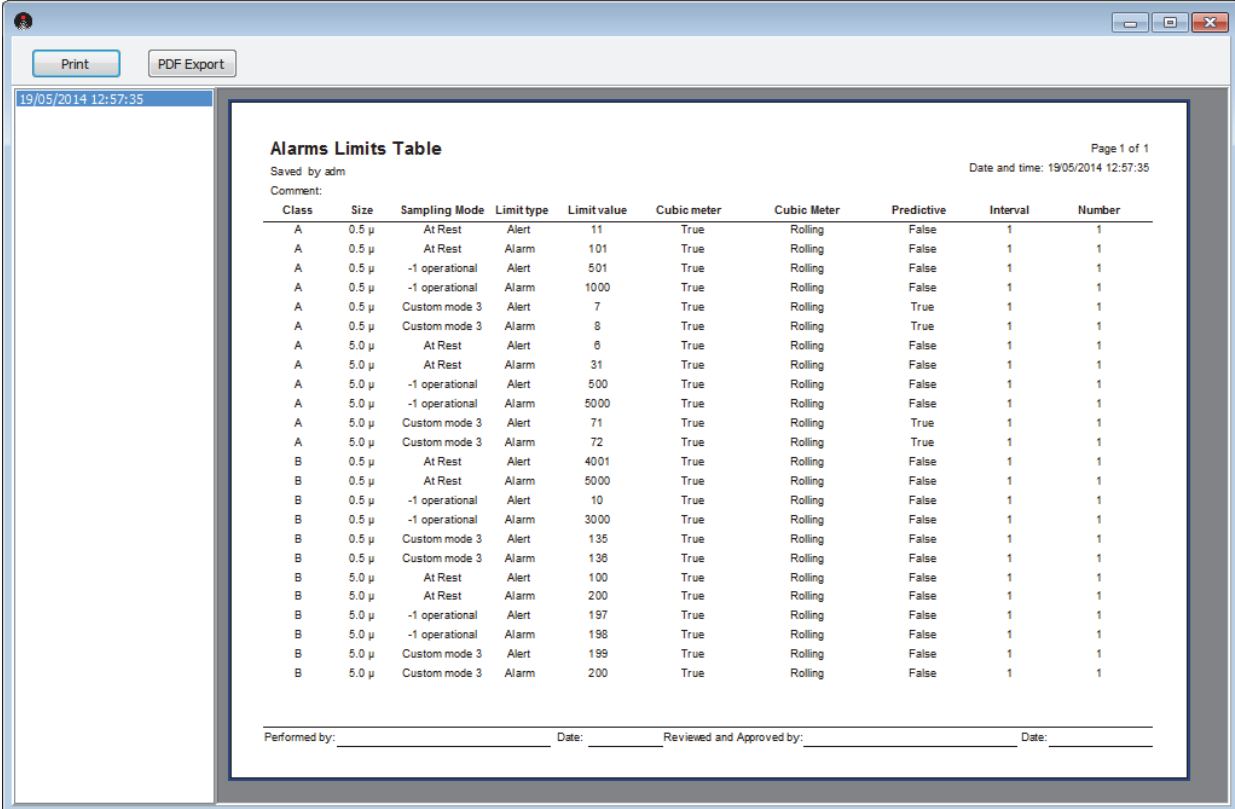
Table 12-3 Components of the **Limits Editor** window (Continued)

Components	Description
 History button	Click this button to open the Limits Editor History window (see Limits Editor History window on page 12-16).
 Configuration button	(Only Administrator users). Click this button to enter the Limits Editor Configuration page.
Values table columns	
Limits Class column	Class identification of particle limit sampling.
Particle Size column	Indicates the size of the particles monitored by the system.
Sampling Mode column	Indicates the type of particle limits.
	Custom types of particle limits can be configured.
	The system has the default types:
	• At Rest • Operational
Limit Type column	• Alert • Alarm
Limit Value column	Value set for each limit.
Cubic Meter Alarm	• ☒ When checked, the alarm is set per cubic meter. • ☐ When not checked, the alarm is set per cubic foot.
Cubic Meter Type column	• Rolling – The system performs a floating window of data per minute to calculate counts per cubic meter (fast trend). • Reset – When the sampling reaches the cubic meter, the system resets the count (total count). • CFX35 – The system multiplies the value of the cubic foot by 35.
Predictive column	When this box is checked ☒ , the algorithm to compute predictive alarms per cubic meter is enabled.
Interval column	The number of samples (Interval) occurring in which the number of events is out of bounds (Number)
Number column	The number of events out of bounds.

Limits Editor History window

>> To open the Limits Editor History window:

- 1 Click  the **Limits** button at the bottom of the Pharmaceutical Net Pro screen.
The **Limits Editor** window appears (see **Figure 12-4** on page 12-13).
- 2 Click  the **History** button at the top of the **Limits Editor** window.
The **Limits Editor History** window appears (see **Figure 12-5**).



The screenshot shows the 'Limits Editor History' window. It has a title bar with standard window controls. Below the title bar are 'Print' and 'PDF Export' buttons. A status bar at the top left shows the date and time '19/05/2014 12:57:35'. The main content area is titled 'Alarms Limits Table' and includes a 'Page 1 of 1' indicator and a 'Date and time: 19/05/2014 12:57:35' stamp. The table is titled 'Alarms Limits Table' and has a 'Comment:' field with the value 'adm'. The table has 10 columns: Class, Size, Sampling Mode, Limit type, Limit value, Cubic meter, Cubic Meter, Predictive, Interval, and Number. The table contains 24 rows of data, grouped by Class (A, B) and Size (0.5 µ, 5.0 µ). The table is followed by a signature line with fields for 'Performed by:', 'Date:', 'Reviewed and Approved by:', and 'Date:'.

Class	Size	Sampling Mode	Limit type	Limit value	Cubic meter	Cubic Meter	Predictive	Interval	Number
A	0.5 µ	At Rest	Alert	11	True	Rolling	False	1	1
A	0.5 µ	At Rest	Alarm	101	True	Rolling	False	1	1
A	0.5 µ	-1 operational	Alert	501	True	Rolling	False	1	1
A	0.5 µ	-1 operational	Alarm	1000	True	Rolling	False	1	1
A	0.5 µ	Custom mode 3	Alert	7	True	Rolling	True	1	1
A	0.5 µ	Custom mode 3	Alarm	8	True	Rolling	True	1	1
A	5.0 µ	At Rest	Alert	6	True	Rolling	False	1	1
A	5.0 µ	At Rest	Alarm	31	True	Rolling	False	1	1
A	5.0 µ	-1 operational	Alert	500	True	Rolling	False	1	1
A	5.0 µ	-1 operational	Alarm	5000	True	Rolling	False	1	1
A	5.0 µ	Custom mode 3	Alert	71	True	Rolling	True	1	1
A	5.0 µ	Custom mode 3	Alarm	72	True	Rolling	True	1	1
B	0.5 µ	At Rest	Alert	4001	True	Rolling	False	1	1
B	0.5 µ	At Rest	Alarm	5000	True	Rolling	False	1	1
B	0.5 µ	-1 operational	Alert	10	True	Rolling	False	1	1
B	0.5 µ	-1 operational	Alarm	3000	True	Rolling	False	1	1
B	0.5 µ	Custom mode 3	Alert	135	True	Rolling	False	1	1
B	0.5 µ	Custom mode 3	Alarm	136	True	Rolling	False	1	1
B	5.0 µ	At Rest	Alert	100	True	Rolling	False	1	1
B	5.0 µ	At Rest	Alarm	200	True	Rolling	False	1	1
B	5.0 µ	-1 operational	Alert	197	True	Rolling	False	1	1
B	5.0 µ	-1 operational	Alarm	198	True	Rolling	False	1	1
B	5.0 µ	Custom mode 3	Alert	199	True	Rolling	False	1	1
B	5.0 µ	Custom mode 3	Alarm	200	True	Rolling	False	1	1

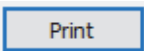
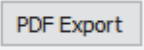
Performed by: _____ Date: _____ Reviewed and Approved by: _____ Date: _____

Figure 12-5 Limits Editor History window displayed in front of the Limits Editor window

- 3 To close the **Limits Editor History** window, click  the **X** button at the top right corner of the window.

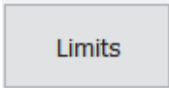
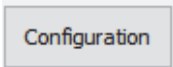
The components of the **Limits Editor History** window are described in **Table 12-4** on page 12-17.

Table 12-4 Components of the **Limits Editor History** window

Components	Description
 Print button	Click this button to open a Print dialog box where you can send a copy of the listing to the printer.
 PDF Export button	Click this button to export the history in PDF format.
Dates list	Click on a date to populate the Alarms Limits table with information from that date. The date you selected displays below the table title.
Alarms Limits table columns Class column Size column Sampling Mode column Limit type column Limit value column Cubic Meter Alarms column Cubic Meter Type column Predictive column Interval column Number column	Class identification of particle limit sampling. The particle size. The Sampling mode. For example: • At Rest • Operational Alarm or Alert . Value of the limit. True or False . • Rolling – The system performs a floating window of data per minute to calculate counts per cubic meter (fast trend). • Reset – When the sampling reaches the cubic meter, the system resets the count (total count). • CF X 35 – The system multiplies the value of the cubic foot by 35. True or False . The interval value. The number of events out of bounds.

Limits Editor Configuration window

>> To open the Limits Editor Configuration window:

- 1 Click  the **Limits** button at the bottom of the Pharmaceutical Net Pro screen.
The **Limits Editor Configuration** window appears (see **Figure 12-4** on page 12-13).
- 2 Click  the **Configuration** button at the top of the **Limits Editor** window.
The **Limits Editor Configuration** window appears (see **Figure 12-6**).

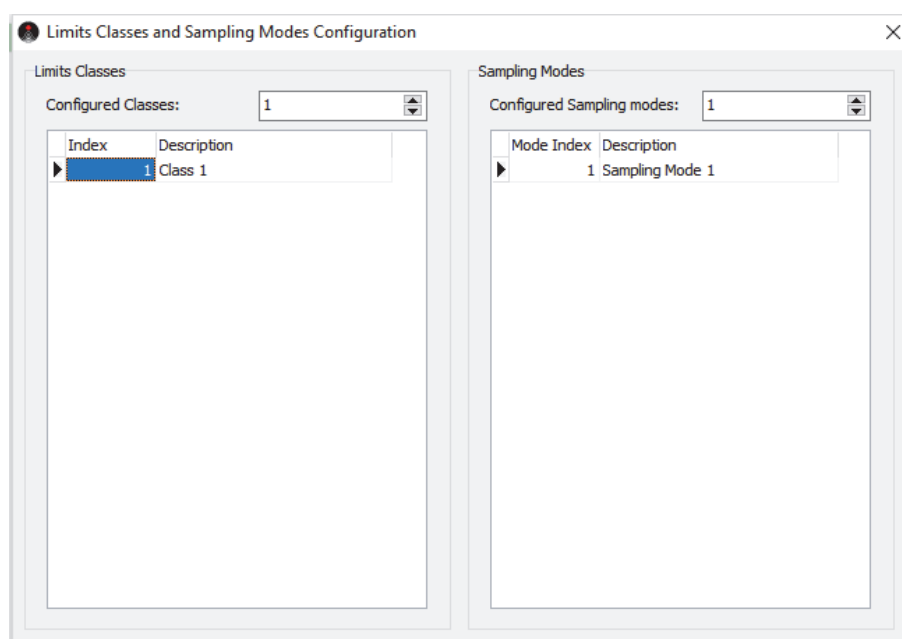


Figure 12-6 Limits Editor Configuration window displayed in front of the **Limits Editor** window

>> To enable a new sensor class:

- 1 In the **Limits Classes** pane of the **Limits Editor Configuration** window (see **Figure 12-6** on page 12-18), place ☒ a check mark in the **Enabled** column for the desired class.
- 2 The name of the class can be modified by typing the new name in the **Description** field, then pressing the **Enter** key.
- 3 To close the **Limits Editor Configuration** window, click  the **X** button at the top right corner of the window.

>> To enable a new sampling mode:

- 1** In the **Sampling Modes** pane of the **Limits Editor Configuration** window (see **Figure 12-6** on page 12-18), place  a check mark in the **Enabled** column for the desired sampling mode.
- 2** The name of the sampling mode can be modified by typing the new name in the **Description** field, then pressing the **Enter** key.
- 3** To close the **Limits Editor Configuration** window, click  the **X** button at the top right corner of the window.

Analog Inputs Limits

Use the **Analog Inputs Limits** to change analog sensor limits.

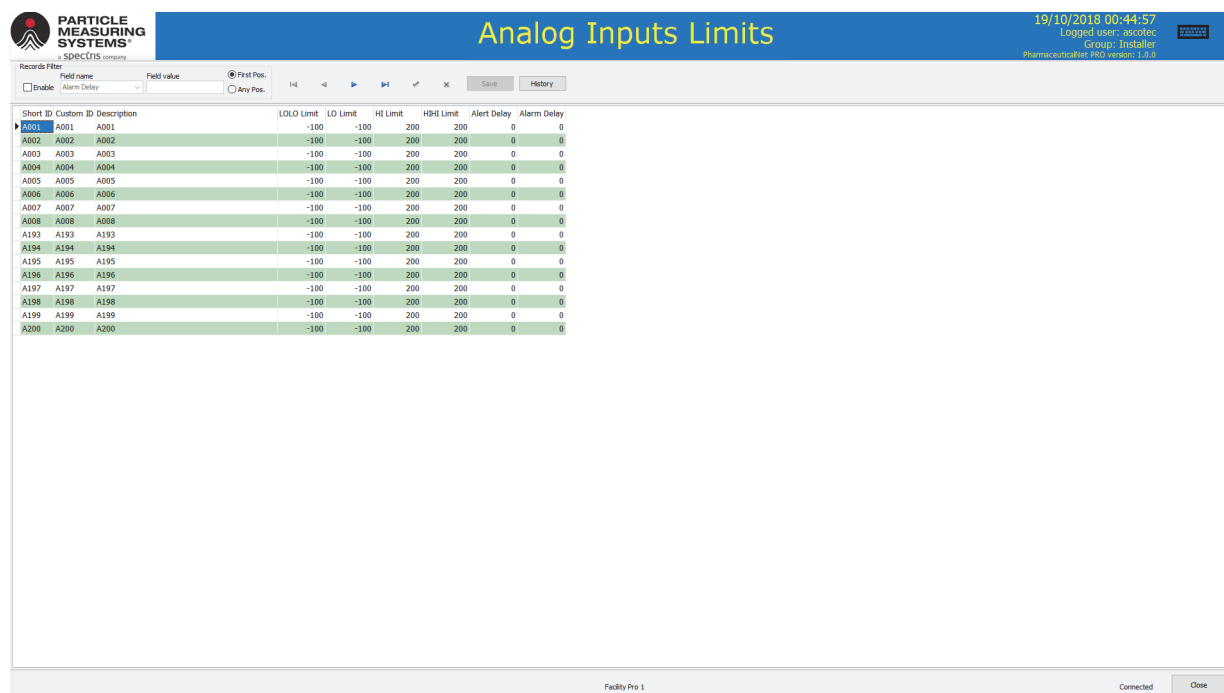


Figure 12-7 Analog Inputs Limits window

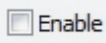
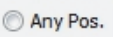
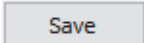
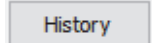
The components of the **Analog Inputs Limits** window are described in **Table 12-5**.

Table 12-5 Analog Inputs Limits window view components

Component		Description
Short ID column		Displays the output's system identification number.
Custom ID column		Displays the output's user-specified identification number.
Description column		Displays a description of the output.
Alert Limits columns		
	LO	Displays the minimum value set for the output's range before the alert is triggered.
	HI	Displays the maximum value set for the output's range before the alert is triggered.
	Delay (sec)	Displays the number of seconds delay before the alert is triggered.
Alarm Limits columns		
	LOLO	Displays the minimum value set for the output's range before the alarm is triggered.
	HIHI	Displays the maximum value set for the output's range before the alarm is triggered.
	Delay (sec)	Displays the number of seconds delay before the alarm is triggered.

Records Filter

Table 12-5 Analog Inputs Limits window view components (Continued)

Component	Description
 Enable box	Check this box to enable the user-specified search criteria.
 Field Name drop down	Use the drop down list of Field names (i.e., LOLO Limit, Description, etc.) to facilitate your search to a specific column.
 Field value	Type the numeric value of the desired field here.
 First Pos.	Use this button to search only for the first positive search result.
 Any Pos.	Use this button to search for all positive search results.
 View First Row button	Click this button to view the first row in the table.
 View Previous Row button	Click this button to view the previous row in the table.
 View Next Row button	Click this button to view the next row in the table.
 View Last Row button	Click this button to view the last row in the table.
 Save button	Click this button to save any changes made in the table.
 History button	Click this button to view a history of previous changes.

>> To change the Custom ID, Description, Limits and Delays:

Double-click on the specific field you want to change and enter the desired value.
The value will be changed after completing the Electronic Signature.

This page is intentionally left blank.

Chapter 13

Aventino Setup Options

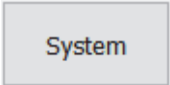
The options of the Aventino Wizard includes the following:

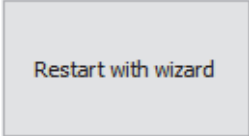
- Setting up a new installation of Pharmaceutical Net Pro on page 13-2
- Upgrading the current version of Pharmaceutical Net Pro with a new version on page 13-12
- Restoring a backup file on page 13-12
- Modifying the current configuration on page 13-13
- Restoring a FacilityPro processor on page 13-13

Opening the Aventino Wizard Application

>> **To open the wizard:**

- 1 At the bottom of the Pharmaceutical Net Pro screen, click

the  **System** button.

- 2 Click  the **Restart with wizard** button.

- 3 Perform the **Electronic Signature**.

- 4 The PC will restart and launch the Aventino Wizard application (see **Figure 13-1** on page 13-2).

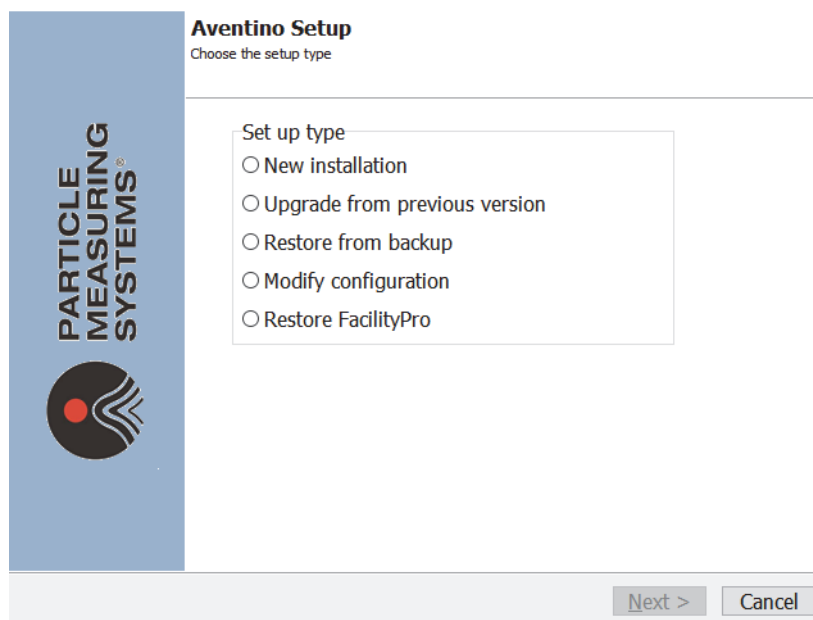


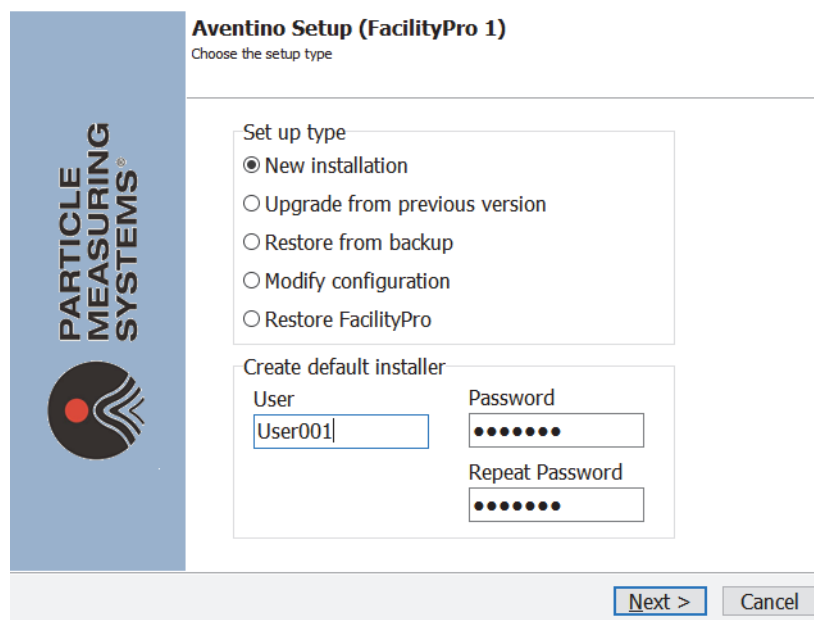
Figure 13-1 Aventino Setup Wizard

New Installation

Choosing the **Set up type** option **New Installation** allows the user to create a new configuration for the FacilityPro processor. This includes creating a new default installer username and password, entering the IP address of the processor(s), and specifying the number of samplers.

>> The steps of the new installation process are as follows:

- 1** After choosing **New installation** from the Aventino Setup Wizard, enter new default installer credentials (see **Figure 13-2** on page 13-3).
- 2** Click **Next**.



Aventino Setup (FacilityPro 1)
Choose the setup type

Set up type

- ☒ New installation
- ☐ Upgrade from previous version
- ☐ Restore from backup
- ☐ Modify configuration
- ☐ Restore FacilityPro

Create default installer

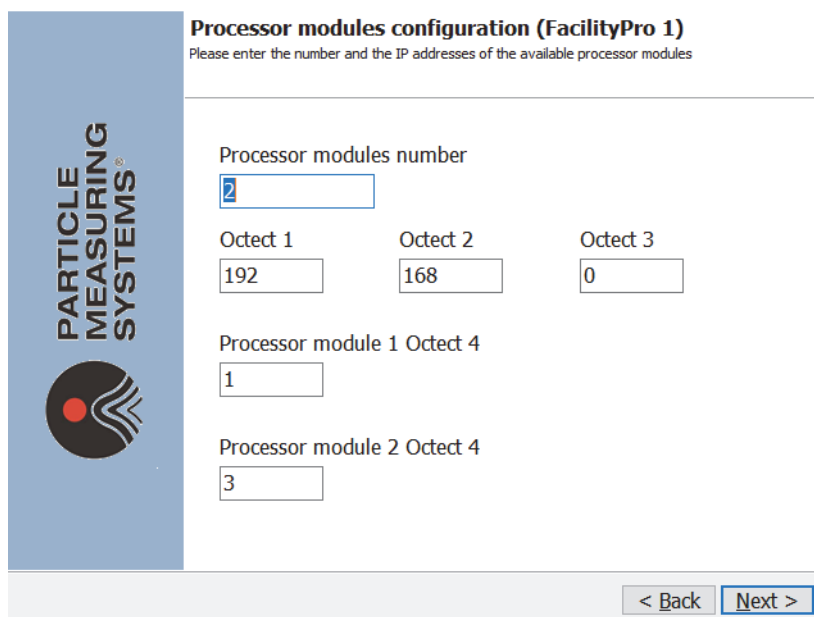
User:

Password:

Repeat Password:

Figure 13-2 New installation and entering user credentials

- 3 Select the number of processor modules in the system (one or two).
- 4 Enter the IP address(es) for each processor (see **Figure 13-3**).
- 5 Click **Next**.



Processor modules configuration (FacilityPro 1)
Please enter the number and the IP addresses of the available processor modules

Processor modules number:

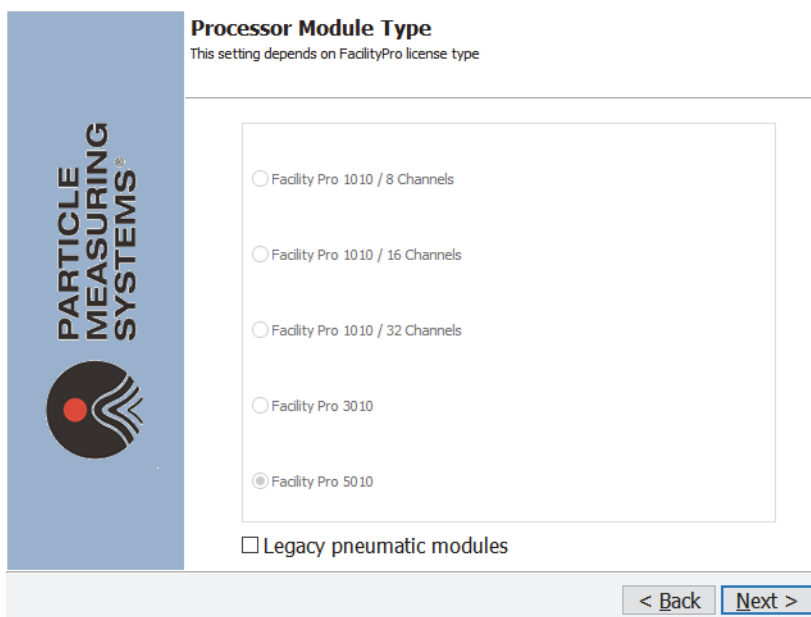
Octect 1: Octect 2: Octect 3:

Processor module 1 Octect 4:

Processor module 2 Octect 4:

Figure 13-3 New installation and entering user credentials

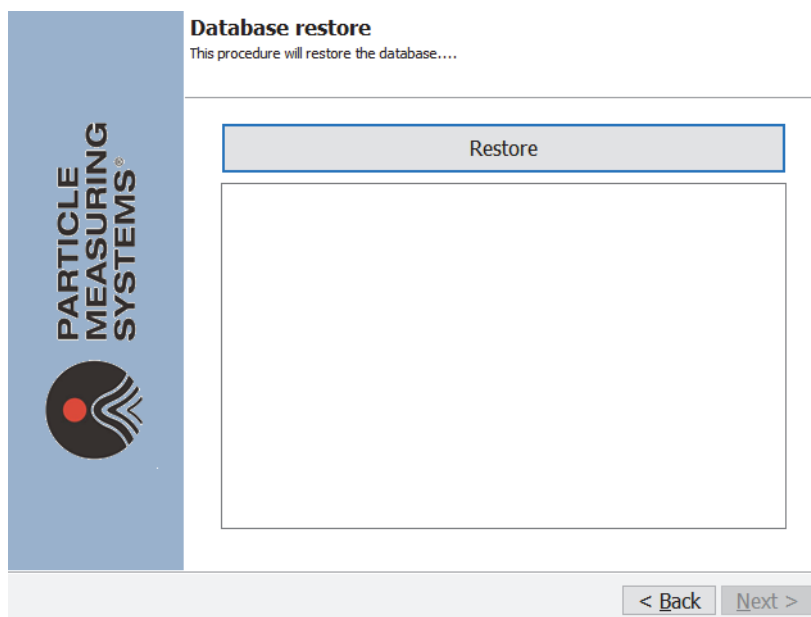
- 6 Depending on your license, the processor module type will already be selected on the next screen (see **Figure 13-4**).
If you are using an old pneumatic/sensor module, checkmark the box beside **Legacy pneumatic modules**.
- 7 Click **Next**.



The screenshot shows a software installation window titled "Processor Module Type" with the subtitle "This setting depends on FacilityPro license type". On the left is a vertical blue bar with the "PARTICLE MEASURING SYSTEMS" logo. The main area contains five radio button options: "Facility Pro 1010 / 8 Channels", "Facility Pro 1010 / 16 Channels", "Facility Pro 1010 / 32 Channels", "Facility Pro 3010", and "Facility Pro 5010". The "Facility Pro 5010" option is selected. Below these options is a checkbox labeled "Legacy pneumatic modules" which is currently unchecked. At the bottom right are "< Back" and "Next >" buttons.

Figure 13-4 Processor module type selected from license

- 8 Click **Restore** to initialize the database with the processor (see **Figure 13-5**).
- 9 Click **Next**.

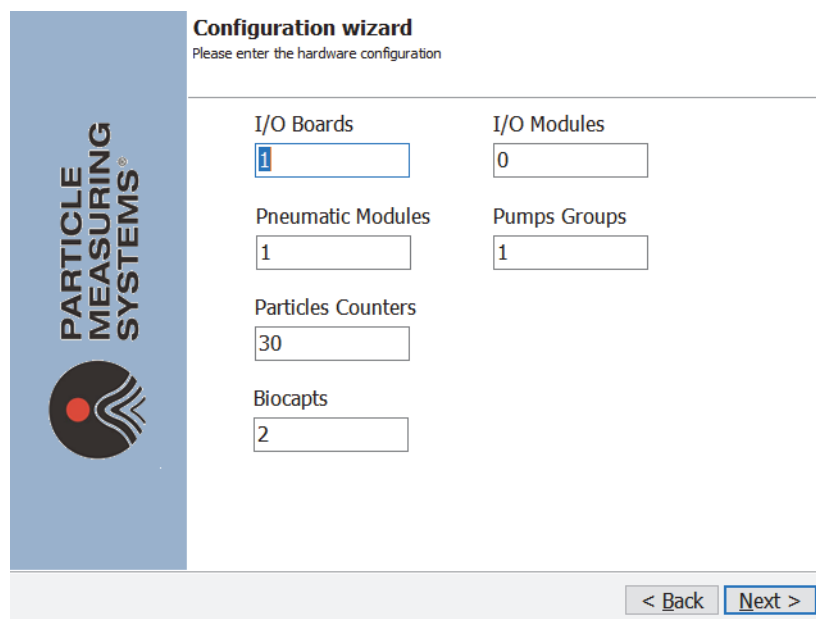


The screenshot shows a software installation window titled "Database restore" with the subtitle "This procedure will restore the database....". On the left is a vertical blue bar with the "PARTICLE MEASURING SYSTEMS" logo. The main area features a large "Restore" button at the top and a large empty rectangular box below it. At the bottom right are "< Back" and "Next >" buttons.

Figure 13-5 Database restore

10 Enter the hardware to be used with the system. This includes all modules, pumps and sensors (see **Figure 13-6**).

11 Click **Next**.

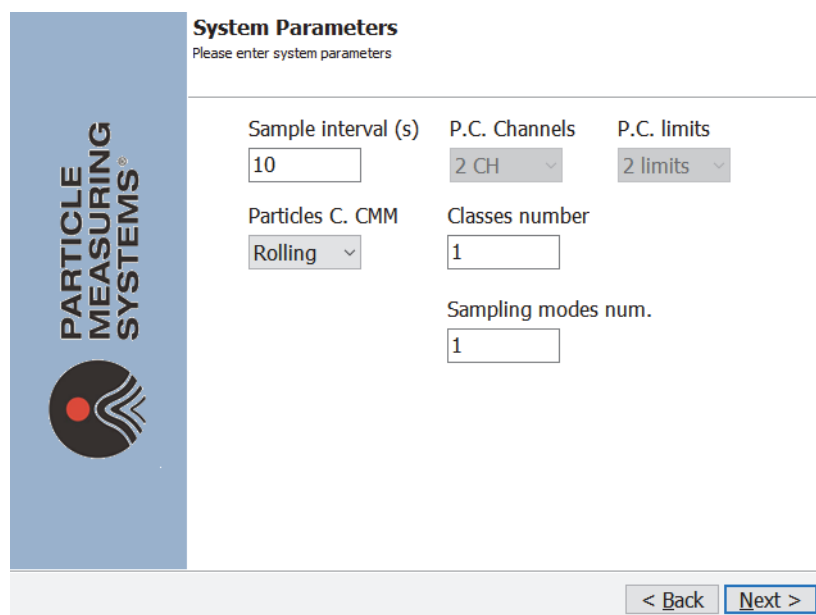


The screenshot shows the 'Configuration wizard' window. On the left is a blue sidebar with the 'PARTICLE MEASURING SYSTEMS' logo. The main area has the title 'Configuration wizard' and the instruction 'Please enter the hardware configuration'. It contains several input fields: 'I/O Boards' (value 1), 'I/O Modules' (value 0), 'Pneumatic Modules' (value 1), 'Pumps Groups' (value 1), 'Particles Counters' (value 30), and 'Biocapts' (value 2). At the bottom right are '< Back' and 'Next >' buttons.

Figure 13-6 System hardware configuration

12 Enter the system parameters, including how often sample readings will be read, number of classes to differentiate samplers and the number of sampling modes (see **Figure 13-7**).

13 Click **Next**.



The screenshot shows the 'System Parameters' window. On the left is a blue sidebar with the 'PARTICLE MEASURING SYSTEMS' logo. The main area has the title 'System Parameters' and the instruction 'Please enter system parameters'. It contains several input fields and dropdown menus: 'Sample interval (s)' (value 10), 'P.C. Channels' (value 2 CH), 'P.C. limits' (value 2 limits), 'Particles C. CMM' (value Rolling), 'Classes number' (value 1), and 'Sampling modes num.' (value 1). At the bottom right are '< Back' and 'Next >' buttons.

Figure 13-7 System sampling parameters

- 14** Enter the number of days between calibration alerts and alarms . Checkmark every associated **Enabled** box for each alarm that you want to apply (see **Figure 13-8**). The default number of days for alarms amounts to once per year. The default number of days for alerts is one month before the alarm.
- 15** Click **Next**.

Calibration alarms configuration
Please set calibration alarms and alerts settings

Sensors		Analog inputs	
Alarm	Alert	Alarm	Alert
365	335	365	335
☐ Enabled	☐ Enabled	☐ Enabled	☐ Enabled

Pneumatic modules

Alarm	Alert
365	335
☐ Enabled	☐ Enabled

< Back Next >

Figure 13-8 Sampler calibration configuration

- 16** If using legacy pneumatic/sensor modules, select the type of module (biological or particle counter) for each one (see **Figure 13-9**).
- 17** Click **Next**.

Legacy Pneumatic Modules Configuration Page
Please Select legacy pneumatic modules type

Legacy pneumatic module	Pneumatic module type
Pneumatic module 1	Biological
Pneumatic module 2	Particles counters

< Back Next >

Figure 13-9 Legacy pneumatic/sensor module configuration

- 18** Configure module channels to be associated with individual samplers (see **Figure 13-10**).

Select the channel and the samplers and click **Connect**. The light blue arrow indicates connection/disconnection is available. The gray arrow indicates no connection/disconnection is available for the components selected.

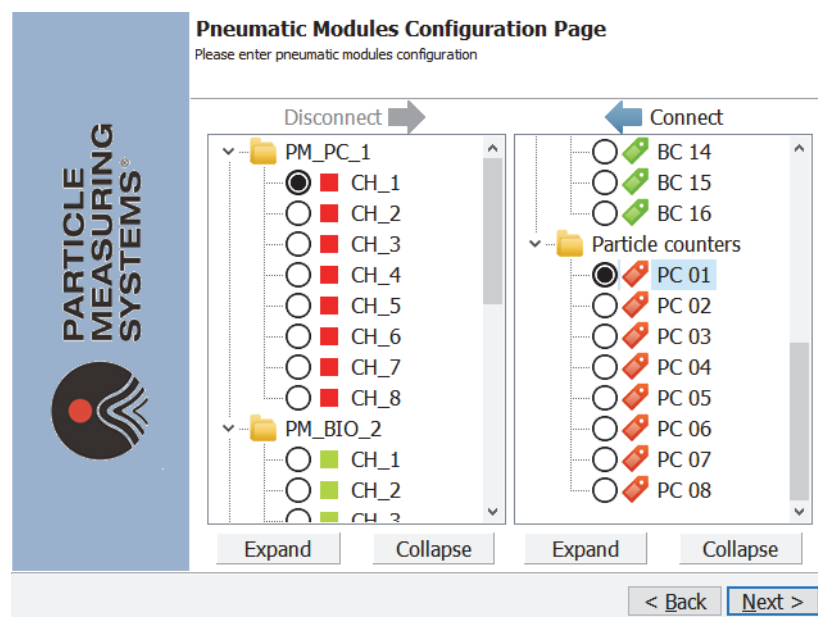


Figure 13-10 Pneumatic module configuration with individual samplers

- 19** To disconnect a sampler, click its associated tag in the left-hand scrolling pane and click **Disconnect**.
- 20** Click **Next**.

21 Set the IP address and associated class for each sampler (see **Figure 13-11**).

22 Click **Next**.

TCP/IP Configuration Page
Please configure TCP/IP addresses for all connected devices.

	Octect 1	Octect 2	Octect 3
	10	0	140

PC	IP Address	Class
PC 01	1	Class 1
PC 02	2	Class 1
PC 03	3	Class 1
PC 04	4	Class 1
PC 05	5	Class 1
PC 06	6	Class 1
PC 07	7	Class 1
PC 08	8	Class 1

< Back Next >

Figure 13-11 Pneumatic module configuration with individual samplers

23 Select the pump group number associated with each pneumatic module (see **Figure 13-12** on page 13-8). The number available is determined with the **Pumps Groups** field in step 10.

24 Click **Next**.

Pump Groups Configuration Page
Please set a pumps group for each pneumatic module

Pneumatic module	Pumps group
Pneumatic module 1	1
Pneumatic module 2	2
Pneumatic module 3	0
Pneumatic module 4	0

< Back Next >

Figure 13-12 Pneumatic module configuration with individual samplers

25 Set the number of pumps in each pump group (see **Figure 13-13**).

26 Click **Next**.

Pumps group	Pumps number in group
Pumps group 1	1
Pumps group 2	2

Figure 13-13 Setting number of pumps in group

27 Associate each I/O board with an I/O module (number of each set in step 10).

28 Set how many analog boards are in each I/O module (see **Figure 13-14**).

29 Click **Next**.

IO board	I/O Module	Analog Boards
IO board 1	I/O Module 1	1
IO board 2	I/O Module 2	0
IO board 3	I/O Module 3	0
IO board 4	I/O Module 4	0

Figure 13-14 I/O board and module configuration

30 The **Configuration Save Page** summarizes all settings made to the system in the previous steps (see **Figure 13-15**). Click **Next**.

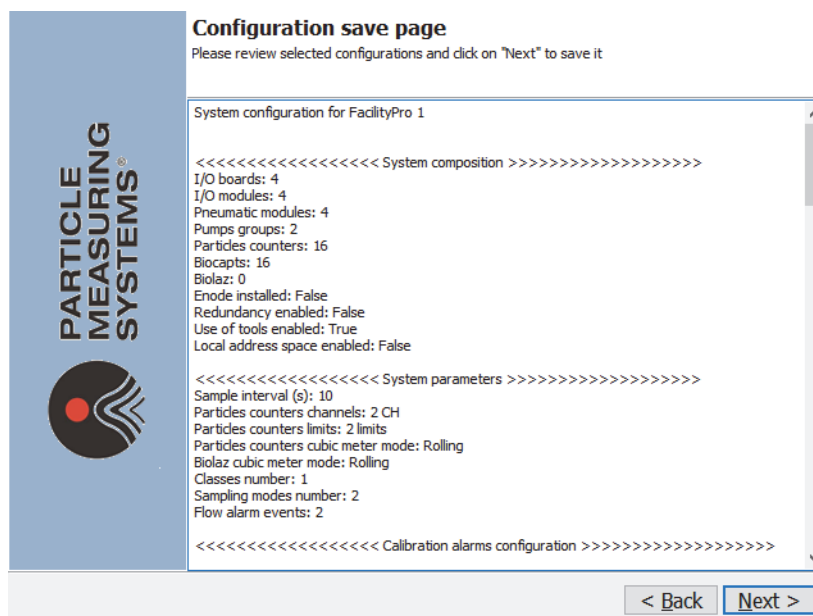


Figure 13-15 Configuration Save Page

31 Click **Transfer**. The FacilityPro processor restarts with the new configuration (see **Figure 13-16**).

32 Click **Next** once progress bar is complete.

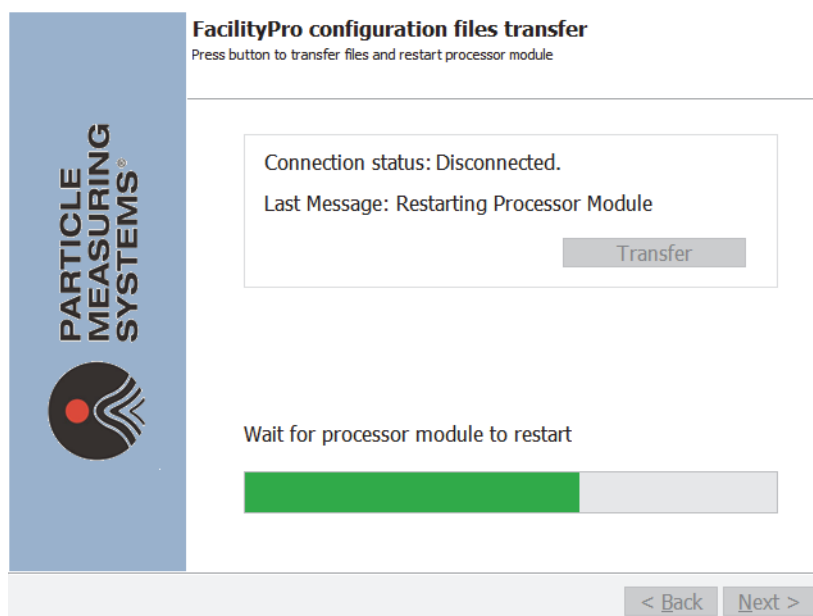


Figure 13-16 FacilityPro configuration file transfer and restart

- 33** Click **Initialize** to reset tags and alarms with the configured settings (see **Figure 13-17**).

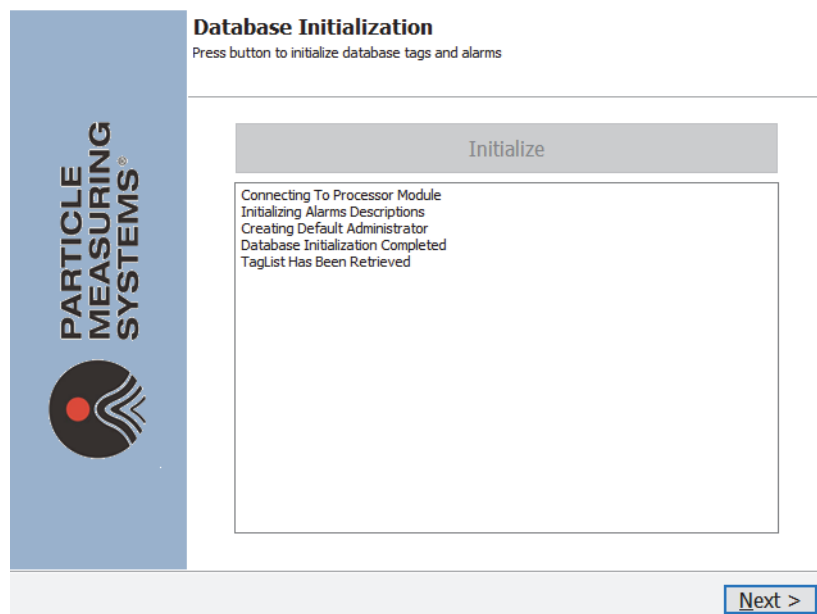


Figure 13-17 Database initialization

- 34** Click **Next**.
- 35** Click **Create Pages**. The Pharmaceutical Net Pro software pages are created with the configured settings (see **Figure 13-18**).
- 36** Click **Finish**. The PC restarts.

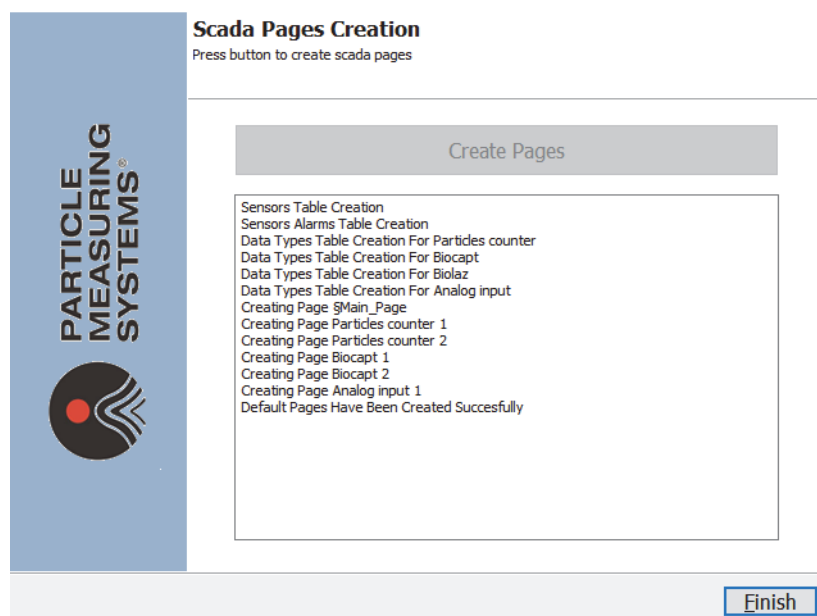


Figure 13-18 Creation of Pharmaceutical Net Pro software pages

Upgrade from Previous Version

Contact your Particle Measuring Systems representative in the event this action needs to be performed.

Restore from Backup

WARNING

The restore procedure completely overwrites existing data!

>> To perform a restore of the database:

- 1 In the Aventino Wizard Setup window (see **Aventino Setup Wizard** on page 13-2), select **Database Restore**.
- 2 Click **Restore**. Select the file to restore from the specified directory and confirm.
- 3 Click Next.
- 4 Click **Test Availability** (see **Figure 13-19**). If a processor module is connected and the license matches OPCLink, you can proceed to the following step by clicking **Next**.

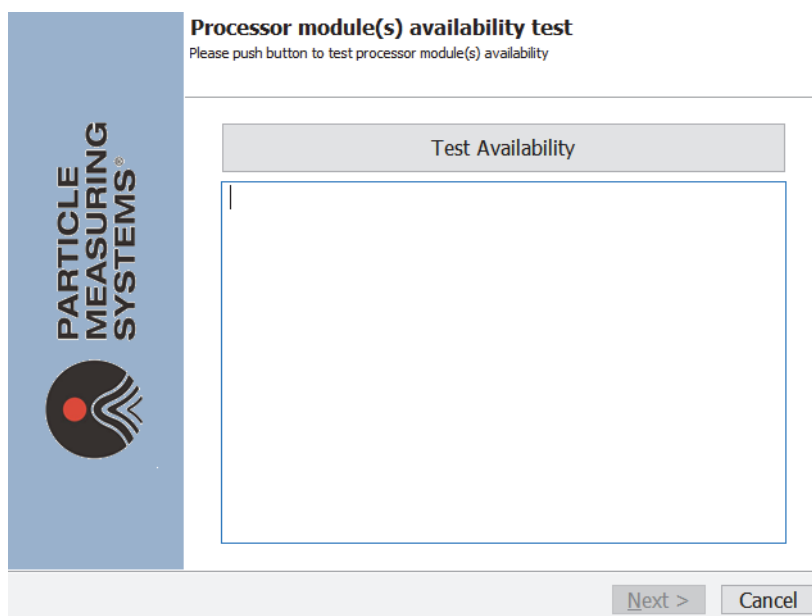


Figure 13-19 Testing processor availability

- 5 Click **Transfer**. The FacilityPro processor restarts with the new configuration (see **Figure 13-16**).
- 6 Click **Next** once the progress bar is complete.
- 7 Click **Initialize** to reset tags and alarms with the configured settings (see **Figure 13-17**).

- 8 Click **Finish** to initialize the database. The PC restarts.

Modify Configuration

This option allows the user to see current settings of the configuration and make adjustments. Steps are identical to **New Installation** on page 13-2.

Restore FacilityPro

>> **To restore the FacilityPro processor to a previous version:**

- 1 Choose **Restore FacilityPro** from the Aventino Setup Wizard (see **Figure 13-1**).
- 2 Click **Next**.
- 3 Click **Test Availability** (see **Figure 13-19**). If a processor module is connected and the license matches OPCLink, you can proceed to the following step by clicking **Next**.
- 4 Click **Transfer**. The FacilityPro processor restarts with the new configuration (see **Figure 13-16**).
- 5 Click **Next** once the progress bar is complete.
- 6 Click **Initialize** to reset tags and alarms with the configured settings (see **Figure 13-17**).
- 7 Click **Finish** to initialize the database. The PC restarts.

This page is intentionally left blank.

Appendix A

Statistical Data Analyzer

The Statistical Data Analyzer is an optional data analysis application. Functions of the software include:

- Imports CSV files provided by History Automation
- Visualizes data in chart form for statistical analysis
- Reports large data volumes for processing
- Extracts processed data in CSV format once per day

Licensing the Application

Upon first running the application, you will be asked to activate Statistical Data Analyzer. You will need a valid License ID and Password.

Running the Application

StatisticalDataAnalyzer.exe is an external tool opened from the Aventino folder. The filepath should be similar to the following:

C:\Aventino\StatisticalDataAnalyzer.exe

Once the executable file has been located, double click it or right click and choose “Open”.

Application Window

Once opened, the following graphical interface appears:

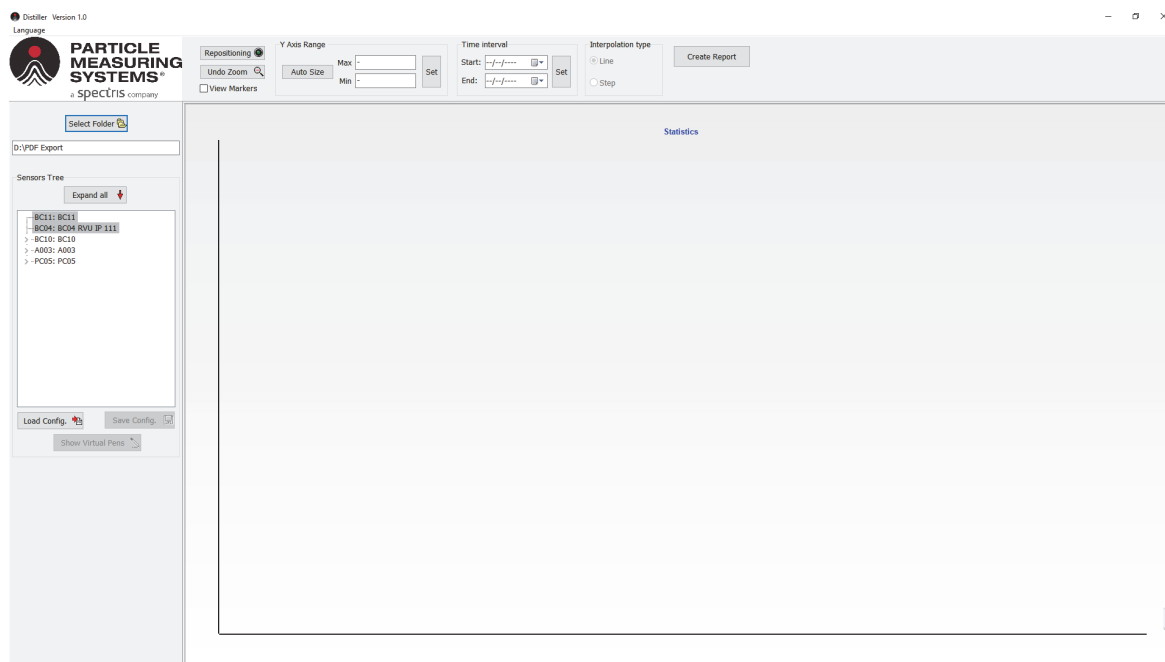


Figure A-1 Statistical Data Analyzer window

In **Figure A-1**, a CSV data file has been automatically chosen by the application. Sensors highlighted in gray within the **Sensors Tree** area have no associated sampling data.

Loading CSV Files

A target folder for saved CSV data files is chosen by clicking **Select Folder**. These CSV files are generated by the History Automation Service and loaded individually to the Statistical Data Analyzer.

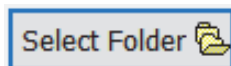


Figure A-2 Select Folder button

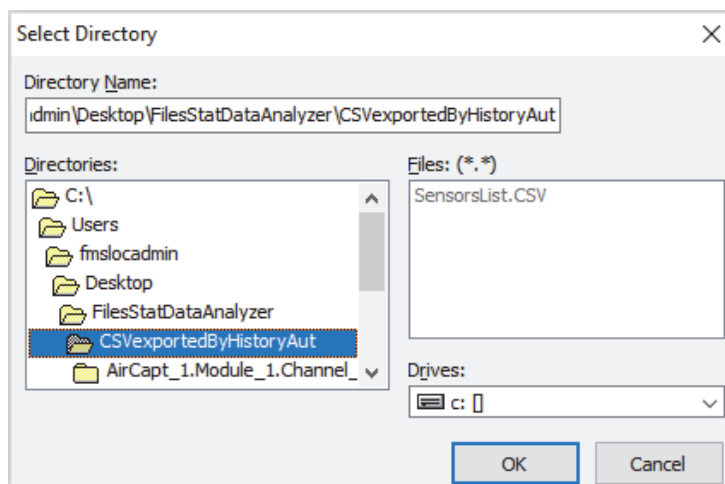


Figure A-3 Select Folder window

In the **Select Folder** window (see **Figure A-3**), a filepath can be typed into the **Directory Name** field, or the **Directories** pane can be used to locate the filepath. Folders can be expanded or collapsed by double-clicking them. The selected drive can be changed with the **Drives** drop-down pane. Files found in the selected directory are shown in the **Files** pane.

The loaded CSV file will always have the name “SensorsList.csv”, and located in the same directory as the sensor module CSV file folders. For example:

- “AirCapt_1.Module_1.Channel_1”

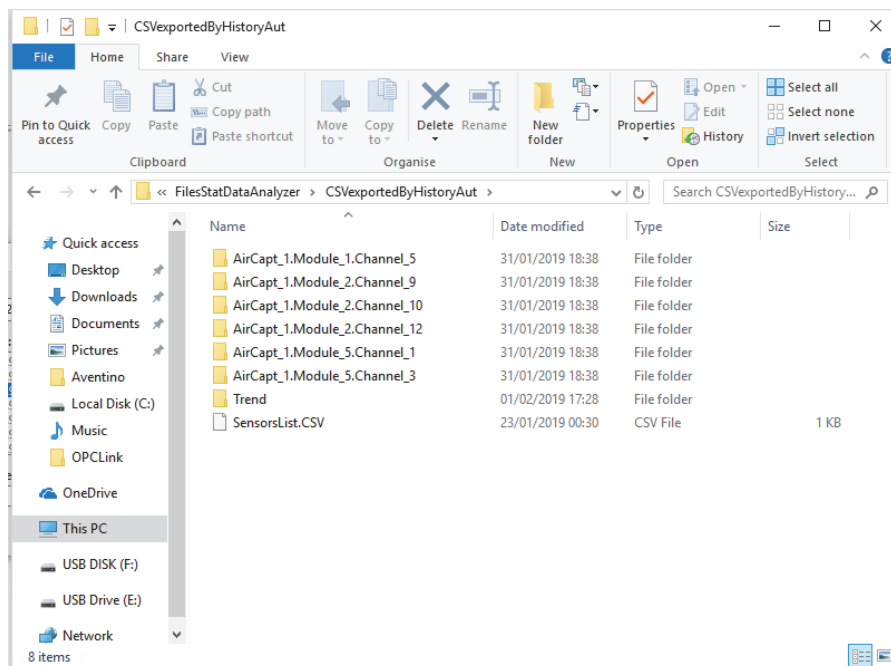


Figure A-4 SensorsList.CSV directory

Display Configuration

Beneath **Sensors Tree**, click the **Expand All** button to display all Global Stat options (Average, Minimum, Maximum and Standard Deviation) for each sensor.

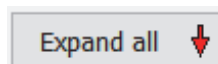


Figure A-5 Expand All button

BioCapt sensors (BC) have both flow and volume global stats, and analog sensors (A) have real value global stats. Particle counter sensors (PC) have cubic feet and cubic meter global stats in addition to stats associated with user-defined sampling modes. The selection(s) made by clicking once will appear in bold and display in the graph.

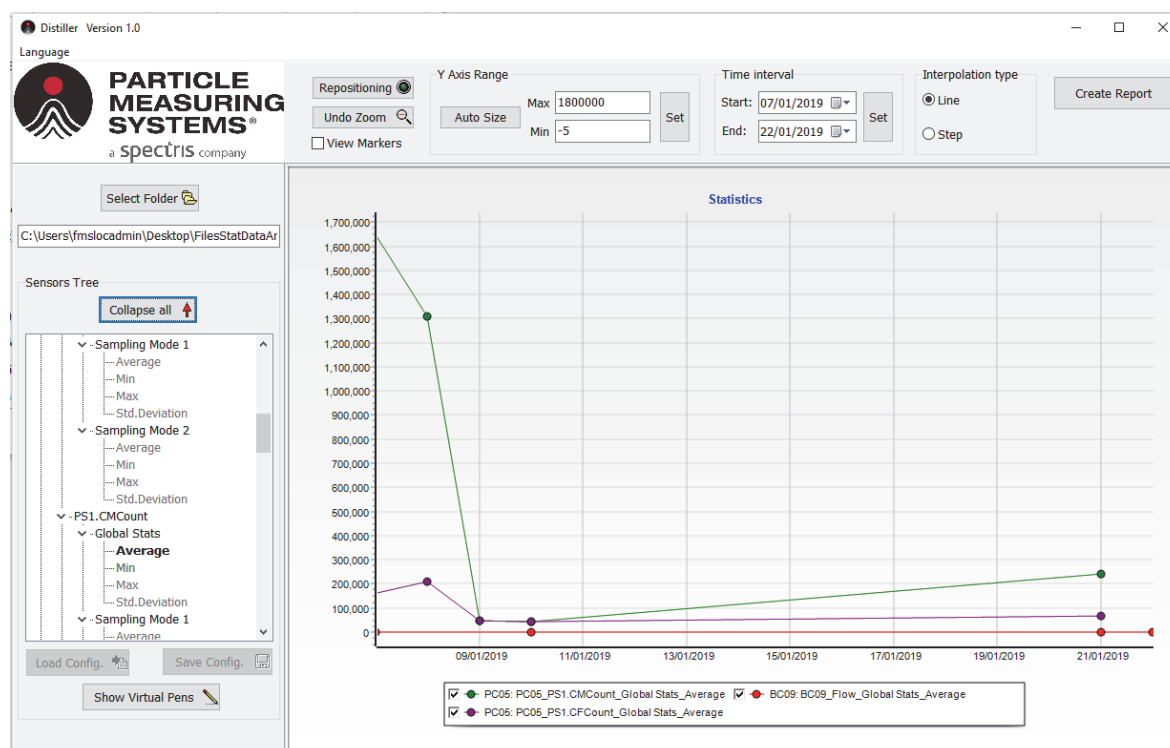


Figure A-6 Global stats display

Clicking on the individual global stats for a sensor generates a graph in the **Statistics** display with the chosen stats.

Axis Adjustment

The **Time interval** is automatically chosen to include all days with data. The **Start** and **End** date can be adjusted with the calendar menu.

The x- and y-axis ranges are automatically set for best-fit to include all data points. The y-axis can be adjusted to limit the view with the **Max** and **Min** fields, and implemented by clicking **Set**. Clicking **Auto Size** resets the y-axis to best-fit.

Zoom

By dragging the right mouse button over the graph, the area selected can be zoomed to cover the total display. Clicking **Undo Zoom** resets the display to regular magnification.

Data values can be zoomed in to the thousandth decimal value.

Interpolation

Data display for Average stats is automatically set to include the **Line** interpolation type (shown in **Figure A-6**). An example of the Step interpolation type is shown in **Figure A-7**.

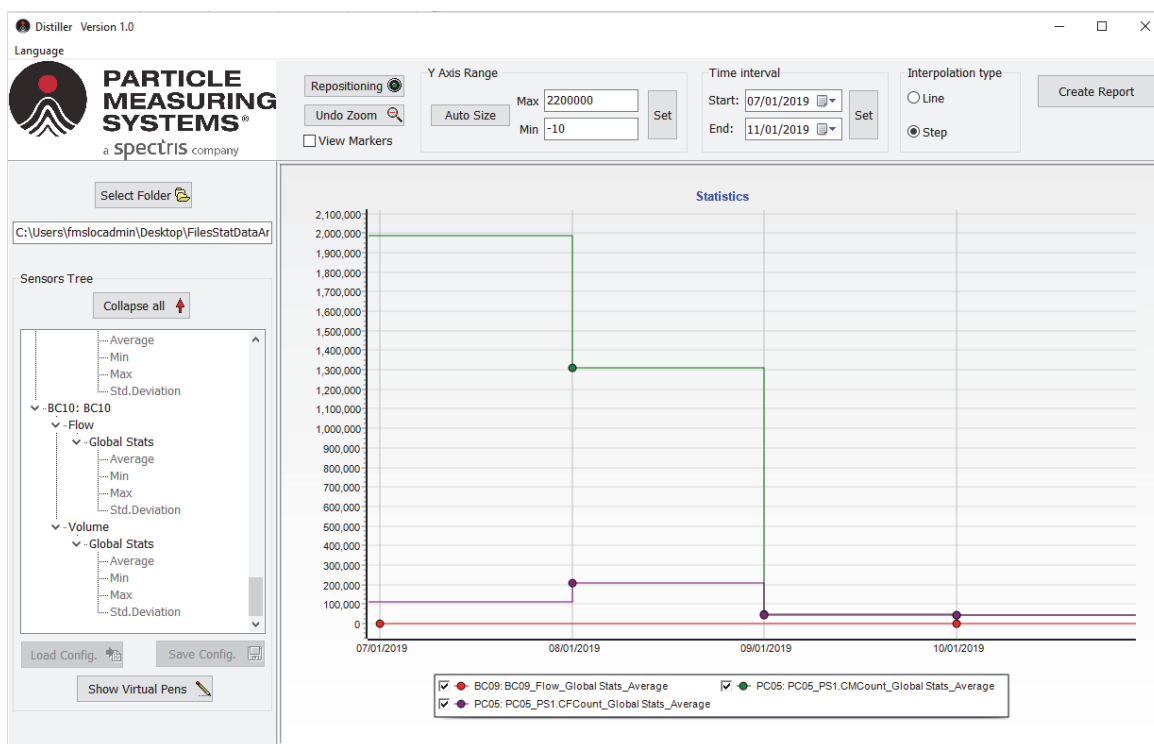


Figure A-7 Step interpolation

Min, Max and Std. Deviation values will be displayed as discrete points with no interpolation.

Saving a Configuration

After making the desired selections to adjust the display, the configuration can be saved for later viewing by clicking the **Save Config.** button. Only one configuration can be saved at once.

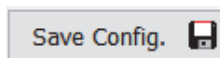


Figure A-8 Save Config. button

The **Repositioning** button resets all adjustments made to the saved configuration after loading.

Loading a Previous Configuratoion

The previous configuration saved can be loaded to the Statistical Data Analyzer by clicking the **Load Config.** button.

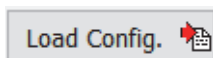


Figure A-9 Load Config. button

Virtual Pens

Horizontal lines can be added to the display to represent alert or action limits. These lines are not part of configuration, but will be generated with the report. To add a line, click the **Show Virtual Pens** button.

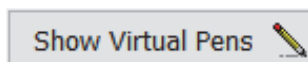


Figure A-10 Show Virtual Pens button

The **Virtual Pens** window appears. The previous configuration can be loaded with the **Load** button and will automatically display configuration parameters for editing. After making adjustments or creating a new configuration, the virtual pens can be saved by clicking **Save**. Click **Reset** to remove all parameters.

Click **Enable** to display the virtual pen in the graphical display. To make any adjustments to the pen (Name, Value and Color), ensure the pen is disabled (unchecked) first.

Enable	Name	Value	Color
☒	Alert	1250000	Blue
☒	Alarm	1500000	Red
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black
☐			Black

Figure A-11 Virtual Pens window

Markers and Sensor Display Toggling

Markers for each data point can be toggled on and off with the **View Markers** checkbox. Individual sensor data that has been selected in the **Sensors Tree** can be toggled on and off by clicking the associated checkbox in the legend area of the Statistics display.

Daily Sensor Chart

A daily sensor chart window can be opened by holding the CTRL button and clicking a data point or interpolation line in the Statistics area. The new window displays the total sensor data for the day associated with the data point.

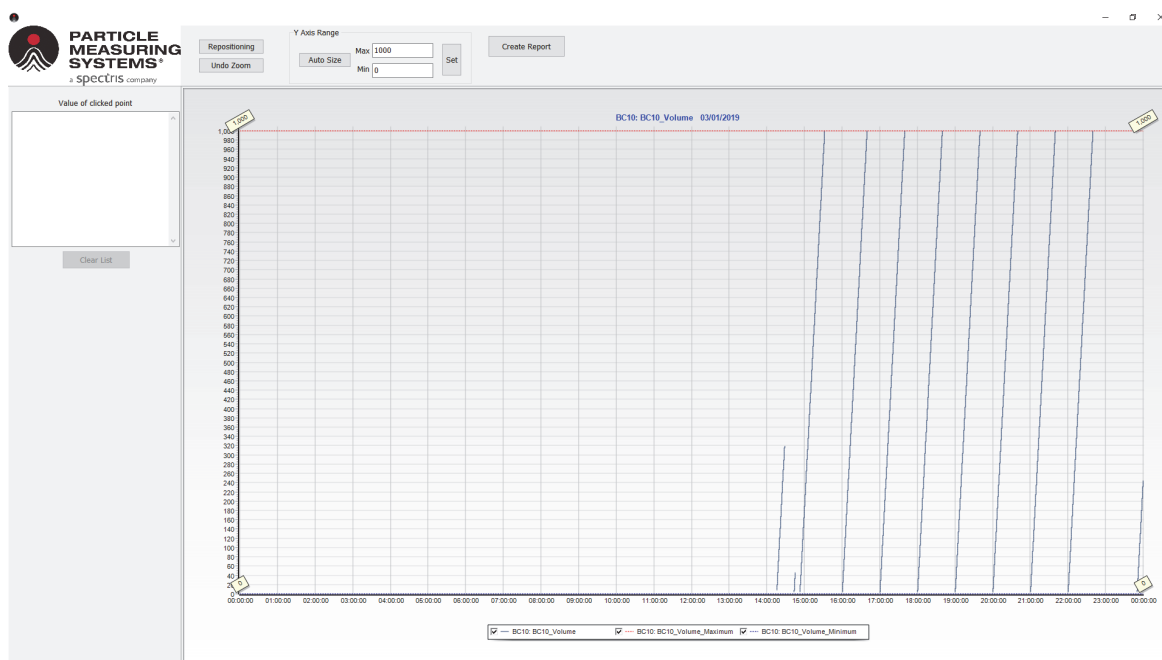


Figure A-12 Daily sensor chart window

Inside this new window, data points can be clicked on and their associated tag name, timestamp and value displayed in the **Value of clicked point** display area. Both this window's graphical display, and the Statistics display can be saved as a report.

Creating a Report

A PDF file containing the Statistics display can be generated by clicking the **Create Report** button. Inside the PDF, the Report Title, comments and name of the user printing the report can be added to the empty fields. After all text additions have been made, the PDF can be saved and exported to a predetermined filepath using the **Export** button located in the top left corner of the window. The filepath is chosen with the History Automation Configuration tool (see **History Automation Configuration** on page 7-13).

Language

The display language can be adjusted to any of the languages supported by FacilityPro SMART NG (see **Change the Display Language** on page 11-11).

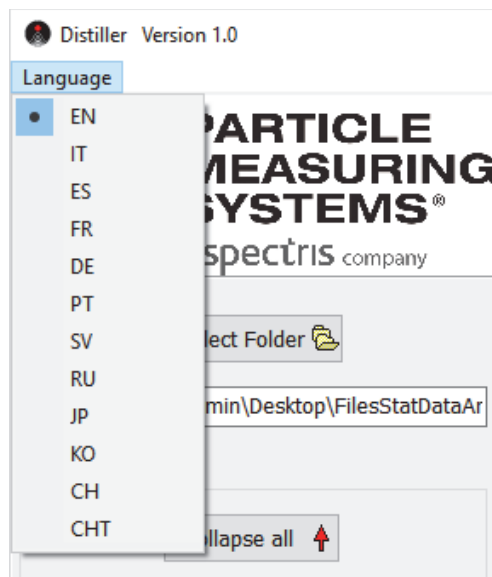


Figure A-13 Display language adjustment

Index

A

Abort button
 Sampling Editor window 5-2, 5-5
 aborting sampling 5-5
 Ack All button 7-2
 Ack selected button 7-2
 acknowledge alarm 7-1
 steps 7-3
 Add user button 11-5
 ADMINISTRATORS group 11-2
 alarms 7-1
 acknowledge 7-1
 steps 7-3
 set particle limits 12-13
 trigger limits, setting 12-13
 Alarms button 3-3, 7-1
 Alarms window 7-1
 components 7-2
 All Values window
 components 12-20
 Analog Limits Editor 12-20
 Analog Sensors button 3-3, 6-9
 Analog Sensors window 6-9
 components 6-10
 Audit report
 electronic signature events 8-1, 8-25,
 8-32, 8-39, 8-41
 automatic backup time, setting 12-2
 Available Tags pane 9-2
 Aventino Wizard 13-1
 modifying the current configuration 13-1
 new installation 13-1
 restoring a backup file 13-1
 restoring a FacilityPro processor 13-1
 upgrading 13-1

B

back up a database 12-8
 backup
 copy files 12-9
 daily, configuring 12-8
 manual 12-9
 Backup button 12-2, 12-9
 backups of database 12-8
 biological modules
 sampling point information 6-7
 Biological Sensors button 3-3, 6-6
 Biological Sensors window 6-6
 components 6-7
 values 6-6
 buttons
 Abort
 Sampling Editor window 5-2, 5-5
 Ack All 7-2
 Ack selected 7-2
 Add user 11-5
 Alarms 3-3, 7-1
 Analog Sensors 3-3, 6-9
 Backup 12-2, 12-9
 Biological Sensors 3-3, 6-6
 Close Application 12-3, 12-9
 Config. 8-2, 8-13
 Config., Report window 8-3
 Configuration 12-15, 12-18
 Sampling Editor window 5-2, 5-7
 Copy Backup Files 12-3, 12-9
 Delete 11-6
 Edit, Sampling Editor window 5-2, 5-5
 Execute Report 8-3, 8-13, 8-20, 8-27,
 8-33, 8-49
 Exit Page 8-3, 8-13
 History 12-15, 12-16
 History, Recipe Editor window 4-2
 Insert
 Sampling Editor window 5-2, 5-4
 Language Switch 12-3, 12-11

- Limits 3-4, 12-13, 12-16, 12-18
- Load, Recipe Editor window 4-2, 4-5
- Log In 3-3, 3-6
- Log-out 3-3, 3-6
- Main 3-5
- Main Page 3-3, 3-5
- manual ON/OFF
 - Particle Sensors window 6-5, 6-8
- New Recipe
 - Recipe Editor window 4-2, 4-5
- Particle Sensors 3-3, 3-4, 6-1
- Recipes 3-3, 4-1, 9-1, 13-1
- Report 3-3, 8-2, 8-21, 8-28
- Reset Buffer 12-3, 12-12
- Restart with wizard 12-3, 12-4, 12-12
- Sampling 3-3, 5-1, 8-28
- Save 12-6, 12-14
- Select 12-2
- Sensors Editor 12-3, 12-5
- Start VNC Viewer 12-3
- System 3-4, 11-3, 12-1, 12-5, 12-8
- System Shutdown 12-4, 12-9, 12-10
- User Event 3-3
- Users Management 12-2

C

- Channel Status Legend, Main Page window 3-5
- channels
 - electronic signature 6-5
 - manual ON/OFF button 6-5
- clients
 - shutdown 12-10
- clients, shutdown 12-9, 12-10
- close
 - FacilityPro SMART application 12-9
- Close Application button 12-3, 12-9
- Config. button 8-2, 8-13
 - Report window 8-3
- Configuration button 12-15, 12-18
 - Sampling Editor window 5-2, 5-7
- configure
 - daily backup 12-8
- copy backup files 12-9
- Copy Backup Files button 12-3, 12-9
- create
 - new user 11-5
 - recipe 4-5
 - report
 - by group in Report window 8-21
 - by sampling in Report window 8-28
 - report from Report window 8-20
 - Sampling Report 8-49
- CSV files A-1
- custom fields
 - for scheduling a sampling 5-6

- Custom fields configuration window 5-6

D

- daily backup, configuring 12-8
- Data Types 8-7
- database
 - backup 12-8
 - restore 12-8, 13-12
- default
 - user groups 11-1
 - functions 11-2
 - usernames and passwords 11-1
- define
 - custom fields for scheduling a sampling 5-6
 - recipes 4-1
- delete
 - recipe 4-6
 - sampling event 5-5
 - user 11-6
- Delete button 11-6

E

- edit
 - recipe 4-5
 - sampling event 5-5
 - sensors definitions 12-7
- Edit button
 - Sampling Editor window 5-2, 5-5
- electronic signature events
 - Audit report 8-1, 8-25, 8-32, 8-39, 8-41
- electronic signature requirement
 - alarm acknowledgement 7-3
 - All Values window, manual ON/OFF button 12-6
 - overview 1-5
 - Particle Sensors window, manual ON/OFF button 6-5, 6-8
 - recipes 4-5, 4-6
 - sampling 5-4, 5-5
 - Sensors Editor 12-7
 - user management 11-5, 11-6
- Electronic Signature window 1-5
- Execute Report button 8-3, 8-13, 8-20, 8-27, 8-33, 8-49
- Exit Page button 8-3, 8-13
- export report
 - in Report window 8-37
 - in Sampling Report window 8-50

F

FacilityPro
 hardware components 1-1
 overview 1-1
 system overview 1-1

G

get started 3-1
groups
 ADMINISTRATORS functions 11-2
 OPERATORS functions 11-2
 POWERUSERS functions 11-2
 SUPERVISORS functions 11-2
user
 default 11-1
 functions 11-2

H

History button 12-15, 12-16
 Recipes Editor window 4-2

I

information about a sampling point 6-1
Insert button
 Sampling Editor window 5-2, 5-4
installation
 Pharmaceutical Net Pro 2-1
 prerequisites 2-1

L

Language Switch button 12-3, 12-11
launching Pharmaceutical Net Pro application
 3-1
Limits button 3-4, 12-13, 12-16, 12-18
Limits Editor 12-13
Limits Editor Configuration window 12-18
Limits Editor History window 12-16
Limits Editor window 12-13
 components 12-14
 viewing controls 12-14
Load button
 Recipes Editor window 4-2, 4-5
log in 3-6
Log In button 3-3, 3-6
Log in window 3-6

Log Out 3-6
logging in 3-6
logging out 3-6
Log-out button 3-3, 3-6

M

Main button 3-5
Main Page button 3-3, 3-5
Main Page window 3-1, 3-2, 3-5
 Channel Status Legend 3-5
 components 3-2
manual backup 12-9
manual ON/OFF button 6-5
 Particle Sensors window 6-5, 6-8
modifying
 user 11-6

N

New Recipe button
 Recipes Editor window 4-2, 4-5
Nobody user 11-1

O

ON/OFF button 6-5
opening
 Alarms window 7-1
 Analog Sensors window 6-9
 Biological Sensors window 6-6
 Custom fields configuration window 5-7
 Limits Editor History window 12-16
 Limits Editor window 12-13
 Main Page window 3-5
 Particle Sensors window 6-1
 Recipes Editor window 4-1, 5-1, 9-1, 13-1
 Report window 8-2
 Sensors Editor window 12-5
 System management window 12-1
OPERATORS group 11-2
overview
 electronic signature requirement 1-5
 software 1-4
 system 1-1

P

particle counters
 sampling point information 6-4, 6-5
particle limits for alarms 12-13
Particle Sensors 6-1

Particle Sensors button 3-3, 3-4, 6-1
Particle Sensors window 6-1, 6-2
 components 6-3
 manual ON/OFF button 6-5, 6-8
 values 6-3
password
 default users 11-1
 Electronic Signature window 4-5, 4-6, 5-3,
 5-5, 6-5, 7-3
Pens Color Selection 8-12
Pharmaceutical Net Pro 3-1
 close the application 12-9
 interface 1-4
 launching 3-1
 log in 3-6
 logging out 3-6
 password
 default users 11-1
 software screen components 3-2
 start-up 3-1
 system tasks 12-1
POWERUSERS group 11-2
print report
 in Report window 8-37
 in Sampling Report window 8-49

R

Real Time Trend 9-1
real-time sampling point information 6-1
Recipes 8-10
recipes 4-1
 create new 4-5
 deleting 4-6
 editing 4-5
 schedule a sampling 5-3
 start a sampling with 5-3
Recipes button 3-3, 4-1, 9-1, 13-1
Recipes Editor window 4-1, 9-1
 components 4-2
Report button 3-3, 8-2, 8-21, 8-28
Report window 8-2
 components
 Select by Group 8-21
 Select by Sampling 8-28
 viewing controls 8-35
reports 8-1
 Audit Report - Select by Group 8-38
 Audit Report - Select by Sampling 8-40
 by group in Report window 8-21
 by sampling, in Report window 8-28
 changing views
 Report window 8-35
 create
 by group in Report window 8-21
 by sampling in Report window 8-28

data intervals 8-26, 8-33
Data Report 8-42, 8-44
data subset 8-25
date/time range 8-24, 8-30
examples 8-38
 Audit Report - Select by Group 8-38
 Audit Report - Select by Sampling 8-40
 Data Report - Select by Group 8-42
 Data Report - Select by Samplings 8-44
 Sampling Report 8-50
 Trend Report 8-46
execute 8-27, 8-33
exporting, in Report window 8-37
exporting, in Sampling Report window 8-50
overview, create from 8-49
overview, create from Report window 8-20
print
 in Report window 8-37
 in Sampling Report window 8-49
raw or summary data 8-26, 8-33
Report window reports 8-1
Sampling 8-48
 create 8-49
Sampling Report 8-50
Sampling Report window reports 8-1
time range 8-24, 8-30
Trend Report 8-47
 calculation parameters 8-26, 8-33
Reset Buffer button 12-3, 12-12
Restart with wizard button 12-3, 12-4, 12-12
restore a backed-up database 13-12
restore database 12-8
Run Time Logic Editor
 Ack. reset delay 10-16
 Available Tags pane 10-2
 Child Rule 10-13
 Editor pane 10-11
 Horn off delay 10-16
 Output Activation 10-18
 Rule Editor 10-12
 Rule Type 10-12
 Sampling on Input Editor 10-14
 TowerLightEditForm 10-15
 Usage for Tag pane 10-11

S

sampling 5-1
 aborting 5-5
 custom fields for scheduling 5-6
 custom fields for scheduling a sampling 5-6
 defining custom fields for 5-6
 delete sampling event 5-5
 edit sampling event 5-5
 information about a sampling point 6-1
 point
 biological modules information 6-7

- particle counters information 6-4, 6-5
 - schedule 5-3
 - select a recipe for 5-3
- Sampling button 3-3, 5-1, 8-28
- Sampling editor dialog box 5-6
 - scheduling a sampling 5-3
- Sampling Editor window 5-1
 - components 5-2
- sampling point information 6-1
- Sampling Report 8-48
 - create 8-49
- Sampling Report window 8-48
- Save button 12-6, 12-14
- schedule a sampling 5-3
- Select button 12-2
- Select sensors in group window 8-17
- Sensor Groups 8-8, 8-9
- Sensor Type 8-4
- Sensors 8-6
- Sensors Alarms 8-5
- Sensors Editor 12-5
- Sensors Editor button 12-3, 12-5
- Sensors Editor window 12-5
 - components 12-6
 - viewing controls 12-6, 12-14
- server
 - overview 1-4
- setting
 - alarm trigger limit 12-13
 - automatic backup time 12-2
 - particle limits 12-13
- shutdown
 - disconnected client 12-10
 - system 12-9, 12-10
- software description 3-1
- Start VNC Viewer button 12-3
- start-up of Pharmaceutical Net Pro application 3-1
- start-up, Pharmaceutical Net Pro 3-1
- StatisticalDataAnalyzer.exe A-1
- SUPERVISORS group 11-2
- system
 - tasks 12-1
- System button 3-4, 11-3, 12-1, 12-5, 12-8
- System management window 12-1
 - components 12-2
- system shutdown 12-9, 12-10
- System Shutdown button 12-4, 12-9, 12-10
- system start-up 3-1

T

- Tags catalog window 8-3
 - Data Type Groups tab 8-19

- Recipes tab 8-10, 8-11, 8-12, 8-13, 8-18, 8-19
- Sensor groups tab 8-4, 8-5, 8-6, 8-7, 8-8, 8-9, 8-16
- Tags Control 8-11
- Trend Report
 - calculation parameters 8-26, 8-33

U

- User Event button 3-3
- user management 11-1, 11-3
- User Management window
 - components 11-4
 - deleting user 11-6
 - opening 11-3
- usernames
 - default users 11-1
- users
 - create new 11-5
 - default 11-1
 - deleting 11-6
 - groups
 - ADMINISTRATORS 11-2
 - default 11-1
 - functions 11-2
 - OPERATORS 11-2
 - POWERUSERS 11-2
 - SUPERVISORS 11-2
 - management 11-3
 - modifying 11-6
 - Nobody 3-1, 11-1
- Users Management button 12-2

V

- viewing controls
 - Limits Editor window 12-14
 - Report window 8-35
 - Sensors Editor window 12-6, 12-14

W

- window
 - Limits Editor 12-13
- windows
 - Alarms 7-1
 - components 7-2
 - opening 7-1
 - All Values
 - components 12-20
 - Analog Sensors 6-9
 - components 6-10
 - opening 6-9

Biological Sensors 6-6
 components 6-7
 opening 6-6
 values 6-6
Custom fields configuration 5-6
 components 5-6
Electronic Signature 1-5
FacilityPro SMART Main Page 3-2
Limits Editor 12-13
 components 12-14
 viewing controls 12-14
Limits Editor Configuration 12-18
Limits Editor History 12-16
 components 12-17
Log in 3-6
Main Page 3-1, 3-5
 Channel Status Legend 3-5
 components 3-2
 opening 3-5
Particle Sensors 6-1, 6-2
 components 6-3
 opening 6-1
 values 6-3
Pharmaceutical Net Pro
 components 3-2
Pharmaceutical Net Pro Main Page 3-2
Recipes Editor 4-1, 9-1
 components 4-2
 opening 4-1, 5-1, 9-1, 13-1
Report 8-2
 components, Select by Group 8-21
 components, Select by Sampling 8-28
 viewing controls 8-35
Sampling Editor 5-1
 components 5-2
Sampling editor dialog box 5-6
 scheduling a sampling 5-3
Sampling Report 8-48
Select sensors in group 8-17
Sensors Editor 12-5
 components 12-6
 viewing controls 12-6, 12-14
System management 12-1
 components 12-2
Tags catalog 8-3
 Data Type Groups tab 8-19
 Recipes tab 8-10, 8-11, 8-12, 8-13,
 8-18, 8-19
 Sensor Groups tab 8-4, 8-5, 8-6, 8-7,
 8-8, 8-9, 8-16
User Management
 components 11-4
 deleting user 11-6
 opening 11-3

Z

zoom in on
 report in Report window 8-35